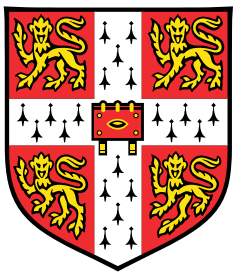
Learning Semiparametric Structural Equation Models

Report 1



Srivathsan Amruth

Supervisor: Prof. Qingyuan Zhao

Department of Physics
Department Of Pure Mathematics and Mathematical Statistics
University of Cambridge

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Declaration

This project report is substantially my own work and conforms to the University of Cambridge's guidelines on plagiarism. Where reference has been made to other research this is acknowledged in the text and bibliography.

Srivathsan Amruth
May 2026

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Abstract

We present SEM2, a probabilistic programming framework for semiparametric structural equation models, developed to allow researchers to specify and train causal models in a fully differentiable setting. We introduce the system design and interface that accommodates flexible parameterisation together with native support for causal interventions. We further present theoretical work on discrete causal graphs that enable partial identification of causal bounds. The associated mathematical and computational challenges of translating this work on discrete graphs into the differentiable SEM2 framework are addressed and allow for theoretical predictions to be validated empirically.

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Chapter 1

Introduction

Structural equation models (SEMs) represent causal mechanisms as directed acyclic graphs and support interventional reasoning [18]. In applied practice, as implemented by widely used software packages [13, 22], these equations are almost universally assumed to be linear and the noise terms Gaussian. This restriction is binding in two ways: nonlinear mechanisms cannot be represented, and the maximum likelihood requires a closed-form density, that is unavailable the moment structural equations or noise distributions are allowed to be nonparametric. Furthermore, in the presence of latent confounding, as in instrumental variable (IV) setting, a further obstruction appears: the observational law may fail to determine the causal effect uniquely. One is then led not to point identification, but to an identified set.

To address this broader class of problems, we developed a probabilistic programming library for SEMiparametric Structural Equation Models, **SEM2**. This report highlights our main contributions, organised around both the development of the SEM2 framework itself and its application to a challenging causal inference problem.

Chapter 3 describes the design of the SEM2 probabilistic programming interface: the formula language, node hierarchy, and differentiable training interface. Chapter 5 provides the mathematical treatment of a finite-support energy-score optimisation framework underlying our computational-statistical method for discrete partial-identification problems. Computational challenges arising from translating this framework into efficient computational implementations are detailed in Chapter 6. We empirically validate this approach on a nonlinear discrete binary IV problem, supporting the predicted behaviour theorised and demonstrating the software's practical utility in empirical research in Chapter 7. Chapter 8 describes future plans.

Chapter 2

Literature Review and Motivation

2.1 Structural Equation Models

Wright [29, 30] proposed representing causal hypotheses as systems of linear equations with uncorrelated errors, and gave graphical rules for reading off predicted correlations from the resulting directed path diagram. Haavelmo and the Cowles Commission [9, 14] formalised the distinction between *structural* parameters, indexing invariant causal mechanisms, and *reduced-form* parameters, indexing the joint distribution of observables. Identification asks whether the former can be recovered from the latter.

Pearl's *Structural Causal Model* (SCM) [18, 21] provides the modern formulation of this programme. A Markovian SCM is a triple $(G, \mathscr{F}, P_U)$ consisting of: a directed acyclic graph (DAG) G whose vertices are the endogenous (modelled) variables; a set of structural equations $\mathscr{F} = \{f_j\}$, one per vertex, of the form $X_j = f_j(\mathrm{PA}_j, U_j)$, where PA_j denotes the parents of X_j in G and U_j is an exogenous noise variable not appearing as a vertex in G; and a product distribution $P_U = \bigotimes_j P_{U_j}$ over mutually independent noise terms. Each equation f_j encodes the causal mechanism by which X_j is determined: given the values of its parents and its private noise, the value of X_j is fixed deterministically. The model is sampled by evaluating all equations in topological order of G.

The central operation is the *intervention*: setting $X_i = x$ means replacing the equation for X_i with the constant x and resampling the remaining variables in topological order. This severs all incoming edges of X_i in G, yielding the interventional distribution $P(X \mid \mathrm{do}(X_i = x))$. Under confounding, this differs from the observational conditional $P(X \mid X_i = x)$, and no conditioning argument recovers it without further assumptions on G. Pearl's do-calculus gives a complete rule system for identifying such distributions from observational data whenever G is known [11, 19].

2.2 The Linear-Gaussian Paradigm and Its Software Ecosystem

In many applications, the structural equations of an SCM take a specific parametric form: each f_j is linear and each noise term ε_j is Gaussian,

$$X_j = \alpha_j + \sum_{i \in \mathrm{PA}_j} \beta_{ji} X_i + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_j^2) \text{ independently.} \tag{2.1}$$

The resulting model is a *linear SEM*. The joint distribution of all endogenous variables is multivariate Gaussian, and structural parameters are estimated by maximum likelihood against the sample covariance matrix. LISREL [13] and the R package `lavaan` [22] are the standard software implementations of this pipeline.

Van Kesteren and Oberski [27] embedded linear SEMs as computation graphs in PyTorch, replacing the covariance-based optimiser with automatic differentiation. Objectives beyond covariance matching follow without reformulating the model: the authors demonstrate least absolute deviation, Bayesian LASSO, and sparse mediation analysis. The structural equations themselves remain linear, and the objective still presupposes a Gaussian joint distribution.

2.3 Limitations of the Linear-Gaussian Specification

Equation (2.1) imposes two constraints that are restrictive in practice.

Linearity. Many structural mechanisms of interest are nonlinear. Returns to education exhibit diminishing marginal returns: an additional year of schooling raises wages substantially at low attainment levels but less so at higher levels [4, 16], a pattern no single linear coefficient captures across the full range. Gene-environment interactions in epidemiology, where a genetic variant modulates the effect of an environmental exposure only above a threshold, are by construction nonlinear [2, 5, 15] and cannot be decomposed into a sum of main effects. More generally, any mechanism involving threshold effects, saturation, or multiplicative interactions among parent variables lies outside the expressive capacity of Equation (2.1).

The likelihood bottleneck. Maximum likelihood requires a tractable closed-form density for the model-induced distribution. When the functions f_j are nonlinear or the noise distributions nonparametric, this density is generally unavailable. In the strict nonparametric limit

the empirical measure is singular with respect to any absolutely continuous reference, and MLE is undefined. Semiparametric estimation therefore requires a density-free discrepancy.

2.4 Data-Driven Models and the Loss of Causal Structure

Neural networks and other black-box predictors relax both constraints by optimising predictive accuracy without imposing any structural form on the data-generating mechanism. A network trained to predict Y from covariates X estimates $\mathbb{E}[Y \mid X = x]$ but cannot compute $\mathbb{E}[Y \mid \mathrm{do}(X = x)]$: the interventional distribution requires knowing which edges are severed, and without an explicit causal graph the do-operator has no object to act on. Black-box models can encode spurious associations that survive training on observational data but would not persist under intervention, and provide no mechanism for detecting this.

2.5 Partial Identification Under Latent Confounding

When the causal graph contains unobserved variables, causal effects need not be *point identified*: multiple structural models may induce the same observational distribution. Consider the instrumental variable (IV) setting, where an unobserved confounder U affects both treatment D and outcome Y, and an instrument Z is assumed to affect Y only through D (the exclusion restriction). The presence of U means that the causal effect of D on Y cannot in general be recovered from observational data alone, though the graph structure does constrain the set of values it can take.

We formalise this *partial identification* where rather than a point estimate, one characterises the *identified set*:

$$\Theta_I(P_X) \;=\; \left\{\, \theta \in \Theta : \Phi(\theta) = P_X \,\right\},$$

where $\Phi : \Theta \to \mathscr{P}(\mathscr{X})$ maps structural parameters to the observational distribution. When $|\Theta_I| = 1$ the effect is point identified; otherwise the inferential output is a set of bounds on the causal estimand.

2.6 SEM2: Advantages and Applications

The preceding sections motivate the need for framework that combines nonlinear structural equations, a density-free estimation objective applicable beyond the Gaussian setting, and support for causal queries including partial identification. On the software side, we need

a probabilistic programming interface that allows researchers to specify and fit arbitrary semiparametric causal graphs using this framework.

We are developing SEM2 to address both. The structural equations f_j can be flexibly parameterised: linear relationships are supported directly, and nonlinear mechanisms (up to the approximation capacity of the chosen architecture) are representable through neural network parameterisation. The estimation objective is generalised from maximum likelihood to the broader *minimum distance estimation* (MDE) framework.

The advantage is the combination of interpretability and flexibility. Unlike classical software, SEM2 is not tied to linear-Gaussian mechanisms; unlike black-box predictors, it retains a graph on which interventions are well defined. This allows for application in settings that are difficult to represent under classical assumptions, including nonlinear causal effects, partial identification, and latent-variable models.

Chapter 3

Computational Model of SEM2

3.1 Design Goals

SEM2 is a probabilistic programming library for specifying, sampling from, and training semiparametric structural equation models. The system is designed around three requirements. First, a researcher should be able to specify a causal graph by writing a formula string, without implementing any estimation or graph-traversal machinery directly. Second, the resulting model must be a differentiable computational object compatible with standard gradient-based optimisers, so that training proceeds without modification regardless of which cost functional is chosen. Third, the model must support interventions natively, realising the do-operator of Section 2.1 at the level of the computational graph.

3.2 The Formula Language

The user-facing interface is a *formula string*: a semicolon-separated sequence of declarations and modifiers, each binding a variable name to either an exogenous node specified by a distribution, or an endogenous node determined by a function of named parent variables. Table 3.1 summarises the syntax.

For generating many separate nodes that follow some distribution, $\mathcal{N}$, we implemented an automatic expansion syntax for ~ to allow concise declaration of multiple exogenous variables:

$$\varepsilon_1\,,\ \varepsilon_2\,,\ \ldots\,,\ u_1\,,\ u_2\,,\ \ldots\ \sim\ \mathcal{N}(\theta)$$

The full formula for a binary instrumental variable model (Figure 3.1) with a latent confounder is written in the language as shown in Listing 3.1. The full pipeline from formula string to a trained, queryable model is summarised in Figure 3.2.

Table 3.1 SEM2 formula syntax. Each formula consists of a node declaration (`~` or `<-`) and a modifier (`+`, `c*`, etc.) specifying its computational behaviour.

`Operator`	**Meaning**	`Example`
`~`	Exogenous Declaration: Root node (value set externally)	`u ~ source(·)`
`Distribution(params)`	Exogenous modifier: Follows a named distribution	`u ~ normal(0,1)`
`<-`	Endogenous Declaration: Function of parent nodes	`y <-` x_{parent}
`+`	Endogenous modifier: Linear combination of parents	`y <- x + z`
`c*`	Endogenous modifier: Fixed (non-learned) weight	`y <- c*x + z`
`n(...)`	Endogenous modifier: Non-linear function of parents	`y <- n(x,z)`
`do(`$\mathscr{C}$`(X))`	Intervention: Overrides named node by constraint	`do(0<D<1)`

Here `c*` denotes multiplication by a user-prescribed constant $c \in \mathbb{R}$, fixing that weight as non-learnable; the canonical case $c = 1$ enforces a unit-weight constraint. The do-operator `do(`$\mathscr{C}$`(X))` applies a constraint $\mathscr{C}(X) = \bigwedge_{i=1}^{n}(X \bowtie_i x_i)$, where $x_i \in \mathscr{X}$ is the support of X and the relational operator, $\bowtie_i \in \begin{cases} \{=,<,>\}, & \text{if } (\mathscr{X},\leq) \text{ is totally ordered,} \\ \{=\}, & \text{otherwise} \end{cases}$, fixing the node's value to any $x \in \mathscr{X}$ satisfying $\mathscr{C}(x)$. In practice: for continuous or discrete ordered variables, constraints may fix an exact value (`do(X=x)`), impose a one-sided bound (`do(X>a)` or `do(X<b)`), or define a range (`do(a<X<b)`); for unordered categorical variables, only exact value constraints (`do(X=x)`) are defined.

```
U, εD , εY ~ uniform(0, 1);
Z ~ bernoulli(0.5) ;
D <- n(Z, U, εD );
Y <- n(D, U, εY );

% Intervention at eval time:
do(Z=1)
```

Listing 3.1 SEM2 formula for a binary IV model. The latent confounder U, exogenous noise terms ε_D, ε_Y, and binary instrument Z are declared with `~`; the endogenous variables D and Y, declared with `<-`, are deterministic nonlinear functions of their parents. At evaluation time, interventions such as `do(Z=1)` are applied to generate interventional samples.

Here U is the latent confounder, ε_D and ε_Y are independent noise terms, and Z is the instrument. Every source of randomness is declared as an exogenous node. The endogenous

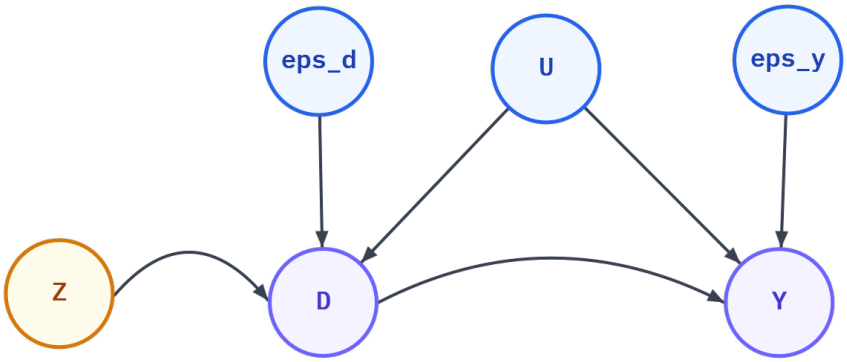


Fig. 3.1 Causal graph for the binary instrumental variable model. Orange: instrument Z, which affects treatment D but not outcome Y directly (exclusion restriction). Purple: endogenous variables D and Y, each determined by their parents and a private noise term. Blue: exogenous root nodes, latent confounder U (affecting both D and Y, creating unobserved confounding) and independent noise terms ε_D, ε_Y.

variables D and Y are deterministic functions of their named parents: given the values of U, ε_D, and Z, the value of D is fully determined by the structural equation f_D, and similarly for Y. This reflects a core design principle: all stochasticity is localised to exogenous nodes, and the forward pass through any endogenous node is deterministic and differentiable.

3.3 Node Abstractions

Each declaration in the formula string becomes one node in the DAG. All nodes share a common abstract base class encoding three responsibilities: storing the current computed value, maintaining the list of parent node names, and registering any learnable parameters with the PyTorch autograd engine. From this base, two abstract subclasses partition the node space:

Exogenous nodes have no parents in the graph. They are the sole sources of randomness in the model: each exogenous node samples from a specified distribution at the start of every forward pass, injecting stochasticity that propagates deterministically through the downstream graph. The current implementation provides a named distribution wrapper compatible with any distribution in the `torch.distributions` family. The design is not

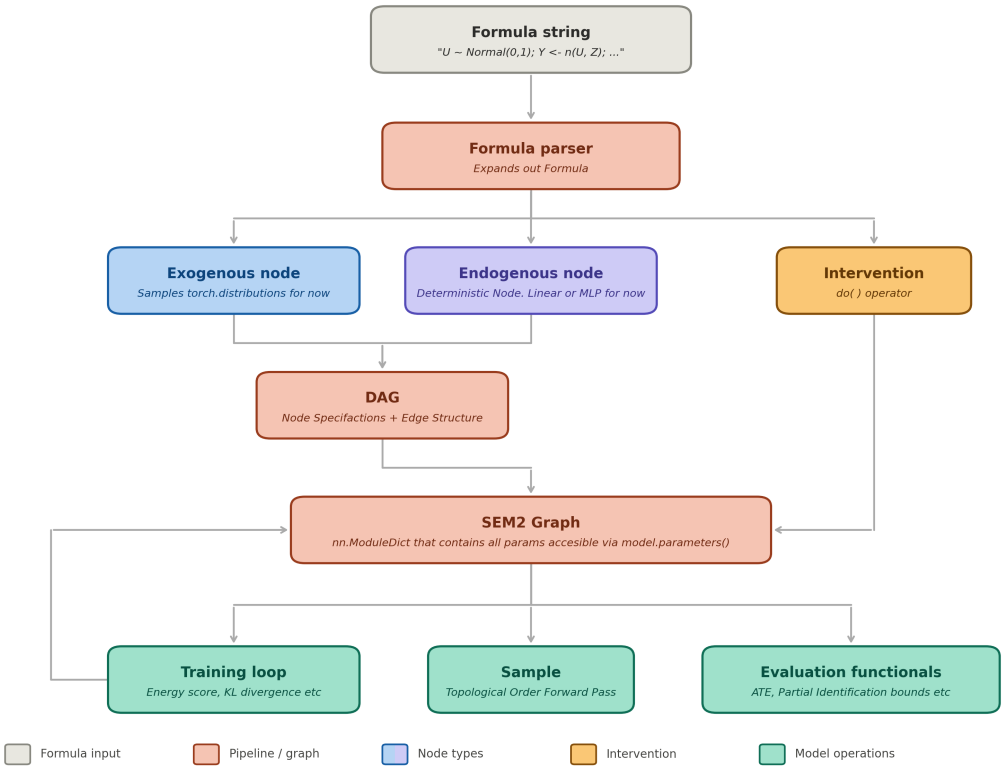


Fig. 3.2 Overview of the SEM2 computational framework. A formula string is parsed into typed specifications, exogenous nodes (blue), endogenous nodes (purple), and interventions (orange), which are assembled into a DAG and registered as a differentiable SEM2 Graph. At runtime the graph supports three operations: sampling via a topological-order forward pass, training via a gradient-based optimiser that feeds parameter updates back into the graph, and evaluation of causal functionals such as the ATE and partial identification bounds.

restricted to parametric noise: because the exogenous interface requires only that the node produce a tensor of the correct shape, arbitrary noise models including nonparametric ones can be accommodated by supplying a custom sampler. Instruments and other contextual variables whose values are set externally are also naturally represented as exogenous nodes.

Endogenous nodes take one or more parent values as input and compute their output deterministically. Three primary concrete endogenous operations are currently implemented as summarised by Table 3.1. The `Linear` modifier operation, controlled by `+` syntax, computes an affine function of its parents, $X_j = W \cdot \mathrm{PA}_j + b$, with W and b as learnable parameters. Individual weights can be fixed to prescribed constants via the `c*` modifier, where $c \in \mathbb{R}$ is specified by the user; like `1*`, useful for imposing unit-weight constraints such as measurement model loadings. The `Nonlinear` operation `n(...)` replaces the affine map with a multi-layer perceptron(MLP) whose depth, width, and activation function are

configurable; this is the operation that gives SEM2 its semiparametric character, allowing any sufficiently regular nonlinear mechanism to be represented. The `n(...)` interface is intended to be extended in future versions to support a wider class of nonlinear functional forms beyond the MLP parameterisation. The separation between exogenous and endogenous nodes is what makes the forward pass fully differentiable: because no endogenous node samples, the gradient of the loss with respect to all learnable parameters is well-defined and computable by reverse-mode automatic differentiation.

Interventions are expressed separately from the model declaration via the `do(`$\mathscr{C}$`(X))` operator, which overrides the value of any named node according to a constraint $\mathscr{C}$, bypassing its structural equation and propagating the fixed value to all downstream nodes in topological order. This is the computational realisation of Pearl's do-operator described in Section 2.1.

3.4 From Formula to Differentiable DAG

Once the formula string is parsed into a collection of typed node objects, the DAG is assembled by reading the parent names from the right-hand side of each endogenous declaration. A topological ordering of the nodes is then computed: an ordering in which every node appears after all of its parents, guaranteeing that parent values are available when each node is evaluated. All nodes are registered together in a single PyTorch `nn.ModuleDict`, which causes all learnable parameters across all nodes to be visible to any standard PyTorch optimiser through a single call to `model.parameters()`. The resulting object is simultaneously a causal model in Pearl's sense, a directed acyclic graph with structural equations, and a differentiable PyTorch module whose parameters can be updated by gradient descent.

3.5 Forward Pass and Interventions

During training, each batch is evaluated by traversing the nodes in topological order. At each node, the cached output values of its parents are collected and passed to the node's computation: a distribution sample for exogenous nodes, an affine map or MLP evaluation for endogenous nodes. The result is stored in the node's value cache for use by downstream nodes. This single traversal produces the joint sample of all variables in the model for the current batch.

Interventions are implemented by fixing the value of one or more nodes before the traversal begins. When a node's value is fixed externally, the traversal respects that value and does not recompute it from the node's structural equation; the remaining nodes are evaluated

as usual in topological order. This is the computational realisation of Pearl's do-operator: severing the incoming edges of the intervened node and replacing its equation with a constant, while leaving the rest of the causal graph intact.

The model can be instantiated and sampled from immediately upon construction, before any training has taken place, since all node types are defined at construction time and the forward pass requires no fitted parameters to execute. After training, the same sampling procedure draws from the fitted model, and interventional samples are obtained by applying the do-operator to the relevant nodes as described above.

3.6 Training

The model as described is a parameterised sampler. Training amounts to choosing a cost functional measuring discrepancy between the model-induced distribution and the empirical distribution of a dataset, then minimising it with respect to the learnable parameters by gradient descent. When all structural equations are differentiable, the forward pass is differentiable end-to-end and standard PyTorch optimisers apply directly.

The choice of cost functional is decoupled from the model specification: the same formula string and DAG can be trained under the energy distance, the maximum likelihood objective if a closed-form density exists, or any other differentiable discrepancy, without altering the model itself. The design, selection, mathematical properties and rationale for these cost functionals are the subject of the coming Chapter 4.

A key observation to bear in mind is that the discrete setting, where endogenous variables take values in a finite state space, breaks this differentiability. Resolving this issue constitutes the central computational challenge addressed in this report, since a principled mechanism for gradient propagation is required in order to generalise the framework beyond continuously differentiable models to a broader class of discrete and hybrid problems. This challenge, and the corresponding methodology, are developed later in Chapters 5 and 6.

Chapter 4

Mathematical Foundations

4.1 Minimum Distance Estimation

Let $\Phi : \Theta \to \mathscr{P}(\mathscr{X})$ denote the forward map of a parametric generative model, sending structural parameters $\theta \in \Theta$ to the induced distribution $\Phi(\theta)$ over the observed variables. Given an empirical distribution P_n constructed from n observations, the *minimum distance estimator* (MDE) is

$$\hat{\theta} \;=\; \arg\min_{\theta\in\Theta} \mathscr{D}\big(\Phi(\theta), P_n\big),$$

where $\mathscr{D}$ is a discrepancy functional on $\mathscr{P}(\mathscr{X})$. Maximum likelihood estimation is recovered as a special case by setting $\mathscr{D}(P,Q) = \mathrm{KL}(Q\|P)$, which requires a tractable density for $\Phi(\theta)$. For generative models where sampling from $\Phi(\theta)$ is straightforward but evaluating its density is not, a sample-based objective is needed. The choice of $\mathscr{D}$ determines both what can be estimated and how efficiently: different choices induce different geometries on $\mathscr{P}(\mathscr{X})$ and hence different optimisation landscapes on Θ. The remainder of this chapter briefly reviews two major families of probability metrics used as MDE objectives, the maximum mean discrepancy and the Wasserstein distance, before Chapter 5 develops the specific loss functional we derive for discrete graph problems in SEM2.

4.2 Kernel Discrepancies and the Maximum Mean Discrepancy

Let $\mathscr{H}_k$ be a reproducing kernel Hilbert space (RKHS) associated to a symmetric positive definite kernel $k : \mathscr{X} \times \mathscr{X} \to \mathbb{R}$. The *maximum mean discrepancy* (MMD) between distributions P and Q is

$$\mathrm{MMD}_k^2(P,Q) \;=\; \mathbb{E}_{X,X'\sim P}[k(X,X')] - 2\,\mathbb{E}_{X\sim P,Y\sim Q}[k(X,Y)] + \mathbb{E}_{Y,Y'\sim Q}[k(Y,Y')],$$

which equals the squared norm $\|\mu_P - \mu_Q\|^2_{\mathscr{H}_k}$ of the difference of kernel mean embeddings $\mu_P = \mathbb{E}_{X\sim P}[k(X,\cdot)] \in \mathscr{H}_k$. When k is *characteristic*, the MMD [8] is a metric on $\mathscr{P}(\mathscr{X})$: $\mathrm{MMD}_k(P,Q) = 0$ if and only if $P = Q$.

A classical result of Schoenberg [23] establishes that the kernel $k(x,y) = -\|x-y\|^\beta$ for $\beta \in (0,2)$ is conditionally negative definite. Substituting into the MMD expression yields, up to a sign convention, the *energy distance* of Székely and Rizzo [26]:

$$\mathscr{D}_E^2(P,Q) \;=\; 2\,\mathbb{E}\|X-Y\|^\beta - \mathbb{E}\|X-X'\|^\beta - \mathbb{E}\|Y-Y'\|^\beta,$$

where $X, X' \overset{\text{iid}}{\sim} P$ and $Y, Y' \overset{\text{iid}}{\sim} Q$. The energy distance is a metric on $\mathscr{P}(\mathscr{X})$ for any $\beta \in (0,2)$, estimable directly from samples without solving any auxiliary optimisation problem. The *energy score* arises as the specialisation to $Q = \delta_{\mathbf{y}_{\text{obs}}}$, a Dirac measure at a single observation. Since $Y = Y' = \mathbf{y}_{\text{obs}}$ almost surely, the term $\mathbb{E}\|Y-Y'\|^\beta = 0$, vanishes and the energy distance reduces to

$$\mathrm{ES}(P, \mathbf{y}_{\text{obs}}) \;=\; 2\,\mathbb{E}_{X\sim P}\|X - \mathbf{y}_{\text{obs}}\|^\beta - \mathbb{E}_{X,X'\sim P}\|X-X'\|^\beta.$$

This is a strictly proper scoring rule [6]: it is uniquely minimised over all predictive distributions P when P equals the true data-generating distribution.

4.3 Optimal Transport and the Wasserstein Distance

A second major family of probability metrics arises from optimal transport [20, 28]. Given a cost function $c : \mathscr{X} \times \mathscr{X} \to \mathbb{R}_{\geq 0}$, the Kantorovich problem seeks the transport plan $\pi^* \in \mathscr{P}(\mathscr{X} \times \mathscr{X})$ with marginals P and Q minimising the expected transport cost:

$$\mathscr{W}_c(P,Q) \;=\; \inf_{\pi \in \Pi(P,Q)} \int_{\mathscr{X}\times\mathscr{X}} c(x,y)\,\mathrm{d}\pi(x,y),$$

where $\Pi(P,Q)$ denotes the set of all couplings of P and Q. Setting $c(x,y) = \|x-y\|^2$ yields the squared Wasserstein-2 distance $\mathscr{W}_2^2(P,Q)$, which metrises weak convergence of probability measures with finite second moments and induces a Riemannian structure on $\mathscr{P}_2(\mathscr{X})$ [1].

Computing $\mathscr{W}_2$ requires solving a linear programme over the coupling space $\Pi(P,Q)$, which is computationally expensive in general. The energy distance requires no transport plan and is estimable in $O(n^2)$ time from n samples. Both are probability metrics admitting gradient-based optimisation, but with different geometric properties (Figure 4.1): the Wasserstein distance is sensitive to the ground metric on $\mathscr{X}$ through the transport cost c, while the energy distance inherits this sensitivity through the kernel $k(x,y) = -\|x-y\|^\beta$.

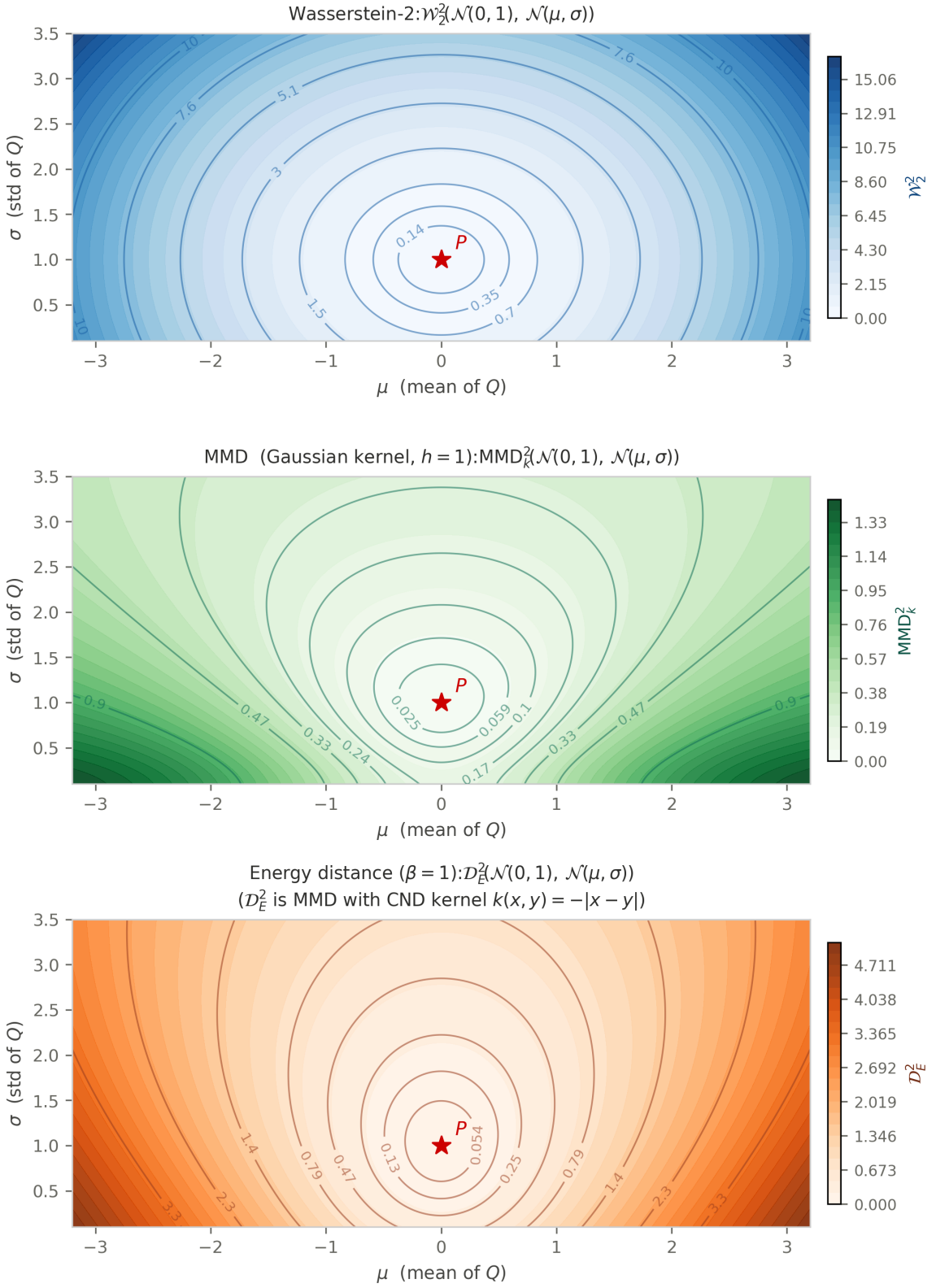


Fig. 4.1 Level sets of three probability discrepancies over the parameter space of $Q = \mathcal{N}(\mu, \sigma^2)$, with $P = \mathcal{N}(0,1)$ fixed (red star). All three attain their minimum at P, but induce geometrically distinct optimisation landscapes on the parameter space. The Wasserstein-2 distance $\mathscr{W}_2^2$ [28] yields elliptical contours, penalising mean and variance shifts comparably. The MMD with a Gaussian kernel $k_h(x,y) = \exp(-|x-y|^2/2h^2)$, $h=1$ [8] saturates at large separations due to the bounded kernel, producing flat contours far from P. The energy distance $\mathscr{D}_E^2$ [25], a special case of MMD with the conditionally negative definite kernel $k(x,y) = -|x-y|$ [24], yields different contours across the full parameter space.

4.4 Deep Learning as a Structural Parameterisation

In a classical linear SEM, each mechanism has the form

$$X_j = \alpha_j + \sum_{i \in \mathrm{PA}_j} \beta_{ji} X_i + \varepsilon_j.$$

In SEM2 this is generalised to

$$X_j = f_{\theta_j}\left(X_{\mathrm{PA}_j}, U_j\right),$$

where f_{θ_j} may be represented by a neural network and U_j is an exogenous noise variable. Multilayer neural networks provide a flexible class of function approximators [7, 10], and automatic differentiation makes it possible to optimise the resulting sample-based losses in modern computational frameworks [17]. This gives SEM2 enough flexibility to represent nonlinear threshold effects, interactions, and non-Gaussian mechanisms while preserving the causal semantics of the graph: interventions are still implemented by replacing selected structural equations, and the remaining variables are generated in topological order. The cost of this flexibility is nonconvex optimisation and dependence on architectural choices.

Chapter 5

Discrete Partial Identification in Causal Graphs

In this chapter, we develop our numerical optimisation method for approximating partial identification bounds in discrete causal graphs and requisite underlying finite-dimensional mathematics. A discrete structural graph induces a smooth map

$$\Phi : \Theta_{\text{feas}} \to \Delta^{K-1}, \qquad \Phi(\theta) = \mathbf{p}(\theta). \tag{5.1}$$

from feasible structural parameters to an observed probability vector. On finite support, the energy score becomes an exact quadratic form in $\mathbf{p}(\theta)$.

The key distinction is between exact fit and penalised relaxation. Exact observational fit imposes

$$\mathbf{p}(\theta) = \mathbf{q},$$

and the sharp identified set is the image of this exact-fit fibre under the causal contrast $a(\theta)$. By contrast, the penalised objective

$$\mathscr{L}_{ES}(\mathbf{p}(\theta), \mathbf{q}) + \lambda a(\theta)$$

trades off observational discrepancy against the causal contrast. A finite λ-sweep therefore traces a supported regularisation path in the (δ, a)-plane, where

$$\delta(\theta) = \mathscr{D}_E^2(\mathbf{p}(\theta), \mathbf{q}).$$

It does not by itself prove sharp bounds, but it gives a controlled numerical route to the tolerance envelopes around the exact-fit fibre.

The results below establish the pieces used later in the empirical chapter: the finite-support matrix form of the energy score, its equivalence to energy distance, the exact-fit formulation of sharp identified sets, the monotonicity of global λ-sweep solutions, and the local KKT expansion showing that near an interior exact fit the causal contrast moves

linearly in λ while the observational loss grows quadratically. These mathematical claims are empirically validated in Chapter 7.

5.1 From Discrete Graphs to Probability Vectors

Let $X_1, \ldots, X_m$ be the variables in a discrete structural graph, possibly including latent variables. For node i, state k, and parent configuration u, write

$$\pi_{i,k|u}(\theta) = \Pr_{\theta}(X_i = k \mid X_{\mathrm{Pa}(i)} = u).$$

For a complete assignment $x = (x_1, \ldots, x_m)$, the joint law factorises as

$$p_\theta(x) = \prod_{i=1}^{m} \pi_{i,x_i|x_{\mathrm{Pa}(i)}}(\theta). \tag{5.2}$$

If some variables are latent, the observed probability vector $\mathbf{p}(\theta)$ is obtained by summing this joint law over the latent coordinates.

Thus the graph induces a structural probability map

$$\Phi : \Theta_{\mathrm{feas}} \to \Delta^{K-1}, \qquad \Phi(\theta) = \mathbf{p}(\theta),$$

where Θ_{feas} is the feasible structural parameter space and Δ^{K-1} is the observed probability simplex. When the elementary probabilities are parameterised by logits, the normalisation constraints are built in. If raw probabilities are used, one must impose

$$\sum_k \pi_{i,k|u} = 1, \qquad \pi_{i,k|u} \geq 0$$

for every node i and parent configuration u. On the relative interior of the feasible set, $\mathbf{p}(\theta)$ is a smooth function of θ. The Jacobian $\mathbf{J}_p(\theta)$ used below is the derivative of this structural probability map.

5.2 The Discrete Energy Score

Let the observed support be

$$\mathscr{S} = \{\mathbf{s}_1, \ldots, \mathbf{s}_K\}.$$

Write

$$\mathbf{p}(\theta) = (p_1(\theta), \ldots, p_K(\theta))^\top \in \Delta^{K-1}$$

for the model probability vector on this support, and let

$$\mathbf{q} = (q_1, \ldots, q_K)^\top \in \Delta^{K-1}$$

be the empirical probability vector. For a single observation,

$$q_k = \mathbf{1}[\mathbf{y}_{\text{obs}} = \mathbf{s}_k].$$

For a sample $\mathbf{y}^{(1)}, \ldots, \mathbf{y}^{(n)}$,

$$q_k = \frac{1}{n} \sum_{r=1}^{n} \mathbf{1}[\mathbf{y}^{(r)} = \mathbf{s}_k].$$

Define the distance matrix

$$D_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|^\beta, \qquad 0 < \beta < 2. \tag{5.3}$$

Throughout this chapter we use the convention

$$\mathscr{L}_{ES}(\mathbf{p}, \mathbf{q}) = \mathbb{E}\|X - Y\|^\beta - \frac{1}{2}\mathbb{E}\|X - X'\|^\beta. \tag{5.4}$$

where $X, X' \overset{\text{iid}}{\sim} \mathbf{p}$ and $Y \sim \mathbf{q}$. Multiplying this loss by 2 gives the alternative normalisation used in some definitions of the energy score. Such a rescaling does not change minimisers and only rescales the penalty weight.

Proposition 1 (Finite-support energy score)**.** *For finite support,*

$$\mathscr{L}_{ES}(\mathbf{p}(\theta), \mathbf{q}) = \mathbf{p}(\theta)^\top \mathbf{D}\mathbf{q} - \frac{1}{2}\mathbf{p}(\theta)^\top \mathbf{D}\mathbf{p}(\theta). \tag{5.5}$$

Proof. Since $X \sim \mathbf{p}$ and $Y \sim \mathbf{q}$,

$$\mathbb{E}\|X - Y\|^\beta = \sum_{i=1}^{K} \sum_{j=1}^{K} p_i q_j D_{ij} = \mathbf{p}^\top \mathbf{D}\mathbf{q}.$$

Similarly,

$$\mathbb{E}\|X - X'\|^\beta = \sum_{i=1}^{K} \sum_{j=1}^{K} p_i p_j D_{ij} = \mathbf{p}^\top \mathbf{D}\mathbf{p}.$$

Substitution gives the claim. □

5.3 Relation to Energy Distance

The squared energy distance is

$$\mathscr{D}_E^2(\mathbf{p},\mathbf{q}) = 2\mathbf{p}^\top \mathbf{D}\mathbf{q} - \mathbf{p}^\top \mathbf{D}\mathbf{p} - \mathbf{q}^\top \mathbf{D}\mathbf{q}. \tag{5.6}$$

Hence

$$\mathscr{D}_E^2(\mathbf{p},\mathbf{q}) = 2\mathscr{L}_{ES}(\mathbf{p},\mathbf{q}) - \mathbf{q}^\top \mathbf{D}\mathbf{q} = -(\mathbf{p}-\mathbf{q})^\top \mathbf{D}(\mathbf{p}-\mathbf{q}), \tag{5.7}$$

where the last equality uses symmetry of $\mathbf{D}$.

By Schoenberg's theorem [23], $\mathbf{D}$ is strictly conditionally negative definite for distinct support points and $0 < \beta < 2$. Thus, for every nonzero $\mathbf{v}$ satisfying

$$\mathbf{1}^\top \mathbf{v} = 0,$$

one has

$$\mathbf{v}^\top \mathbf{D}\mathbf{v} < 0. \tag{5.8}$$

Since $\mathbf{p}-\mathbf{q}$ has zero sum,

$$\mathscr{D}_E^2(\mathbf{p},\mathbf{q}) \geq 0,$$

with equality if and only if $\mathbf{p} = \mathbf{q}$. The squared energy distance is therefore a proper discrepancy on the simplex. Moreover,

$$\mathscr{L}_{ES}(\mathbf{p},\mathbf{q}) = \frac{1}{2}\mathscr{D}_E^2(\mathbf{p},\mathbf{q}) + \frac{1}{2}\mathbf{q}^\top \mathbf{D}\mathbf{q}.$$

Since the second term is independent of $\mathbf{p}$, minimising $\mathscr{L}_{ES}$ is equivalent to minimising $\mathscr{D}_E^2$.

For later use, define

$$\delta(\theta) = \mathscr{D}_E^2(\mathbf{p}(\theta),\mathbf{q}). \tag{5.9}$$

Then

$$\delta(\theta) \geq 0, \qquad \delta(\theta) = 0 \iff \mathbf{p}(\theta) = \mathbf{q}.$$

It is also convenient to write

$$\ell(\theta) = \mathscr{L}_{ES}(\mathbf{p}(\theta),\mathbf{q}) - \mathscr{L}_{ES}(\mathbf{q},\mathbf{q}) = \frac{1}{2}\delta(\theta). \tag{5.10}$$

The shifted loss ℓ has the same gradients and minimisers as $\mathscr{L}_{ES}$, but vanishes at exact observational fit.

5.4 Exact Fit and Sharp Identified Sets

Let $a : \Theta_{\text{feas}} \to \mathbb{R}$ be a scalar causal contrast. The local analysis below uses only differentiability of a. In the binary-treatment experiments, we use the average treatment effect (ATE)

$$a(\theta) = \mathbb{E}_\theta[Y \mid \text{do}(D = 1)] - \mathbb{E}_\theta[Y \mid \text{do}(D = 0)]. \tag{5.11}$$

The sign convention in the penalised sweep is tied to this definition. If an implementation reports the opposite contrast, the interpretation of increasing and decreasing effects must be reversed. With bounded outcomes, a is bounded on the feasible probability space. For binary Y, the ATE lies in $[-1, 1]$.

The exact-fit set is

$$\mathscr{R} = \{\theta \in \Theta_{\text{feas}} : \mathbf{p}(\theta) = \mathbf{q}\} = \{\theta \in \Theta_{\text{feas}} : \delta(\theta) = 0\}. \tag{5.12}$$

The sharp identified set for the causal contrast is

$$\mathscr{I} = a(\mathscr{R}) = \{a(\theta) : \theta \in \mathscr{R}\}. \tag{5.13}$$

Its sharp outer bounds are

$$\underline{a} = \inf_{\theta \in \Theta_{\text{feas}}} a(\theta) \quad \text{subject to} \quad \mathbf{p}(\theta) = \mathbf{q}, \tag{5.14}$$

$$\overline{a} = \sup_{\theta \in \Theta_{\text{feas}}} a(\theta) \quad \text{subject to} \quad \mathbf{p}(\theta) = \mathbf{q}. \tag{5.15}$$

When the infimum and supremum are attained, these are equality-constrained minimum and maximum programs.

Figure 5.1 illustrates the construction. If $\mathscr{I}$ is a singleton, the causal contrast is point identified. If $\mathscr{I}$ contains more than one value, the causal contrast is partially identified. Nonidentification of the structural parameter θ alone is not enough for partial identification of a; the contrast must vary over the exact-fit set.

5.5 The Penalised λ Sweep

For $\lambda \in \mathbb{R}$, define

$$F_\lambda(\theta) = \mathscr{L}_{ES}(\mathbf{p}(\theta), \mathbf{q}) + \lambda a(\theta). \tag{5.16}$$

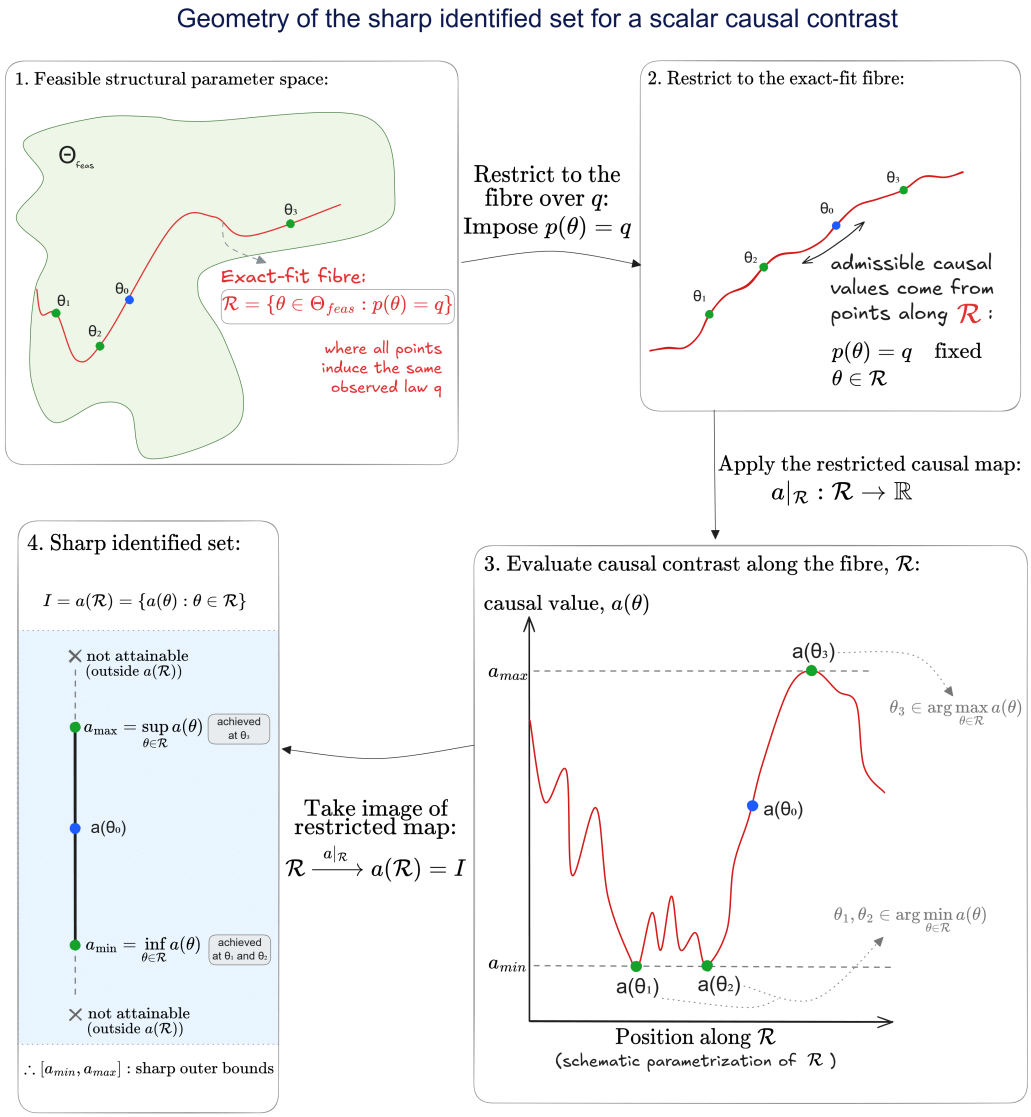


Fig. 5.1 From feasible structural parameters to sharp causal contrast bounds. The exact-fit set $\mathscr{R} = \{\theta \in \Theta_{\text{feas}} : \mathbf{p}(\theta) = \mathbf{q}\}$ consists of all parameter values that reproduce the observed law. These parameter values are observationally indistinguishable, but they may imply different causal values $a(\theta)$. The full sharp identified set is $a(\mathscr{R})$, and the sharp outer bounds are $\underline{a} = \inf_{\theta \in \mathscr{R}} a(\theta)$ and $\overline{a} = \sup_{\theta \in \mathscr{R}} a(\theta)$. The drawing shows $\mathscr{R}$ as one connected curve only for clarity. In general, $\mathscr{R}$ may have several connected components, and $a(\mathscr{R})$ may be a union of separated ranges rather than the whole interval $[\underline{a}, \overline{a}]$.

Equivalently, after subtracting the constant $\mathscr{L}_{ES}(\mathbf{q},\mathbf{q})$,

$$F_\lambda(\theta) \equiv \ell(\theta) + \lambda a(\theta) = \frac{1}{2}\delta(\theta) + \lambda a(\theta), \tag{5.17}$$

where equivalence means equality up to an additive constant. With the ATE convention above, positive λ penalises large values of a, while negative λ rewards them (Figure 5.2). Thus the λ-sweep trades off observational discrepancy against the causal contrast. For finite λ, it does not impose exact observational fit unless the optimum happens to satisfy $\delta(\theta) = 0$.

For the penalised objective to have a finite global minimum for each finite λ, the objective must be bounded below. In the finite discrete setting, probability constraints place the elementary probabilities in products of simplices. With bounded outcomes, the causal contrast is bounded, so this condition is satisfied on compact feasible parameterisations.

5.6 Tolerance Envelopes and Supported Trade-offs

For $\varepsilon \geq 0$, define the tolerance set of causal values

$$\mathscr{I}_\varepsilon = \{a(\theta) : \delta(\theta) \leq \varepsilon, \theta \in \Theta_{\text{feas}}\}. \tag{5.18}$$

When exact fit is attainable,

$$\mathscr{I}_0 = \mathscr{I}.$$

The lower and upper tolerance envelopes are

$$a_-(\varepsilon) = \inf\{a(\theta) : \delta(\theta) \leq \varepsilon, \theta \in \Theta_{\text{feas}}\}, \tag{5.19}$$

$$a_+(\varepsilon) = \sup\{a(\theta) : \delta(\theta) \leq \varepsilon, \theta \in \Theta_{\text{feas}}\}. \tag{5.20}$$

The use of $\delta(\theta) \leq \varepsilon$ is essential. These are tolerance balls, not fixed-distance shells. If one used $\delta(\theta) = \varepsilon$ instead, the exact-fit set $\{\delta = 0\}$ would not be contained in the relaxed set. The nesting property would be lost.

The tolerance sets are nested. If $\varepsilon_1 \leq \varepsilon_2$, then

$$\{\theta : \delta(\theta) \leq \varepsilon_1\} \subseteq \{\theta : \delta(\theta) \leq \varepsilon_2\}.$$

Consequently,

$$\mathscr{I}_{\varepsilon_1} \subseteq \mathscr{I}_{\varepsilon_2}.$$

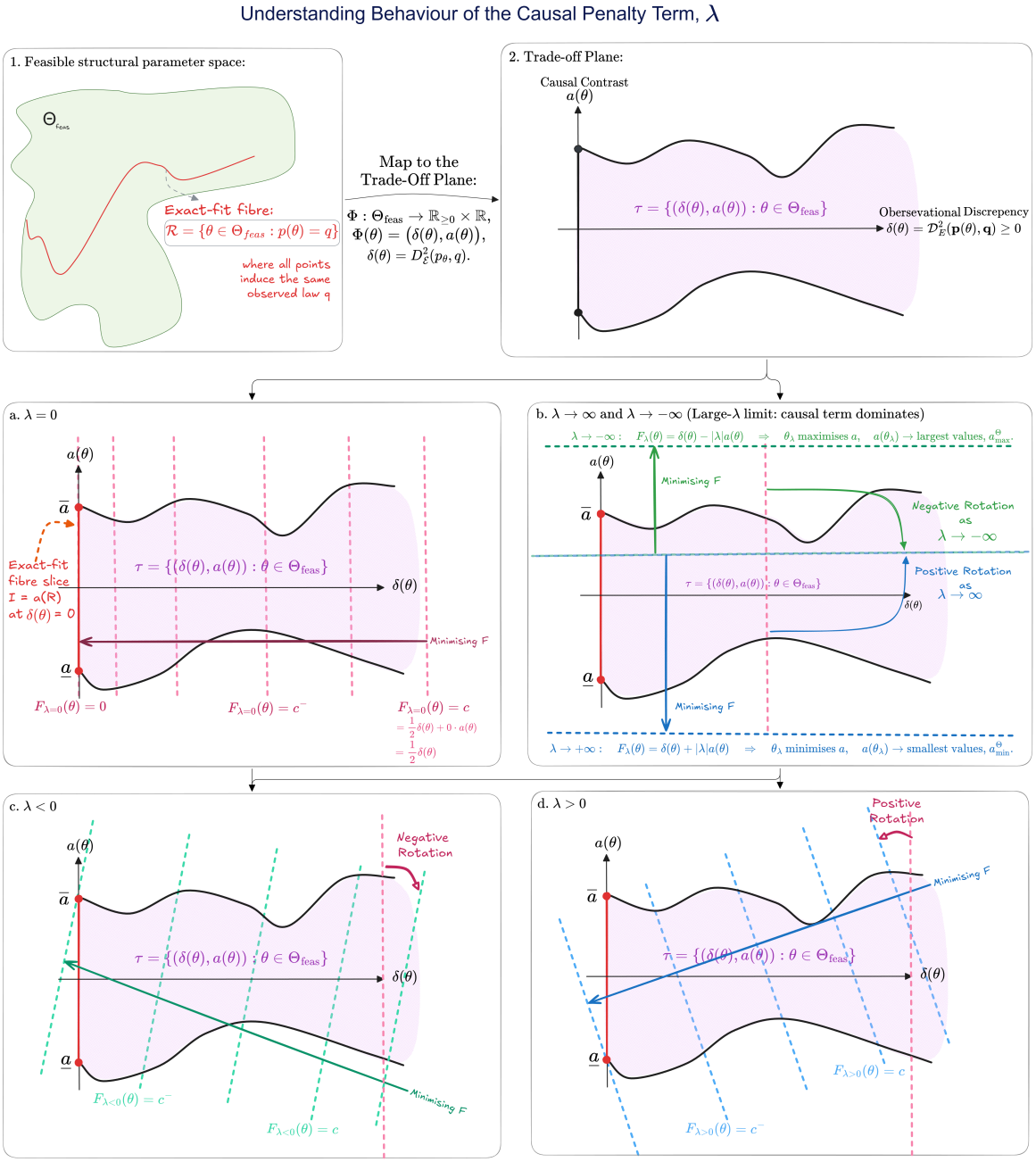


Fig. 5.2 Geometry of the λ-penalised objective, $F_\lambda(\theta) = \delta(\theta) + \lambda a(\theta)$. Feasible structural parameters are mapped to the trade-off plane by $\Phi(\theta) = (\delta(\theta), a(\theta))$, where $\delta(\theta) = D^2_{\mathscr{E}}(p_\theta, q)$ is the observational discrepancy and $a(\theta)$ is the causal contrast. For $\lambda \neq 0$, the level sets satisfy $F_\lambda(\theta) = c \iff a = \frac{c-\delta}{\lambda}$, so their slope in the (δ, a)-plane is $\frac{da}{d\delta} = -\frac{1}{\lambda}$. Equivalently, $\nabla_{(\delta,a)} F_\lambda = (1, \lambda)$, so descent moves in the direction $(-1, -\lambda)$: for $\lambda < 0$ it moves left and upward towards better fit and larger a, while for $\lambda > 0$ it moves left and downward towards better fit and smaller a. The exact-fit fibre $\mathscr{R} = \{\theta : \delta(\theta) = 0\}$ projects to the identified interval $[\underline{a}, \bar{a}]$, and as $\lambda \to 0_-$ and $\lambda \to 0_+$ one-sided values of λ select its upper and lower endpoints respectively. Once again, the above drawings are for visualisation and the image $\Phi(\Theta_{\text{feas}})$ need not have a smooth boundary (nothing asserts the connectedness or convexity of the feasible image), and the exact-fit fibre $\mathscr{R}$ need not be connected.

In terms of outer envelopes,

$$a_-(\varepsilon_2) \le a_-(\varepsilon_1), \qquad a_+(\varepsilon_2) \ge a_+(\varepsilon_1). \tag{5.21}$$

Thus increasing the allowed observational discrepancy can only expand the outer causal envelope. It does not mean that every feasible point with larger δ has a larger causal value. The monotonicity is a property of the extremal envelopes, not of all points in the (δ, a) plane.

Let θ_λ be a global minimiser of

$$\ell(\theta) + \lambda a(\theta) = \frac{1}{2}\delta(\theta) + \lambda a(\theta),$$

and set

$$\varepsilon_\lambda = \delta(\theta_\lambda).$$

If $\lambda > 0$, then θ_λ attains the lower tolerance envelope at its own discrepancy level:

$$a(\theta_\lambda) = a_-(\varepsilon_\lambda). \tag{5.22}$$

Indeed, for any θ with $\delta(\theta) \le \varepsilon_\lambda$, global optimality gives

$$\frac{1}{2}\delta(\theta) + \lambda a(\theta) \ge \frac{1}{2}\varepsilon_\lambda + \lambda a(\theta_\lambda).$$

Therefore

$$\lambda\{a(\theta) - a(\theta_\lambda)\} \ge \frac{1}{2}\{\varepsilon_\lambda - \delta(\theta)\} \ge 0.$$

Since $\lambda > 0$,

$$a(\theta) \ge a(\theta_\lambda).$$

Hence θ_λ attains $a_-(\varepsilon_\lambda)$. Similarly, if $\lambda < 0$, then

$$a(\theta_\lambda) = a_+(\varepsilon_\lambda). \tag{5.23}$$

Thus the weighted-sum sweep samples supported points on the lower envelope for positive λ, and supported points on the upper envelope for negative λ. It need not recover the entire global frontier, especially when the feasible trade-off set

$$\tau = \{(\delta(\theta), a(\theta)) : \theta \in \Theta_{\text{feas}}\}. \tag{5.24}$$

is nonconvex, disconnected, or has unsupported efficient points.

This also clarifies the relation to exact sharp bounds. Suppose the exact-fit lower bound $\underline{a}$ is attained by some $\theta_- \in \mathscr{R}$. Then for every global minimiser with $\lambda > 0$,

$$a(\theta_\lambda) \leq \underline{a}.$$

Likewise, if the exact-fit upper bound $\overline{a}$ is attained by some $\theta_+ \in \mathscr{R}$, then for every global minimiser with $\lambda < 0$,

$$a(\theta_\lambda) \geq \overline{a}.$$

Finite nonzero λ can therefore push the selected causal value outside the exact sharp bounds by sacrificing observational fit (Figure 5.3).

A mixture contamination gives a concrete way to view the tolerance ball. If

$$\mathbf{q}_\eta = (1-\eta)\mathbf{q} + \eta\mathbf{r},$$

then

$$\mathbf{q}_\eta - \mathbf{q} = \eta(\mathbf{r} - \mathbf{q}),$$

and hence

$$\mathscr{D}_E^2(\mathbf{q}_\eta, \mathbf{q}) = \eta^2 \mathscr{D}_E^2(\mathbf{r}, \mathbf{q}).$$

Mixture contamination therefore moves along rays inside an energy-distance ball. The tolerance formulation keeps all parameter values whose induced observational law lies within such a ball, not only those lying on one contamination path.

5.7 Monotonicity in λ

There is a global monotonicity property for exact global minimisers of the penalised problem. Let

$$\theta_1 \in \operatorname*{arg\,min}_{\theta \in \Theta_{\text{feas}}} \{\ell(\theta) + \lambda_1 a(\theta)\},$$

and

$$\theta_2 \in \operatorname*{arg\,min}_{\theta \in \Theta_{\text{feas}}} \{\ell(\theta) + \lambda_2 a(\theta)\},$$

with $\lambda_1 < \lambda_2$. By optimality of θ_1 at λ_1,

$$\ell(\theta_1) + \lambda_1 a(\theta_1) \leq \ell(\theta_2) + \lambda_1 a(\theta_2).$$

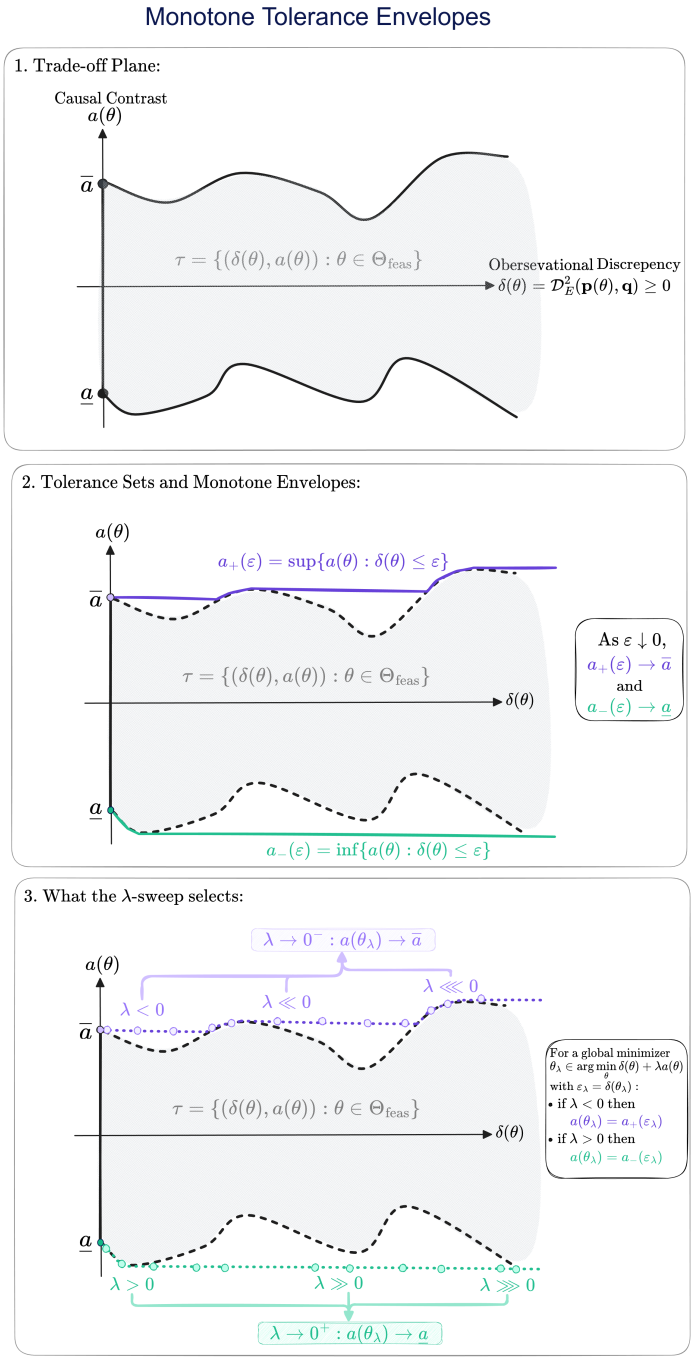


Fig. 5.3 Tolerance-envelope geometry of the λ-sweep. The feasible structural models are represented in the trade-off plane by $\tau = \{(\delta(\theta), a(\theta)) : \theta \in \Theta_{\text{feas}}\}$, $\delta(\theta) = D^2_{\mathscr{G}}(p_\theta, q)$, where $\delta(\theta)$ is the observational discrepancy and $a(\theta)$ is the causal contrast. For each tolerance level $\varepsilon \geq 0$, define $a_+(\varepsilon) = \sup\{a(\theta) : \delta(\theta) \leq \varepsilon\}$, $a_-(\varepsilon) = \inf\{a(\theta) : \delta(\theta) \leq \varepsilon\}$. As $\varepsilon \downarrow 0$, these envelopes converge to the upper and lower endpoints of the exact-fit identified set, $a_+(\varepsilon) \to \bar{a}$ and $a_-(\varepsilon) \to \underline{a}$. For a global minimiser $\theta_\lambda \in \arg\min_\theta \{\delta(\theta) + \lambda a(\theta)\}, \varepsilon_\lambda = \delta(\theta_\lambda)$, negative λ selects the upper tolerance envelope $a(\theta_\lambda) = a_+(\varepsilon_\lambda)$, while positive λ selects the lower tolerance envelope $a(\theta_\lambda) = a_-(\varepsilon_\lambda)$. Thus the one-sided limits $\lambda \to 0^-$ and $\lambda \to 0^+$ recover the upper and lower identified bounds, respectively. Again, the drawing here is just for visualisation, in non-convex, non-smooth problems the λ-sweep need not trace every point of the envelopes.

By optimality of θ_2 at λ_2,

$$\ell(\theta_2) + \lambda_2 a(\theta_2) \leq \ell(\theta_1) + \lambda_2 a(\theta_1).$$

Adding the two inequalities cancels the discrepancy terms and gives

$$(\lambda_2 - \lambda_1)\{a(\theta_2) - a(\theta_1)\} \leq 0.$$

Since $\lambda_2 - \lambda_1 > 0$,

$$a(\theta_2) \leq a(\theta_1). \tag{5.25}$$

Thus the global solution correspondence is monotone: increasing λ cannot increase the causal value selected by a global minimiser.

There is also a one-sided discrepancy monotonicity. Let

$$\Delta a = a(\theta_2) - a(\theta_1), \qquad \Delta \ell = \ell(\theta_2) - \ell(\theta_1).$$

The two optimality inequalities imply

$$\Delta \ell + \lambda_1 \Delta a \geq 0, \qquad \Delta \ell + \lambda_2 \Delta a \leq 0.$$

If $0 < \lambda_1 < \lambda_2$, then $\Delta a \leq 0$, and hence

$$\Delta \ell \geq -\lambda_1 \Delta a \geq 0.$$

Thus, on the positive branch, increasing λ moves the selected point toward smaller a and weakly larger observational discrepancy. On the negative branch, if $\lambda_1 < \lambda_2 < 0$, then the same inequalities imply

$$\Delta \ell \leq 0.$$

Thus moving from a more negative λ toward zero weakly reduces the observational discrepancy. Equivalently, increasing $|\lambda|$ on either branch weakly increases the observational discrepancy for global minimisers.

These monotonicity results do not imply that $\lambda \mapsto \theta_\lambda$ is continuous. In polyhedral or nonconvex settings, the global minimiser may switch between distinct faces, active sets, or connected components. Numerical local optimisers can also violate the global monotonicity if they get trapped in different local minima.

Large-$|\lambda|$ behaviour. The large-$|\lambda|$ limits concern the full feasible model, not the exact-fit set. As $\lambda \to +\infty$, global minimisers are pushed toward minimisers of a over Θ_{feas}. As $\lambda \to -\infty$, they are pushed toward maximisers of a over Θ_{feas}. Thus large $|\lambda|$ values need not correspond to sharp identified bounds, since they may sacrifice observational fit.

5.8 Local KKT Expansion of the Penalised Path

The previous sections described global properties of exact minimisers. We now derive the local behaviour of a smooth solution branch near an interior exact observational fit. This local calculation explains the linear movement of the causal contrast and the quadratic growth of the observational loss tested in the λ-sweeps of Sections 7.1.2 and 7.1.4.

Let $C(\theta) \leq 0$ denote inequality constraints not already enforced by the parameterisation. Equality constraints, such as normalisation, are assumed either to be built into the coordinates or handled by working in local coordinates on the constraint manifold.

The KKT stationarity condition for the inequality-constrained penalised problem is

$$\nabla_\theta \mathscr{L}_{ES}(\theta) + \lambda \nabla_\theta a(\theta) + \mathbf{J}_C(\theta)^\top \mu = 0, \tag{5.26}$$

with

$$C(\theta) \leq 0, \qquad \mu \geq 0, \qquad \mu_i C_i(\theta) = 0.$$

In the interior regime all inequalities are inactive and $\mu = 0$.

Let θ_0 be an interior minimiser of the unpenalised loss, and assume

$$\mathbf{p}(\theta_0) = \mathbf{q}.$$

Set

$$\mathbf{J} = \mathbf{J}_p(\theta_0), \qquad g = \nabla_\theta a(\theta_0).$$

Since

$$\nabla_{\mathbf{p}} \mathscr{L}_{ES} = \mathbf{D}\mathbf{q} - \mathbf{D}\mathbf{p},$$

we have

$$\nabla_\theta \mathscr{L}_{ES} = \mathbf{J}_p(\theta)^\top \mathbf{D}(\mathbf{q} - \mathbf{p}(\theta)).$$

For $\theta = \theta_0 + d\theta$,

$$\mathbf{p}(\theta_0 + d\theta) = \mathbf{q} + \mathbf{J} d\theta + O(\|d\theta\|^2).$$

Therefore

$$\nabla_\theta \mathscr{L}_{ES}(\theta_0 + d\theta) = -\mathbf{J}^\top \mathbf{D}\mathbf{J} d\theta + O(\|d\theta\|^2).$$

Define

$$\mathbf{G} = -\mathbf{J}^\top \mathbf{D}\mathbf{J}. \tag{5.27}$$

On directions which change the induced observational distribution, $\mathbf{G}$ is positive definite. If $\mathbf{G}$ is nonsingular on the relevant tangent space, the first-order stationarity equation is

$$\mathbf{G} d\theta + \lambda g = 0,$$

and hence

$$\boxed{d\theta = -\lambda \mathbf{G}^{-1} g.} \tag{5.28}$$

Thus positive λ moves in the negative energy-natural gradient direction of the causal contrast, while negative λ moves in the corresponding ascent direction.

The induced observational distribution satisfies

$$\mathbf{p}(\theta_0 + d\theta) = \mathbf{q} - \lambda \mathbf{J}\mathbf{G}^{-1} g + O(\lambda^2). \tag{5.29}$$

The causal contrast satisfies

$$a(\theta_0 + d\theta) = a(\theta_0) + g^\top d\theta + O(\lambda^2),$$

so

$$\boxed{a(\theta_0 + d\theta) = a(\theta_0) - \lambda g^\top \mathbf{G}^{-1} g + O(\lambda^2).} \tag{5.30}$$

Since $\mathbf{G}$ is positive definite in the identifiable directions,

$$g^\top \mathbf{G}^{-1} g \geq 0.$$

It is strictly positive unless the first-order causal contrast is flat in the energy metric. Thus, to first order, positive λ decreases a, and negative λ increases a.

The observational fit changes only at second order:

$$\mathscr{L}_{ES}(\theta_0 + d\theta) - \mathscr{L}_{ES}(\theta_0) = \frac{1}{2} d\theta^\top \mathbf{G} d\theta + O(\|d\theta\|^3).$$

Substituting $d\theta = -\lambda \mathbf{G}^{-1} g$ gives

$$\boxed{\mathscr{L}_{ES}(\theta_0 + d\theta) - \mathscr{L}_{ES}(\theta_0) = \frac{1}{2}\lambda^2 g^\top \mathbf{G}^{-1} g + O(\lambda^3).} \tag{5.31}$$

Thus, near an interior exact fit, the causal contrast moves linearly in λ, while the energy score grows quadratically.

5.9 Degeneracy, Invisible Directions, and Active Constraints

The matrix

$$\mathbf{G} = -\mathbf{J}^\top \mathbf{D} \mathbf{J}$$

is positive semidefinite in general. It is singular when there are nonzero directions u such that

$$\mathbf{J}u = 0.$$

Such directions are locally invisible to the observational distribution. Moving infinitesimally along u does not change $\mathbf{p}(\theta)$ to first order. These directions imply nonidentification of the structural parameterisation, but they imply partial identification of the causal contrast only if a varies along them:

$$\nabla_\theta a(\theta_0)^\top u \neq 0.$$

If a is constant along the exact-fit fibre, structural parameters may be nonidentified while the causal contrast remains point identified.

The nonsingular KKT formula should therefore not be confused with the partially identified case. If the exact-fit fibre has nontrivial tangent directions, then

$$\ker \mathbf{J} \neq \{0\},$$

and $\mathbf{G}$ is singular. If $g^\top u \neq 0$ for some $u \in \ker \mathbf{J}$, then the causal contrast changes along an observationally invisible direction. In that case, the inverse formula is not the right object; one must optimise over the fibre, work on a quotient by invisible directions, or specify a pseudoinverse convention.

The interior formula is also valid only while the perturbed parameters remain in the relative interior of the feasible set. Suppose that an active set $\mathscr{A}$ of inequalities is fixed at a boundary point, and set

$$\mathbf{A} = \mathbf{J}_{C_{\mathscr{A}}}(\theta_0).$$

The linearised KKT system is

$$\begin{pmatrix} \mathbf{G} & \mathbf{A}^\top \\ \mathbf{A} & 0 \end{pmatrix} \begin{pmatrix} d\theta \\ \mu_{\mathscr{A}} \end{pmatrix} = \begin{pmatrix} -\lambda g \\ 0 \end{pmatrix}. \tag{5.32}$$

The second row enforces $\mathbf{A}d\theta = 0$, so the path is tangent to the active boundary. When the relevant inverses exist, the projected direction is

$$d\theta = -\lambda \left[\mathbf{G}^{-1} - \mathbf{G}^{-1}\mathbf{A}^\top(\mathbf{A}\mathbf{G}^{-1}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{G}^{-1}\right] g. \tag{5.33}$$

This formula is valid only for an active set whose multipliers satisfy $\mu_{\mathscr{A}} \geq 0$. When this fails, the active set changes.

5.10 Energy Metric Interpretation

The local calculation has a geometric interpretation. Let

$$T\Delta^{K-1} = \{\mathbf{v} \in \mathbb{R}^K : \mathbf{1}^\top \mathbf{v} = 0\}$$

be the tangent space of the simplex. Conditional negative definiteness gives

$$\mathbf{v}^\top \mathbf{D}\mathbf{v} < 0 \qquad \text{for } 0 \neq \mathbf{v} \in T\Delta^{K-1}.$$

Hence $-\mathbf{D}$ is positive definite on $T\Delta^{K-1}$. Around $\mathbf{q}$,

$$\mathscr{L}_{ES}(\mathbf{q} + d\mathbf{p}, \mathbf{q}) - \mathscr{L}_{ES}(\mathbf{q}, \mathbf{q}) = \frac{1}{2} d\mathbf{p}^\top (-\mathbf{D}) d\mathbf{p}.$$

Thus the energy score is locally convex on the simplex. Pulling this quadratic form back through the structural probability map gives

$$\mathbf{G}_\theta = \mathbf{J}_p(\theta)^\top (-\mathbf{D}) \mathbf{J}_p(\theta). \tag{5.34}$$

This is a pullback energy form on parameter space. It is positive semidefinite in general and becomes a genuine Riemannian metric only after restricting to directions that change the observational law, or equivalently after quotienting out observationally invisible directions.

With respect to this form, the local update

$$d\theta = -\lambda \mathbf{G}_\theta^{-1} \nabla_\theta a(\theta)$$

is natural-gradient descent on a when $\lambda > 0$, and natural-gradient ascent on a when $\lambda < 0$, on the identifiable quotient where the inverse is defined. This is analogous to Fisher natural gradient for likelihood objectives, but the metric is induced by energy distance rather than by KL divergence.

The tolerance sets have the same geometric interpretation. Since

$$\delta(\theta) = \mathscr{D}_E^2(\mathbf{p}(\theta), \mathbf{q}),$$

the set

$$\mathscr{T}_\varepsilon = \{\theta : \delta(\theta) \leq \varepsilon\}. \tag{5.35}$$

is the preimage under Φ of an energy-distance ball around $\mathbf{q}$:

$$\mathscr{T}_\varepsilon = \Phi^{-1}\big(\{\mathbf{p} : \mathscr{D}_E^2(\mathbf{p}, \mathbf{q}) \leq \varepsilon\}\big). \tag{5.36}$$

Thus $\mathscr{T}_\varepsilon$ is a thickening of the exact-fit fibre

$$\mathscr{R} = \Phi^{-1}(\mathbf{q}).$$

Locally near an exact fit,

$$\delta(\theta_0 + d\theta) = d\theta^\top \mathbf{G} d\theta + O(\|d\theta\|^3). \tag{5.37}$$

If $\mathbf{G}$ is singular, this local set is an ellipsoidal cylinder: it is bounded in directions that change the observational law and flat along observationally invisible directions. In this sense, the tolerance set is both a tube around the exact-fit fibre and, in local product coordinates, a cylinder with the fibre directions as its axis. These flat directions are precisely where partial identification of the causal contrast can occur.

Chapter 6

Computational Implementation and Challenges for Discrete Graphs

Gradient-based training of SEM2 models requires a differentiable forward pass, as noted in Section 3.6. When the causal graph contains discrete nodes, as is common in categorical or ordinal settings, this requirement is violated: non-differentiable operations such as discrete sampling and argmax have zero or undefined derivatives with respect to their inputs almost everywhere. A differentiable implementation must therefore represent the discrete causal structure without changing its probabilistic semantics.

SEM2 implements four approaches: continuous relaxation with penalty regularisation, exact marginalisation on the probability simplex, score-function estimation, and Gumbel-softmax reparameterisation. Continuous relaxation passes through non-physical intermediate states. Score-function estimators are unbiased but high variance. Gumbel-softmax introduces a temperature-dependent bias-variance trade-off. For the inital experiments of Chapter 7, we use exact marginalisation, which preserves the discrete state space and gives exact gradients under the distributional parameterisation.

6.1 Exact Marginalisation on the Probability Simplex

Exact marginalisation represents each discrete variable by its probability law rather than by a sampled state. A categorical variable with state space $\{1,\dots,K\}$ is represented by a vector $\mathbf{p} \in \Delta^{K-1}$. The forward pass enumerates joint assignments of the discrete variables, evaluates the structural equations on those assignments, and weights each output by its probability. The resulting computation is deterministic and differentiable. The gradient is exact with respect to the distributional parameterisation and introduces no continuous relaxation of the discrete state space.

Figure 6.1 gives the basic geometric picture. A discrete variable is lifted from a finite set of outcomes to a point on the simplex. Optimisation then acts on probability vectors rather than on sampled states.

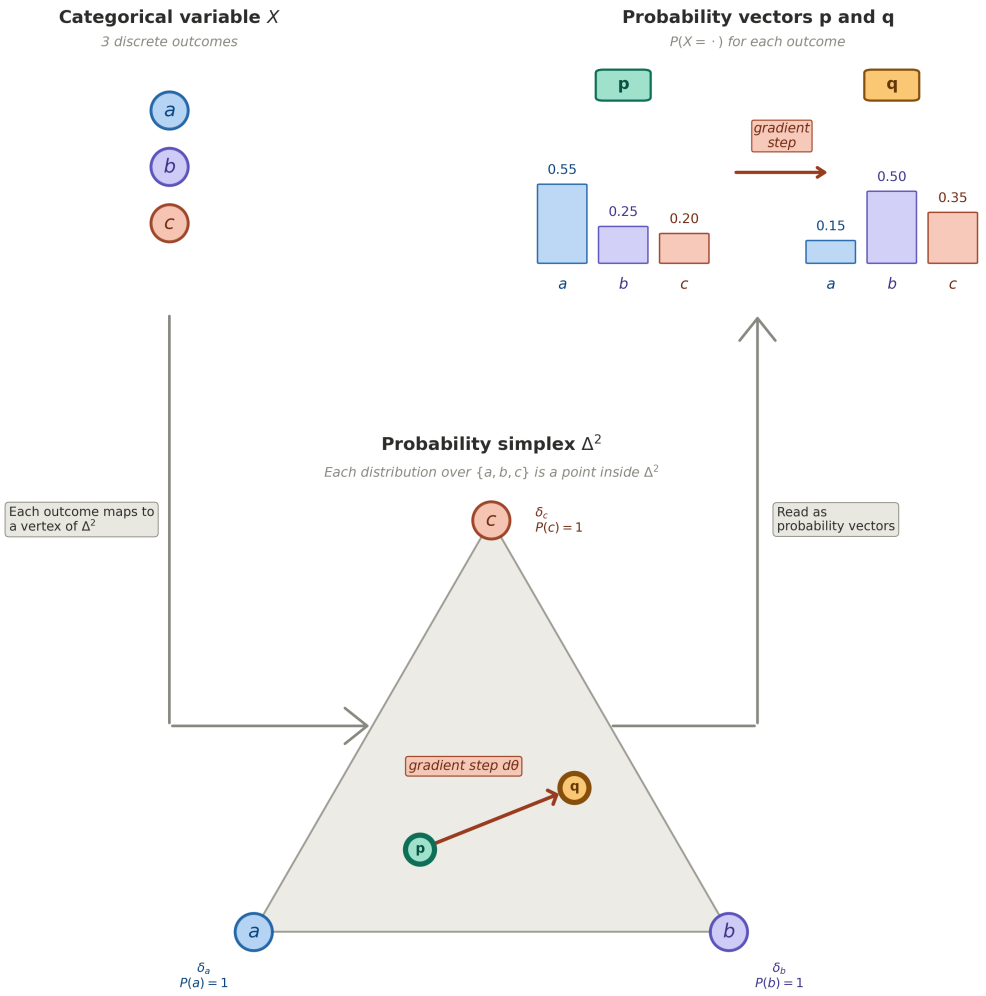


Fig. 6.1 Exact marginalisation replaces a sampled categorical state by a probability vector on the simplex. For a three-state variable, each Dirac mass is a vertex of Δ^2, while a general predictive law is an interior point. Gradient updates act on the probability vector.

For a subgraph containing N discrete nodes each with K states, the induced joint law lives on Δ^{K^N-1}, and exact enumeration requires summing over all K^N joint configurations. With $N = 20$ binary nodes, this reaches $2^{20} \approx 10^6$ configurations; with $K = 5$ states and $N = 10$ nodes, $5^{10} \approx 10^7$. The method is therefore well-suited to small discrete subgraphs, as in the binary IV problem of Chapter 7, but not as a general strategy for large discrete graphs. More sophisticated exact inference methods may exploit additional graphical structure, but they are not used here.

Figure 6.2 visualises this growth for several fixed state counts. The plotting window is clipped so that the continued growth leaves the frame rather than being compressed by the vertical scale.

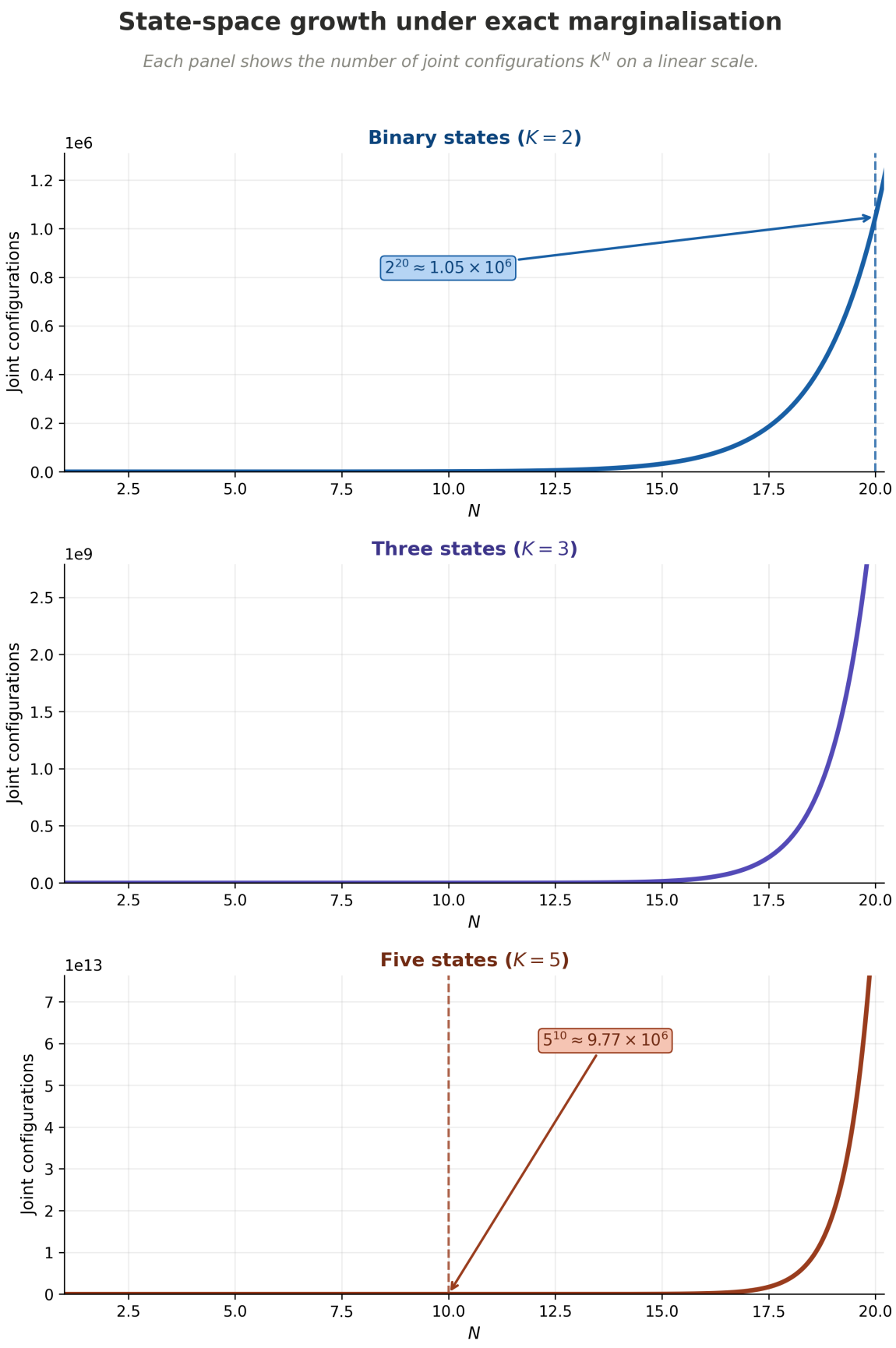


Fig. 6.2 Growth of the enumerated state space under exact marginalisation. Each panel shows K^N on a linear scale for a fixed number of states K. The plotting window is clipped so that the curves leave the frame, making the continued growth visually explicit. The annotations mark the examples $2^{20} \approx 1.05 \times 10^6$ and $5^{10} \approx 9.77 \times 10^6$ used in the text.

If some variables are latent, the forward pass produces a joint law over both observed and latent nodes. The energy-score loss must then be computed on the observed marginal. The implementation therefore sums over latent states before evaluating the loss, producing the predicted law over the observed variables. Including latent coordinates in the loss would compare distributions on the wrong space and hence give the wrong gradient.

6.2 Mini-Batch versus Precomputed Empirical Targets

The energy-score objective uses an empirical target vector $\mathbf{q}$. In the discrete IV experiment this target is computed separately within each instrument stratum $Z = i$, giving an empirical law for $P(D, Y \mid Z = i)$. We have two ways to do this:

One option is to estimate $\mathbf{q}$ from the current mini-batch. If few batch samples fall in a stratum, the resulting frequency vector is noisy. The loss then uses a different observational target at each step. This does not change the definition of the estimator, but it introduces target noise which can mask the approximately linear relationship between λ and the recovered causal contrast predicted by the local analysis of Section 5.8.

The alternative is to precompute $\mathbf{q}$ once from the full observational dataset and keep it fixed during training. With the energy-score convention of Section 5.2, the gradient has the form

$$\nabla_\theta \mathscr{L}_{ES} = \mathbf{J}_p(\theta)^\top \mathbf{D}\big(\mathbf{q} - \mathbf{p}(\theta)\big)$$

at the level of the finite-support objective. In the implementation, fixing $\mathbf{q}$ removes one source of stochasticity: the observational target no longer changes from step to step. Training is still stochastic because it uses mini-batches, but the target distribution is no longer noisy. This is the point relevant to the linear analysis of Chapter 5. The cost of precomputation is a single pass over the full dataset prior to training. Both strategies are evaluated empirically in Chapter 7.

6.3 Numerical Stability via Log-Space Probability Accumulation

The joint probability of a configuration is computed as a product of conditional probabilities:

$$p(x_1, \ldots, x_N) = \prod_{i=1}^{N} \pi_{i,x_i|x_{\mathrm{Pa}(i)}}(\theta).$$

Each factor lies in $(0,1)$, so the product decreases geometrically in N. In float32 arithmetic, the smallest positive normal value is approximately 1.18×10^{-38}. This is already reached by

$$0.5^{126} = 2^{-126} \approx 1.18 \times 10^{-38}.$$

For graphs with many nodes or highly concentrated conditionals, the running product can underflow to zero before the traversal is complete, and the corresponding gradient signal is lost.

The implementation avoids this by accumulating in log-space:

$$\log p(x_1, \ldots, x_N) = \sum_{i=1}^{N} \log \pi_{i,x_i|x_{\mathrm{Pa}(i)}}(\theta).$$

When probabilities are produced by logits, the preferred implementation is to use log-softmax values directly. When explicit probabilities are available, the code applies a numerical floor ε before taking logarithms. The running quantity is then a sum of log-probabilities rather than a product of probabilities. Interventions are handled by masking non-matching configurations, equivalently assigning them log-probability $-\infty$, before marginalisation.

If downstream computations require ordinary probabilities, a final exponentiation is still needed and extremely small probabilities may underflow at that last stage. The point of log-space accumulation is to prevent intermediate products from vanishing during the graph traversal; for larger graphs one would keep later computations in log-space or normalise using log-sum-exp.

Figure 6.3 compares the two representations. The direct probability product collapses rapidly toward machine precision, while the running log-probability remains finite over a much larger range.

The numerical floor perturbs very small probabilities slightly and should be viewed as an implementation-level stabilisation rather than a mathematical part of the estimator.

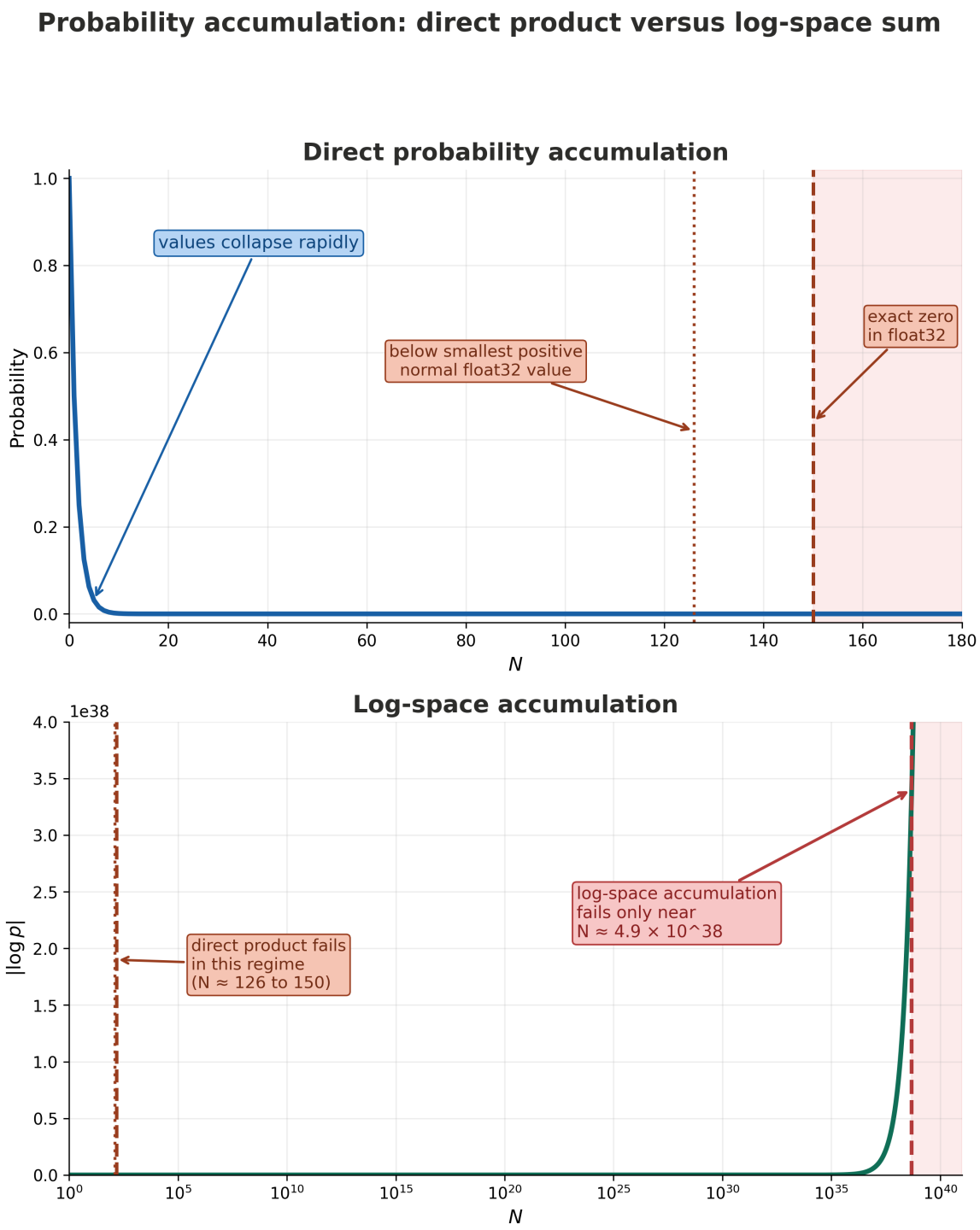


Fig. 6.3 Probability accumulation for a repeated factor of 0.5. Top: on the ordinary probability scale, the direct product drops below the smallest positive normal float32 value near $N = 126$ and reaches exact zero at $N = 150$. Bottom: log-space accumulation stores $\log p_N = -N \log 2$ instead of p_N. Thus $N = 126$ marks entry into the subnormal float32 range, while $N = 150$ marks underflow below the smallest positive subnormal value. Log-space accumulation avoids this intermediate underflow until the stored log magnitude exceeds the largest finite float32 value, at roughly $N \approx \frac{3.4\times10^{38}}{\log 2} \approx 4.9 \times 10^{38}$. If the log-probability is exponentiated back to ordinary probability, underflow can still occur; the benefit is preserving the computation during accumulation.

6.4 Context Deduplication for Discrete Parent Nodes

A naive enumerated forward pass evaluates each structural equation for every joint configuration. This is redundant if two configurations agree on the discrete parents of node v as they give the same input to the structural equation for v.

Let $\mathscr{X}_{\text{enum}}$ denote the enumerated global state space and write

$$S = |\mathscr{X}_{\text{enum}}|$$

for the number of global configurations. In the homogeneous case of N discrete nodes with K states each, $S = K^N$.

Let $\text{Pa}_{\text{disc}}(v)$ denote the discrete parents of v. The number of parent contexts that occur in the enumeration is

$$M_v = \left|\left\{x_{\text{Pa}_{\text{disc}}(v)} : x \in \mathscr{X}_{\text{enum}}\right\}\right|.$$

If parent r has K_r states, then

$$M_v \leq \prod_{r \in \text{Pa}_{\text{disc}}(v)} K_r,$$

with equality when all parent configurations occur.

The implementation extracts these unique parent contexts, evaluates the structural equation once per context, and then gathers the result back to all compatible global configurations using the inverse index returned by the deduplication step. This preserves exactness: no approximation is introduced, because every global configuration receives the same conditional probability vector it would have received under the naive evaluation.

The cost of evaluating the structural equation is therefore reduced from $O(S \cdot d)$ to $O(M_v \cdot d) + O(S)$, where d is the cost of one structural-equation evaluation. The first term is the cost of evaluating f_v on the unique parent contexts, and the second term is the cost of gathering the outputs back to the full enumerated state space.

The saving is most pronounced when a node has few discrete parents. If $\text{Pa}_{\text{disc}}(v) = \varnothing$, then $M_v = 1$ and a single evaluation suffices regardless of the size of the global state space. As the graph grows, this reduction compounds across nodes.

Figure 6.4 illustrates this method for an example node D with two discrete parents A and B. The rows $(A,B,D) = (0,0,0)$ and $(0,0,1)$ are distinct global configurations, but they share the same parent context $(A,B) = (0,0)$. Thus f_D need only be evaluated once for that parent context, returning the full conditional vector over the states of D.

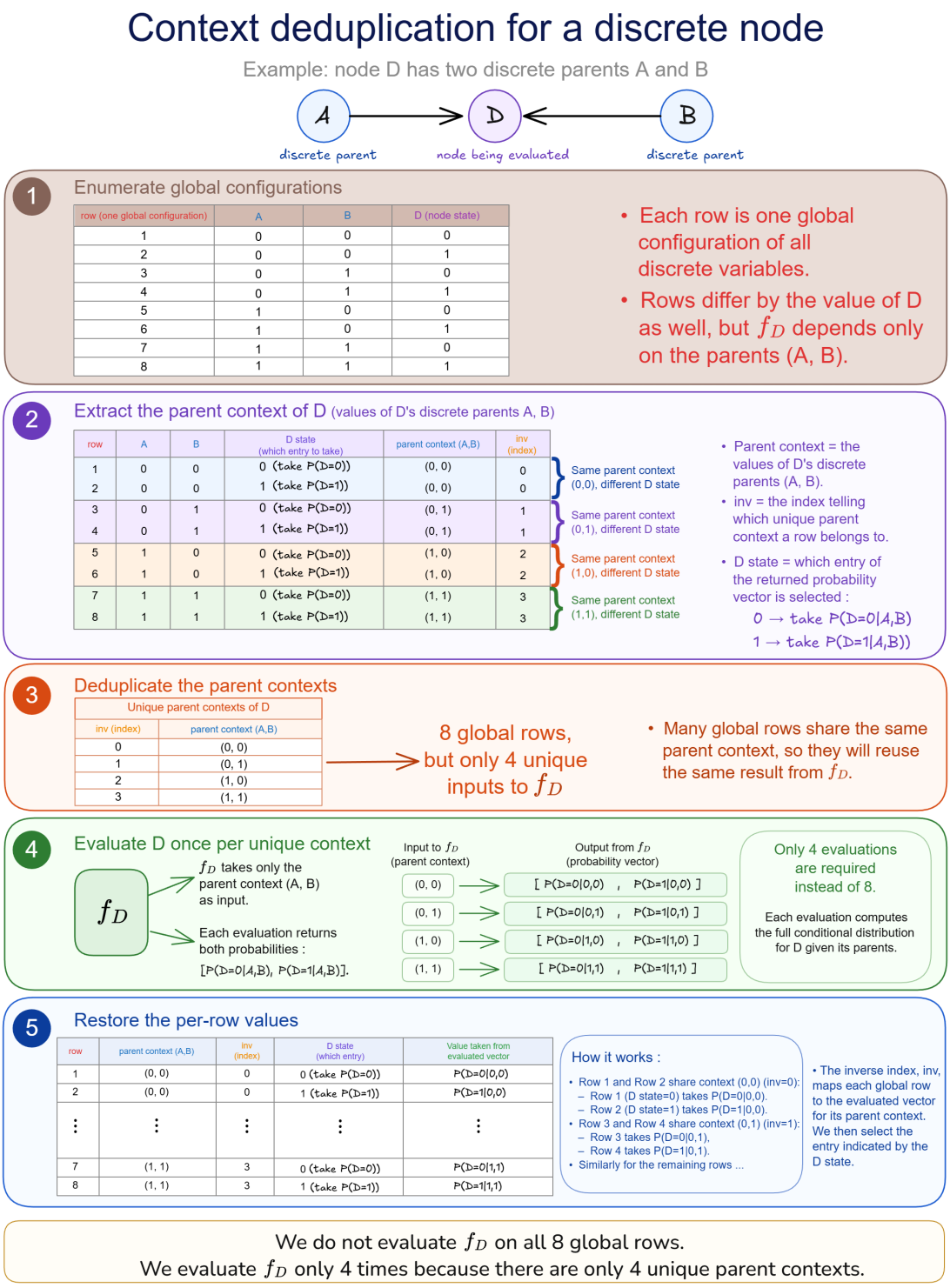


Fig. 6.4 Context deduplication for a discrete node. In the example, D has two discrete parents A and B. Although there are eight global configurations of (A,B,D), the structural equation for D depends only on the parent context (A,B). The implementation therefore extracts the unique parent contexts, evaluates f_D once per context, restores the resulting probability vectors to all global rows using the inverse index, and finally selects the entry corresponding to the required state of D.

Chapter 7

Initial Results for Binary IV test cases

This chapter reports the initial empirical test of SEM2 on the setting that is most directly connected to the initial computational and mathematical analysis of Chapters 5 and 6: a discrete binary instrumental-variable problem with latent confounding. Simpler point-identified linear sanity checks are useful software tests, but they are deferred to Report 2. The purpose here is to stress test the difficult cases first: exact marginalisation, finite-support energy-score training, precomputed empirical targets, and the λ-sweep geometry recovery.

The target is not a point estimate. The observational law of (Z,D,Y) is compatible with many structural models, and these models can imply different ATEs. The relevant object is therefore the sharp identified set. The experiment below asks whether the discrete SEM2 implementation can recover this partial-identification structure from data and whether the empirical λ-sweep exhibits the behaviour predicted in Chapter 5.

7.1 Binary IV with Latent Confounding

7.1.1 Problem Setup

The main experiment is a non-linear binary instrumental-variable model, detailed by Table 7.2 for reproducibility.

The target structural equations are

$$D = \mathbb{1}\left\{Z\sqrt{U} + (1-Z)U^2 > \varepsilon_D\right\}, \qquad Y = \mathbb{1}\left\{D(1-\sqrt{U}) + (1-D)U > \varepsilon_Y\right\}.$$

The nonlinearity enters through the threshold functions U^2, $\sqrt{U}$, and $1-\sqrt{U}$, while the latent variable U induces unobserved confounding between D and Y. This rules out a linear SEM solution by construction. Since U is unobserved, the law of (Z,D,Y) is compatible with many structural models, and these models can imply different average treatment effects. For this simulator, the true interventional ATE can be computed directly. Since $\varepsilon_Y \sim \mathrm{Unif}(0,1)$,

$$\mathbb{P}\{a > \varepsilon_Y\} = a \qquad \text{for } a \in [0,1].$$

Under $do(D=0)$,

$$Y(0) = \mathbf{1}\{U > \varepsilon_Y\},$$

so

$$\mathbb{E}[Y \mid do(D=0)] = \int_0^1 u\,du = \frac{1}{2}.$$

Under $do(D=1)$,

$$Y(1) = \mathbf{1}\{1-\sqrt{U} > \varepsilon_Y\},$$

so

$$\mathbb{E}[Y \mid do(D=1)] = \int_0^1 (1-\sqrt{u})\,du = \frac{1}{3}.$$

Hence

$$\mathrm{ATE}_{\text{true}} = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6} \approx -0.167.$$

This value is the causal effect of the particular simulator used to generate the data. However, the observational law of (Z,D,Y) does not point-identify this quantity. The learning task is therefore to recover the Balke-Pearl [3] identified set compatible with the observed conditional law $\mathbb{P}(D,Y \mid Z)$, rather than to recover only the simulator-specific ATE.

Table 7.1 Exact conditional observational law implied by the simulator.

(D,Y)	$Z=0$	$Z=1$
$(0,0)$	$5/12$	$7/30$
$(0,1)$	$1/4$	$1/10$
$(1,0)$	$2/7$	$1/2$
$(1,1)$	$1/21$	$1/6$

Writing

$$p_{zdy} = P(D=d, Y=y \mid Z=z),$$

the Balke-Pearl [3] formulas applied to Table 7.1 give

$$\begin{aligned}
\mathscr{I}_{\mathrm{BP}} &= \left[\max\left\{p_{000}+p_{111}-1,\ldots\right\},\ \min\left\{1-p_{001}-p_{110},\ldots\right\}\right] \\
&= \left[\max\left\{\frac{5}{12}+\frac{1}{6}-1,\ldots\right\},\ \min\left\{1-\frac{1}{4}-\frac{1}{2},\ldots\right\}\right] \\
&= \left[-\frac{5}{12},\frac{1}{4}\right] \approx [-0.417,\ +0.250].
\end{aligned}$$

The plotted green band uses the Monte Carlo target table generated by the experiment, giving the numerical benchmark

$$[-0.4165, +0.2481].$$

Table 7.2 Binary IV experiment specification.

Component	Specification
Graph	Observed instrument Z, treatment D, and outcome Y, with latent confounder U affecting both D and Y. (Figure 3.1)
SEM2 formula	Listing 3.1 shows the schematic SEM2 specification used for these experiments following the computational model defined in Chapter 3.
Observed target	The target distribution is represented by the two conditional tables $\mathbf{q}_z(d,y) = \mathbb{P}(D=d, Y=y \mid Z=z)$, one for $z=0$ and one for $z=1$.
Fitted mechanisms	D and Y are represented by categorical neural mechanisms with softmax outputs. The learned model uses noise dimension 8, 3 hidden layers, and hidden width 1024.
Training objective	For each penalty value λ, a separate model is trained as per Chapter 5 with $\mathscr{L}(\theta) = \mathscr{L}_{\mathrm{ES}}(\theta) + \lambda \widehat{\mathrm{ATE}}(\theta)$, where $\mathscr{L}_{\mathrm{ES}}$ is the finite-support energy-score loss.
Optimisation	AdamW with learning rate 10^{-3}, zero weight decay, cosine warmup schedule, batch size 512, and 256 optimisation steps per epoch. Experiments are run for 2, 8, and 32 epochs.
Penalty grid	A dense grid of nonzero λ-values, ranging from -0.2 to 0.2.
Compared targets	The precomputed version fixes $\mathbf{q}$ before training. The mini-batch version re-estimates $\mathbf{q}$ from the current batch (as per Section 6.2).
Evaluation	Each trained model is evaluated using 10^5 fresh samples.

Training uses the discrete implementation of Chapter 6. Each run enumerates the discrete state space, marginalises latent coordinates before computing the loss, and trains with the finite-support energy score. Two ways of constructing the empirical target $\mathbf{q}$ are compared. The precomputed strategy estimates $\mathbf{q}$ once from the full dataset and keeps it fixed. The mini-batch strategy estimates $\mathbf{q}$ from the current batch. Chapter 6 predicts that the first should give a cleaner λ-path, because the observational target no longer changes from step to step.

7.1.2 Dense λ Sweep

Figure 7.1 shows the dense sweep at 2, 8, and 32 epochs and Table 7.3 makes the visual conclusion of Figure 7.1 explicit. For each strategy and epoch, models are trained independently across positive and negative values of λ. The plotted quantity is the recovered ATE.

The sign agrees with Eq. (5.25). The objective is

$$\mathscr{L}_{ES} + \lambda a,$$

so positive λ penalises large ATE values and pushes the selected effect downward. Negative λ pushes it upward. This is the direction observed in all three dense sweeps.

Table 7.3 Numerical bound recovery in the dense λ-sweep. The Balke–Pearl benchmark used for the error columns is the empirical target benchmark $[-0.4165, +0.2481]$ computed from the generated target table. Bounds are obtained by fitting the outer positive and negative λ branches and extrapolating to $\lambda = 0$. The precomputed target gives stable bounds from epoch 8 onward: its lower and upper bound errors are both below approximately 0.02. The mini-batch target is less stable, especially on the lower branch at epochs 2 and 8, because the empirical target changes from batch to batch. By epoch 32 the mini-batch lower branch improves substantially, but the upper branch still overshoots the benchmark. These results support the computational analysis in Chapter 6: fixing the empirical target $\mathbf{q}$ reduces target-induced gradient noise and produces a cleaner λ-path.

Target construction	Epochs	Lower bound	Upper bound	Lower error	Upper error
Precomputed $\mathbf{q}$	2	-0.438	+0.265	**0.022**	**0.016**
Mini-batch $\mathbf{q}$	2	-0.206	+0.181	0.211	0.067
Precomputed $\mathbf{q}$	8	-0.424	+0.264	**0.008**	**0.015**
Mini-batch $\mathbf{q}$	8	-0.191	+0.263	0.225	**0.015**
Precomputed $\mathbf{q}$	32	-0.424	+0.257	0.007	**0.009**
Mini-batch $\mathbf{q}$	32	-0.416	+0.281	**0.000**	0.033

The precomputed strategy matches the predicted behaviour most clearly. At epoch 2 the outer linear region is already visible. At epoch 8 the path is stable and the extrapolated bounds are close to the analytical set. Epoch 32 gives little additional improvement, suggesting that the precomputed sweep has essentially converged. The mini-batch strategy follows the same broad geometry but with more scatter. This is not a failure of the energy score itself; it is the target-noise problem described in Section 6.2. A changing batch estimate of $\mathbf{q}$ perturbs the gradient field and makes the linear region harder to identify.

The bound extraction method is deliberately simple. The outer part of each sweep is treated as approximately linear and extrapolated to $\lambda = 0$. The quality of this heuristic depends more on the noise level of the sweep than on the exact choice of line-fitting rule. A clean precomputed path gives a usable input to many possible extraction procedures, whereas a noisy mini-batch path makes the same task unstable.

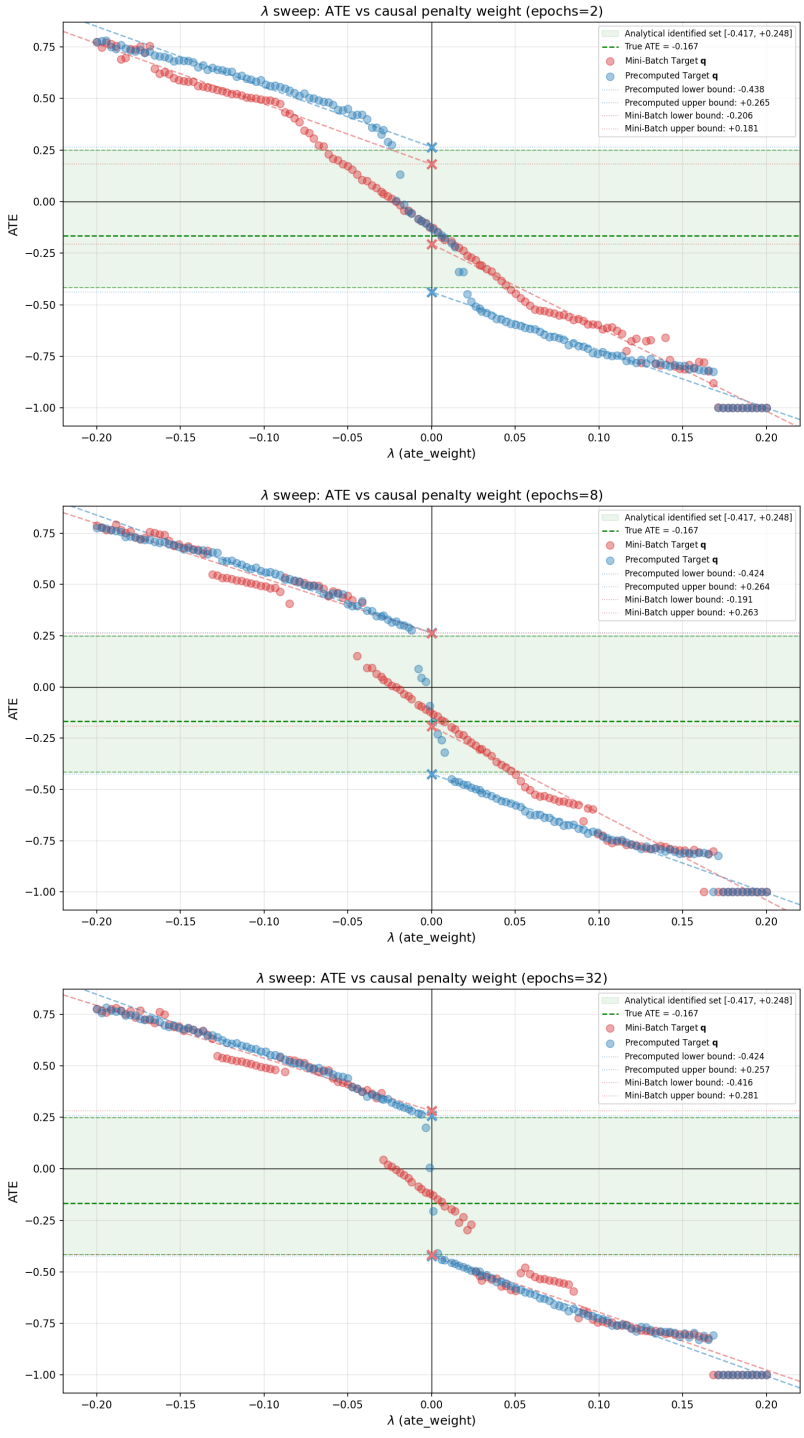


Fig. 7.1 Dense λ-sweep at epochs 2, 8, and 32. Blue points use the precomputed empirical target $\mathbf{q}$; red points use a mini-batch estimate of $\mathbf{q}$. The green band is the Balke–Pearl identified set $[-0.417, +0.248]$, and the dashed green line is the true simulator ATE, shown only as a reference. The sweep direction agrees with Eq. (5.25): increasing λ decreases the selected ATE. The precomputed target produces a clean locally linear path from epoch 2 and stabilises by epoch 8 with bounds near $[-0.424, +0.257]$. The mini-batch target exhibits larger scatter and needs more training to approach reliable bounds.

7.1.3 The (δ, a) Trade-off Set

Chapter 5 defines the trade-off set

$$\tau = \{(\delta(\theta), a(\theta)) : \theta \in \Theta_{\text{feas}}\},$$

as in Eq. (5.24). The λ-sweep minimises a linear functional over this set, and therefore selects supported points on the tolerance envelopes described in Eqs. (5.22)–(5.23). Figure 7.2 plots the empirical version of this geometry.

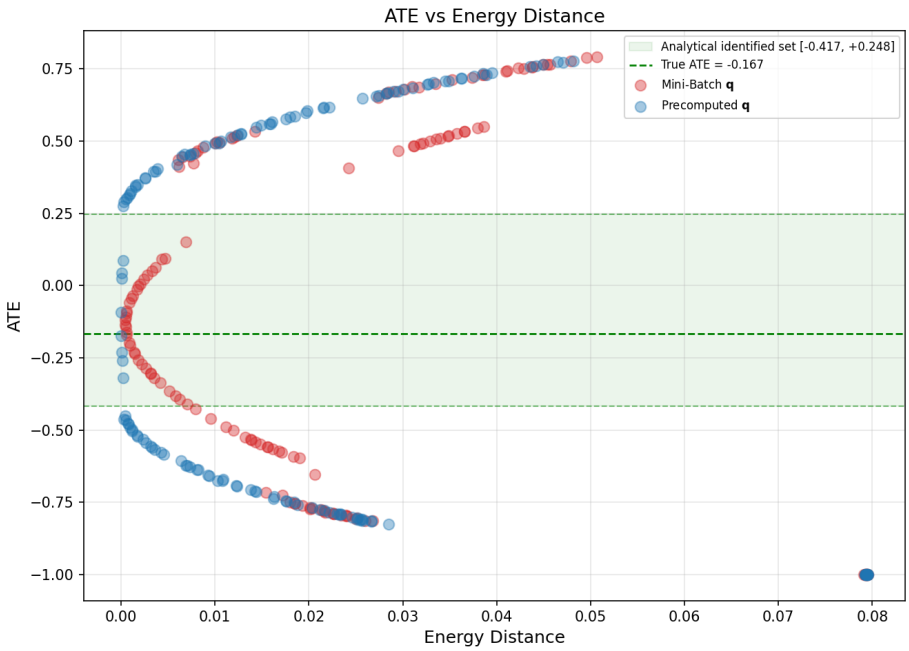


Fig. 7.2 Empirical trade-off set τ at epoch 8. The horizontal axis is the energy-distance discrepancy δ; the vertical axis is the recovered ATE. The precomputed **q** sweep traces a cleaner supported frontier, while the mini-batch target produces a noisier cloud. This is the tolerance-envelope view of Section 5.6: allowing larger observational discrepancy expands the attainable causal envelope, but individual points in τ need not be monotone in δ.

This plot is useful because it separates two ideas that can be confused in the λ-plot. Larger $|\lambda|$ usually means the optimiser accepts more observational discrepancy, but the theory is not a pointwise monotonicity claim about all pairs (δ, a). It is an envelope claim: the set of allowed values expands as the tolerance $\delta \leq \varepsilon$ increases.

7.1.4 Energy Distance versus λ

The local KKT expansion predicts that, near an interior exact fit,

$$\mathscr{L}_{ES}(\theta_0 + d\theta) - \mathscr{L}_{ES}(\theta_0) = O(\lambda^2),$$

with coefficient given in Eq. (5.31). Figure 7.3 checks this directly.

The precomputed target again gives the clearest empirical confirmation of the theory. The energy distance is small near the unpenalised fit and grows roughly quadratically as $|\lambda|$ increases. Deviations occur in the tails, where the local interior expansion no longer applies. This is consistent with the active constraint discussion of Section 5.9.

7.1.5 Sparse Sweeps and Computational Cost

The dense sweep diagnoses the geometry, but a practical method should not require a dense grid of penalty values. Figure 7.4 and Table 7.4 shows sparse sweeps using only a small number of λ values. This is a direct test of whether the cleaner precomputed path can reduce computational cost.

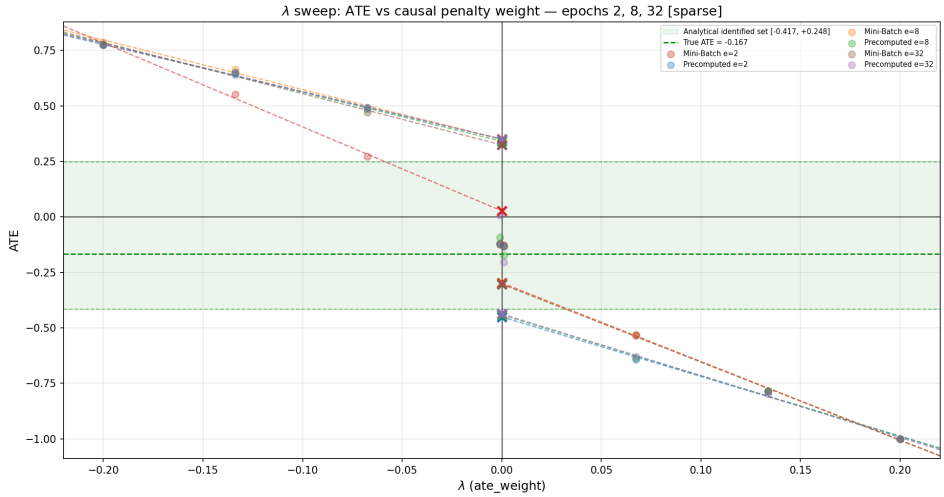


Fig. 7.4 Sparse λ-sweeps across epochs. The sparse grid is less precise than the dense sweep, but the precomputed target preserves the main directional behaviour and gives useful preliminary bounds with far fewer training runs. The mini-batch strategy is more sensitive to the reduced grid because each point carries additional target-estimation noise.

At epoch 8, the precomputed sparse sweep gives bounds comparable to the dense result. Even at epoch 2, the precomputed run gives a correctly signed and useful preliminary interval. This does not replace a dense sweep when final bounds are needed, but it suggests a practical

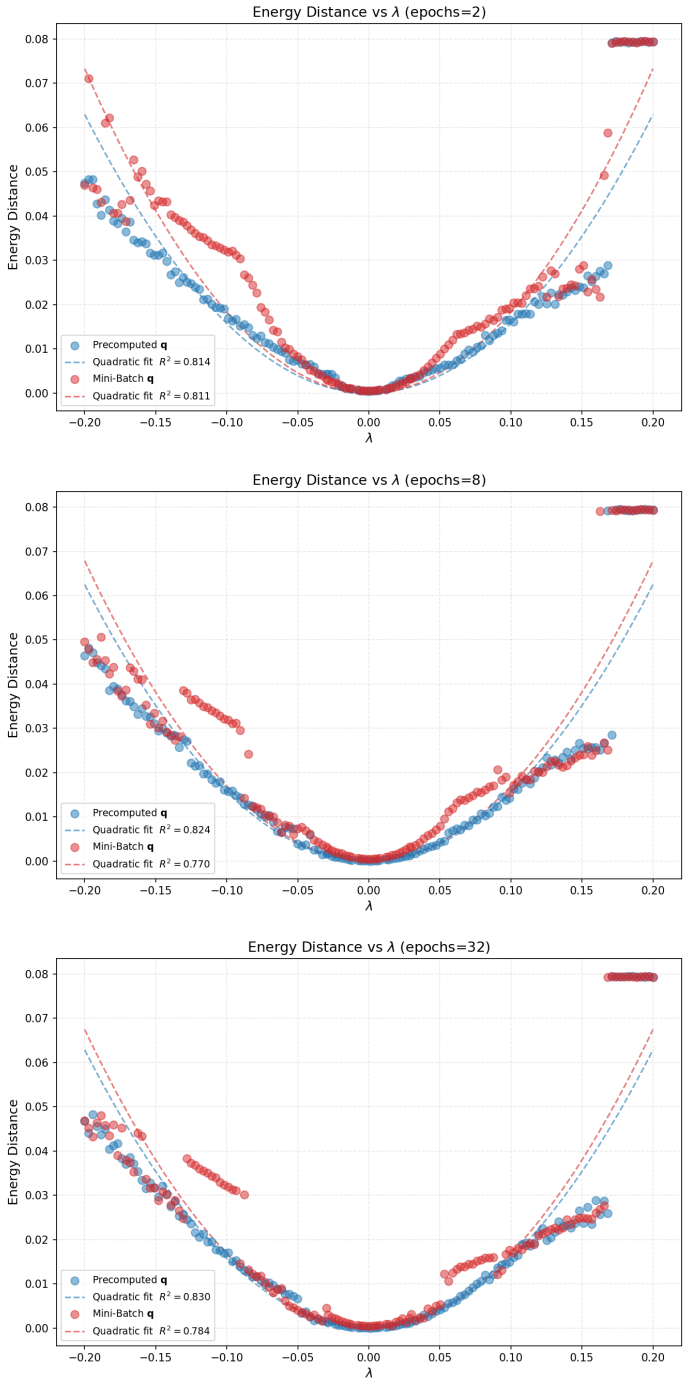


Fig. 7.3 Energy distance versus λ for the dense sweep at epochs 2, 8, and 32. The local expansion of Eq. (5.31) predicts quadratic growth near an interior exact fit. The precomputed target follows this shape most cleanly, with quadratic fit quality improving from epoch 2 to epoch 32. The mini-batch target is noisier and displays a saturation plateau at large positive λ, indicating the onset of a boundary regime in which further increases in the penalty no longer substantially change the induced distribution.

Table 7.4 Sparse λ-sweep bound recovery. The sparse plot uses four negative and four positive λ-values: -0.2000, -0.1337, -0.0675, -0.001, 0.001, 0.0675, 0.1337, and 0.2000. The benchmark interval is the Monte Carlo Balke–Pearl interval $[-0.4165, +0.2481]$ computed from the generated target table. The sparse sweep gives less accurate numerical bounds than the dense sweep. This is expected as it uses only four points on each side of the penalty path, so the extrapolation to $\lambda = 0$ is more sensitive to optimisation noise and branch curvature. The precomputed target remains more reliable on the lower branch, while the upper branch is systematically overestimated in the sparse setting. For this reason, the sparse sweep is best viewed as a low-cost diagnostic, whereas the dense sweep provides the main numerical evidence for bound recovery.

Target construction	Epochs	Lower bound	Upper bound	Lower error	Upper error
Precomputed **q**	2	-0.451	+0.350	**0.034**	**0.102**
Mini-batch **q**	2	-0.297	+0.027	0.119	0.221
Precomputed **q**	8	-0.442	+0.341	**0.025**	**0.093**
Mini-batch **q**	8	-0.300	+0.350	0.116	0.102
Precomputed **q**	32	-0.437	+0.350	**0.020**	0.102
Mini-batch **q**	32	-0.303	+0.324	0.113	**0.076**

workflow: use a coarse precomputed sweep to locate the relevant branch, then refine with additional λ values or longer training only where needed.

7.2 Takeaways

The binary IV experiment supports the central claims of the report. The dense λ-sweep confirms the sign and monotonicity predicted by Chapter 5: with the convention $\mathscr{L}_{ES} + \lambda a$, positive λ pushes the ATE downward and negative λ pushes it upward. The (δ, a)-plot displays the trade-off set τ of Eq. (5.24) and gives the empirical version of the tolerance-envelope picture. The energy-distance plots support the local quadratic prediction of Eq. (5.31). Across all three diagnostics, precomputing **q** gives a substantially cleaner path than estimating **q** from each mini-batch, confirming the implementation analysis of Chapter 6.

Chapter 8

Plans for Report 2

With this, Report 1 establishes our core contributions: the design and implementation of the SEM2 probabilistic programming language, the finite-support energy-score treatment of discrete partial-identification problems, the associated computational challenges encountered in bringing this method into the SEM2 framework, and empirical validation of the theory using SEM2. In the process, several additional research directions surfaced from this work. However, realistically, the remaining three months of the MPhil project are not sufficient to address every possible direction that has emerged. The plan for Report 2 is therefore to focus on a selected tractable subset of these directions, exploring them in good depth with the goal of presenting a rigorous treatment within the remaining time.

8.1 Simulation Studies and Software Demonstration

The primary deliverable of Report 2 is an extended simulation study demonstrating SEM2 across a wider range of causal graph structures. This will compare the discrete training strategies deferred from Chapter 6, expanded testing to additional causal settings such as regression discontinuity and proximal causal inference, and further comparisons against alternative software implementations where applicable, particularly in settings where analytical bounds are not available.

In the meantime, discussions are underway with practitioners attending the Isaac Newton Institute's Causal Inference Programme (INI-CIF) [12]. The consolidated knowledge from these conversations is expected to surface applied test cases for which SEM2 may be useful. Where feasible within the project timeline, representative cases from these discussions will be incorporated into the Report 2 simulation study, though no specific outcomes can be guaranteed at this stage given the time required to assess what is theoretically and computationally tractable.

8.2 Theoretical Developments

We are working in the meantime on the statistical theory underlying the MDE framework, including things like $\sqrt{n}$-consistency of the plug-in estimator $\theta(\hat{P})$ of causal functionals with respect to the MMD metric, and a one-step debiasing correction via the MMD influence function. Connecting this to the SEM2 implementation requires understanding what regularity conditions the neural network parameterisation must satisfy for these results to apply. These are genuinely difficult open problems requiring input from multiple specialists; a complete resolution is unlikely within the remaining timeline of the MPhil. If meaningful developments arise in the background, the relevant regularity conditions and assumptions will be presented in Report 2.

The deeper treatment of the MMD and Wasserstein frameworks sketched in Section 4 is tentatively planned. Whether this is included will depend on the final direction taken in the simulation study. Only theory directly relevant to the test problems will be presented in detail; for everything else, external references and related work will be cited where appropriate.

8.3 Software Development

The SEM2 codebase is also undergoing architectural refactors and is being actively extended beyond the current implementation, informed by feedback from discussions with visiting researchers at INI-CIF, to support new emerging directions of our framework.

Ultimately, the aim is for SEM2 to develop into a broadly useful open framework for researchers across disciplines, including philosophy, computer science, sociology, physics, psychology, statistics, epidemiology, political science, and economics, to be able to conduct better causal analysis.

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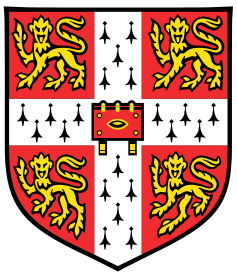
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Learning Semiparametric Structural Equation Models

Report 2



Srivathsan Amruth

Supervisor: Prof. Qingyuan Zhao

Department of Physics
Department Of Pure Mathematics and Mathematical Statistics
University of Cambridge

This dissertation is submitted for the degree of
Master of Philosophy (Scientific Computing)

Fitzwilliam College August 2026

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Chapter 1

Subsequent Development of SEM2

1.1 From a difficult benchmark to a general solver

Report 1 established the central SEM2 workflow: a structural model is written as a probabilistic-programming formula, converted into a differentiable directed acyclic graph, and fitted through the distribution that the graph induces on the observed variables. Its main experiment deliberately addressed a difficult discrete instrumental-variable problem with latent confounding and partial identification. In that setting, SEM2 converted an analytically defined identification problem into a computational search over structural models compatible with the observed distribution.

Report 2 asks a broader question where it investigates whether the same structural representation can serve as a practical modelling framework across problems that are normally handled by separate domain-specific software. The evaluation therefore moves from a single proof of concept to a suite of applied benchmarks drawn from analytical chemistry [34], social science [1], engineering [21, 27], and policy evaluation [7, 22, 26].

These applications appear statistically different, but can be recast in SEM2. The main development of Report 2 is a model-aware architecture in which the same graph and formula language can support exact linear calculations, likelihood-based latent-variable learning, and differentiable generative fitting. Here, we test whether this common interface recovers the quantities produced by widely used practitioner tools.

1.2 Further development

The original `SEM2` class provides a general differentiable graph for nonlinear structural mechanisms. During Report 2, this interface was extended with model-specific classes and operators that exploit the mathematical structure declared in the formula. The resulting package distinguishes linear-gaussian, linear non-gaussian, nonlinear, discrete, and mixed models using both their structural equations and exogenous distributions. The general nonlinear implementation is retained through `SEM2`, exact linear operations are exposed

through `SEM2Linear`, and the finite-support implementation developed in Report 1 remains available through `SEM2Discrete`.

This distinction improves both statistical and computational performance. For a linear SEM,

$$V = BV + c + \varepsilon,$$

where B is the path-coefficient matrix, c contains exogenous means and intercepts, and Ω is the covariance matrix of the exogenous inputs, the model-implied moments are

$$\mu = (I - B)^{-1}c, \qquad \Sigma = (I - B)^{-1}\Omega(I - B)^{-\mathsf{T}}.$$

These are the standard model-implied mean and covariance expressions for a linear structural equation model [3]. The interfaces `coef_matrix()`, `mean_vector()`, and `cov_matrix()` construct these quantities directly from the parsed graph while remaining differentiable in the learned coefficients. Linear-model fitting therefore no longer requires repeated simulation to approximate the covariance matrix. This removes Monte Carlo variation from every optimisation step and gives a deterministic representation of the model-implied observed distribution.

An exact gaussian `LogLikelihood` operator was also added for linear-gaussian SEMs. It evaluates the observed-data likelihood using Cholesky factorisation and triangular solves, providing a stable full-batch objective for numerical maximum-likelihood estimation. The same formula can therefore be fitted through exact covariance algebra or through numerical likelihood optimisation, allowing the implementation to be checked against independently derived classical solutions.

The probabilistic-programming semantics were tightened at the same time. The declaration `unobs(...)` explicitly determines which nodes are latent, while all remaining nodes are treated as observed. Fixed and learned coefficients are distinguished directly in the formula, and the graph exposes a consistent interface for coefficient extraction, sampling, and intervention. Consequently, the statistical model being fitted can be read and audited from the formula rather than reconstructed from implementation code.

The same graph representation is retained when exact gaussian calculations are unavailable. Non-gaussian latent-source models can use the distributions and coefficient matrix declared by the graph to form an appropriate likelihood. Nonlinear implicit models can instead be sampled differentiably and fitted by distributional discrepancies such as Sinkhorn divergence. SEM2 therefore exploits exact algebra where it exists without restricting the package to models for which a closed-form observed density is available. Development followed a test-driven workflow.

1.3 A unified applied validation programme

The application suite is designed as an external validation programme rather than a collection of unrelated demonstrations. Each study asks whether the target quantity extracted from a SEM2 graph agrees with the answer obtained from a recognised specialist method in that field. Table 1.1 gives the high-level structure; the complete formulas, simulation settings, and numerical evidence are presented in the corresponding application chapters.

Table 1.1 Applied validation programme for SEM2. Each chapter introduces a different structural regime while retaining the same formula-defined graph interface.

Application	Applied question	SEM2 regime	Reference tool
Calibration (Chapter 2)	Can the package recover instrument sensitivity in a controlled linear response model?	Exact linear moments and gaussian maximum likelihood	OLS
Political Democracy (Chapter 3)	Can one graph represent latent constructs, measurement equations, structural paths, and residual covariances?	Linear-gaussian latent-variable SEM	`lavaan`
ICA (Chapter 4)	Can SEM2 recover independent non-gaussian sources from observed linear mixtures?	Linear non-gaussian latent-source likelihood	FastICA
RDD (Chapter 5)	Can a nonlinear structural model recover a policy-threshold effect and the local outcome distributions?	Side-specific nonlinear generators fitted by Sinkhorn divergence	`rdrobust`

OLS [9], `lavaan` [33], FastICA [20], and `rdrobust` [6] are designed for different statistical tasks and expose different interfaces. SEM2 recasts each task as a structural graph, fits it using the information available in that model class, and extracts the relevant target through one package. Agreement with these references therefore checks that the graph represents the scientific quantity that practitioners intend to estimate.

1.4 Contribution of Report 2

The central contribution of Report 2 is the development and validation of SEM2 as a flexible, general-purpose solver. Seemingly unrelated problems like instrument calibration, latent social-science measurement, engineering source separation, and policy evaluation at a threshold, are represented through one probabilistic-programming language and one underlying graph abstraction. A user can now solve an array of problems by changing the structural formula and using the statistically appropriate fitting operator with SEM2, rather than moving between different software systems and model representations.

Report 2 shows that our single package can reproduce or closely track the target quantities obtained from several specialised daily-driven practitioner tools across increasingly demanding model classes while preserving a common modelling, sampling, coefficient, and intervention interface. The detailed evidence is developed in Chapters 2, 3, 4, and 5. The final discussion considers the remaining trade-offs, particularly computational cost, uncertainty quantification, and the advantages that specialised methods retain in their native settings. Together, these findings validate and support our SEM2 package as novel multipurpose statistical solver while acknowledging its current limitations and identifying priorities for further development.

Chapter 2

Calibration Study I: Recovering Instrument Sensitivity in a Linear Response Model

2.1 Applied problem and purpose

The first study is motivated by a simple calibration problem in analytical chemistry [13]. In instrumental calibration, known calibration samples are measured so that a mathematical relationship can be established between the known analyte quantity and the corresponding instrumental response [5, 11, 28]. This allows the instrument response from an unknown sample to be converted into an estimated analyte amount or concentration [34]. In spectrophotometry, the Beer–Lambert law gives a particularly simple linear response model: under ideal conditions, absorbance is proportional to concentration and path length [23].

This chapter uses that setting as a stylised controlled input-response calibration problem. The aim is to verify the most basic path-coefficient interpretation [25, 37] in SEM2 before moving to richer applied problems involving latent variables, non-gaussian sources and treatment thresholds.

Let C_i denote the known concentration of a calibration sample, and let A_i denote the measured absorbance response. A first-order calibration model is

$$A_i = \alpha + \beta C_i + c\varepsilon_{Y,i},$$

where β is the instrument sensitivity: the expected change in absorbance per unit increase in concentration. After centring the concentration and absorbance variables,

$$X_i = C_i - \bar{C}, \qquad Y_i = A_i - \bar{A},$$

the calibration model reduces to the one-edge structural equation model

$$X_i = a\varepsilon_{X,i}, \qquad Y_i = \beta X_i + c\varepsilon_{Y,i}.$$

Here $\varepsilon_{X,i}$ and $\varepsilon_{Y,i}$ are independent standard gaussian variables in the simulation design.

The equation for X_i should be interpreted as a Monte Carlo mechanism for generating centred exogenous calibration inputs. It is not a claim that physical concentrations are normally distributed or can be negative since negative simulated values of X_i simply correspond to calibration inputs below the mean concentration after centring. The substantive assumption is that the calibration input is externally controlled and independent of the absorbance measurement noise. Under this design, ordinary least squares [9] provides the natural benchmark for estimating the calibration slope β.

2.2 SEM representation

Figure 2.1 shows the corresponding structural graph for the simulation. The variables X and Y are observed, while the noise variables ε_X and ε_Y are latent. In the calibration interpretation, X is the centred known input concentration and Y is the centred measured response. SEM2 fits the observed joint distribution of (X,Y) while treating the exogenous noise variables as unobserved structural components.

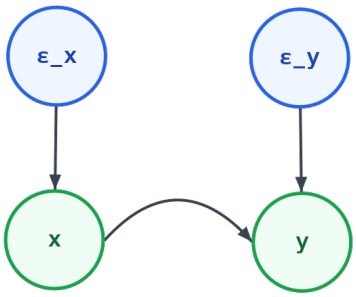


Fig. 2.1 Structural graph for the one-edge calibration study. The lower nodes X and Y are observed variables, representing centred calibration input and centred instrument response. The upper nodes ε_X and ε_Y are latent exogenous noise variables. The target parameter is the path coefficient β on the edge $X \to Y$, interpreted here as the calibration sensitivity.

There are two SEM2 routes for obtaining the slope in this example. Since the model is linear and gaussian, the coefficient can be read from the model-implied covariance. From

$$\mathrm{Cov}(X,Y) = \beta \mathrm{Var}(X),$$

it follows that

$$\beta = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}.$$

The sample version of this expression is exactly the OLS slope with an intercept. This is still a SEM2 result: it uses the linear-gaussian covariance equations of the structural model. To check that the package optimisation route gives the same answer, the numerical SEM2 gaussian maximum-likelihood estimation (MLE) fit is also run over repeated seeds.

A successful calibration experiment should therefore show two patterns. First, SEM2 should recover the same calibration slope as OLS in a single simulated calibration data set. Second, across repeated simulations and sample sizes, the closed-form SEM2 and numerical MLE SEM2 estimates should track the OLS estimate rather than drifting to a different coefficient.

2.3 Experimental setup

The SEM2 formula used for the experiment is shown in Listing 2.1. In the mathematical notation, the latent noise variables are denoted by ε_X and ε_Y. In the SEM2 formula, these are written as the code-level variable names `eps_x` and `eps_y`. The absence of a fixed numeric coefficient before `x` in `y <- x + eps_y` means that the path coefficient from X to Y is learned. The declaration `unobs(eps_x,eps_y)` means that the two noise variables are latent. Since `x` and `y` are not listed in `unobs(...)`, they are treated as observed variables.

```
eps_x, eps_y ~ Normal(0,1);
x <- eps_x;
y <- x + eps_y;
unobs(eps_x,eps_y)
```

Listing 2.1 SEM2 specification for the one-edge calibration study. The exogenous noise variables eps_x and eps_y correspond to ε_X and ε_Y in the mathematical notation. They are latent, while x and y are observed. The learned coefficient on x in the equation for y is the SEM2 estimate of the calibration slope β.

Table 2.1 summarises the simulation design, target parameter, benchmark, and fitting routes. The simulator uses $\beta = 2$, so all estimates in this chapter should concentrate around this value.

Table 2.1 Experimental setup for the one-edge linear-regression sanity check. The purpose of this experiment is to verify that SEM2 recovers the same path coefficient as ordinary least squares in the simplest linear-Gaussian structural model.

Component	Specification
Graph	One observed exogenous node X and one observed outcome node Y, with the structural edge $X \to Y$.
Data-generating process	$X = \varepsilon_X$, $Y = 2X + \varepsilon_Y$, with independent $\varepsilon_X, \varepsilon_Y \sim N(0,1)$.
Observed variables	X and Y.
Latent variables	The exogenous noise variables ε_X and ε_Y.
Target parameter	The structural path coefficient $\beta = 2$ on the edge $X \to Y$.
OLS benchmark	Ordinary least squares with an intercept, fitted separately for each simulated data set.
Closed-form SEM2 regime	The path coefficient is recovered from the model-implied covariance identity $\beta = \mathrm{Cov}(X,Y)/\mathrm{Var}(X)$.
MLE SEM2 regime	SEM2Linear is fitted by the centred Gaussian log-likelihood using LBFGS maximum-likelihood estimation.
$\beta = 2$ recovery setting	1,000 independent (seeds) simulated data sets, each with $n = 2{,}000$ observations, for the repeated-seed coefficient comparison.
Sample-size scaling setting	Sample sizes ranging from $n = 40$ to $n = 32{,}000$ on a dense grid, with 1,500 repeated simulations for each n.
Evaluation metrics	Estimate of β, mean absolute difference from OLS, mean absolute error relative to $\beta = 2$, and empirical standard deviation across repeated simulations.

2.4 Single-dataset visual check

Figure 2.2 shows one simulated calibration data set. The plot is included to make the result familiar before reporting repeated-seed summaries. The true simulator line, OLS fitted line, closed-form SEM2 line, and numerical MLE SEM2 line are all shown on the same axes.

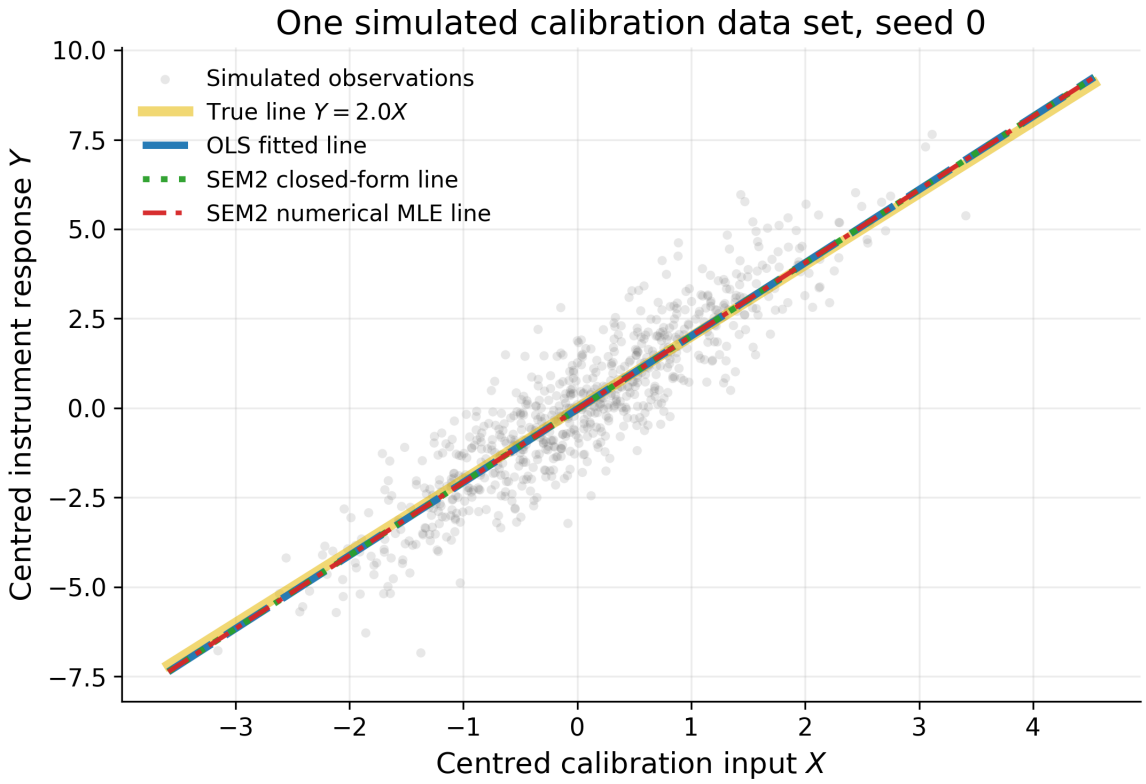


Fig. 2.2 Scatter plot for one simulated calibration data set. The translucent gold line is the simulator value $Y = 2X$, while the blue, green, and red lines show the OLS, closed-form SEM2, and numerical MLE SEM2 fits. The fitted lines are drawn with distinct dash patterns so that the near-overlap remains visible. The fitted lines nearly overlap, which is the expected behaviour in this linear-gaussian calibration setting.

The visual conclusion from Figure 2.2 is that all three fitted lines recover the same linear calibration relationship. This is the expected result because, in the one-edge linear-gaussian model, the SEM2 covariance equations and the OLS normal equations identify the same slope.

2.5 Repeated-seed comparison

A single data set is not sufficient evidence of stability, so the comparison is repeated over 1,000 simulated calibration data sets. Figure 2.3 plots the fitted coefficient from each method for each simulation seed. The horizontal gold reference line is the true value $\beta = 2$.

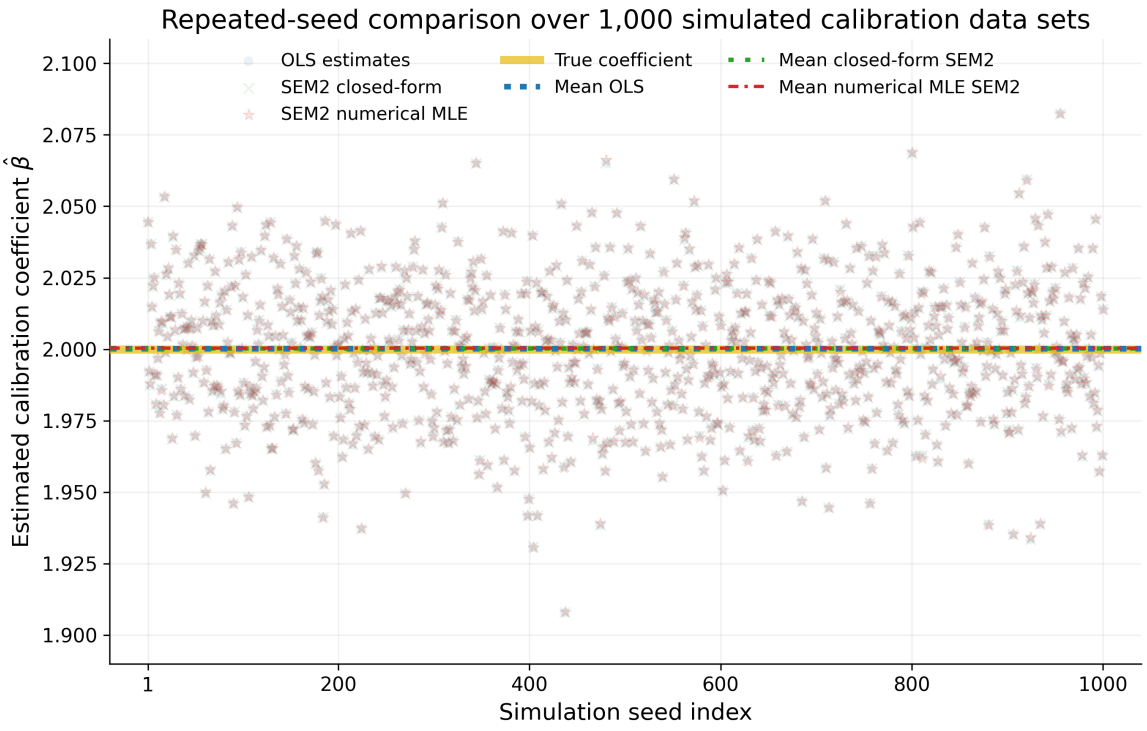


Fig. 2.3 Repeated-seed comparison for the one-edge calibration model over 1,000 simulated data sets. Blue circles show OLS estimates, green crosses show the closed-form SEM2 estimates, and red stars show the numerical MLE SEM2 estimates. The translucent gold horizontal line is the true structural coefficient $\beta = 2$. The coloured reference lines use distinct dash patterns and plotting order, with no numerical offset, because the actual mean estimates are almost identical. The three methods overlap closely, which is the expected behaviour in this linear-gaussian setting.

Table 2.2 gives the corresponding numerical summaries for increasing seed prefixes. The true value is included as a reference column. The closed-form SEM2 estimate is numerically identical to the OLS slope in this model, as predicted by the covariance identity above.

Table 2.2 Seed-stability summary for the linear-regression sanity check. The closed-form SEM2 and OLS estimates agree across the 10-, 50-, 100-, and 1,000-seed prefixes, showing that the fitted SEM2 path coefficient is the same statistical object as the ordinary regression slope in this simple model.

Seeds	True Value (β)	OLS estimate	SEM2 Closed-Form estimate	Mean $\lvert\hat{\beta}_{\text{closed}} - \hat{\beta}_{\text{OLS}}\rvert$	Mean $\lvert\hat{\beta}_{\text{closed}} - \beta\rvert$
10	2	2.0085 ± 0.0228	2.0085 ± 0.0228	0.000000	0.0201
50	2	2.0055 ± 0.0201	2.0055 ± 0.0201	0.000000	0.0169
100	2	2.0046 ± 0.0225	2.0046 ± 0.0225	0.000000	0.0189
1000	2	2.0002 ± 0.0232	2.0002 ± 0.0232	0.000000	0.0186

2.6 Numerical SEM2 MLE cross-check

The closed-form SEM2 route verifies the linear-gaussian covariance calculation. However, it is also important to check that the package's numerical MLE implementation recovers the same coefficient. Table 2.3 therefore compares OLS, closed-form SEM2, and numerical MLE SEM2 over the repeated-seed experiment.

Table 2.3 Numerical SEM2Linear MLE cross-check for the one-edge regression model. The numerical MLE fit is compared with OLS and the closed-form SEM2 solution across increasing seed prefixes. The true structural coefficient is $\beta = 2$.

Seeds	True Value (β)	OLS estimate	Closed-form SEM2 estimate	MLE SEM2 estimate	Mean $\lvert\hat{\beta}_{\text{MLE}} - \hat{\beta}_{\text{closed}}\rvert$	Mean $\lvert\hat{\beta}_{\text{MLE}} - \beta\rvert$
10	2	2.0085 ± 0.0228	2.0085 ± 0.0228	2.0087 ± 0.0227	0.000276	0.0200
50	2	2.0055 ± 0.0201	2.0055 ± 0.0201	2.0057 ± 0.0202	0.000289	0.0170
100	2	2.0046 ± 0.0225	2.0046 ± 0.0225	2.0048 ± 0.0225	0.000268	0.0189
1000	2	2.0002 ± 0.0232	2.0002 ± 0.0232	2.0003 ± 0.0232	0.000274	0.0186

Table 2.4 gives the corresponding optimiser diagnostics. The most important quantities are the MLE–closed-form gap and the LBFGS closure-call count. A closure call is one objective-and-gradient evaluation used internally by LBFGS during its line search, so the mean and maximum closure-call counts describe optimisation effort rather than statistical sample size. The small MLE–closed-form gap shows that the numerical optimiser recovers the same coefficient as the exact SEM2 covariance calculation.

The numerical MLE SEM2 estimates agree with both OLS and the closed-form SEM2 solution. This confirms that the implemented optimisation route recovers the same path coefficient in the simplest linear-gaussian case.

2.7 Sample-size scaling diagnostic

The repeated-seed comparison above fixes the sample size at $n = 2{,}000$. A complementary diagnostic is to vary the sample size and check whether the fitted coefficient becomes less variable at the expected rate. For each sample size n, the experiment is repeated R times, producing R independent simulated data sets of size n. Each repeated data set gives one fitted coefficient,

$$\hat{\beta}_1(n), \hat{\beta}_2(n), \ldots, \hat{\beta}_R(n).$$

Table 2.4 Numerical diagnostics for the SEM2 MLE cross-check. LBFGS is a quasi-Newton optimiser; a closure call is one evaluation of the objective and gradient during the line search. The closure-call counts therefore measure optimisation effort rather than statistical sample size.

Diagnostic	Value	Interpretation
Mean $\lvert\hat{\beta}_{\mathrm{MLE}} - \hat{\beta}_{\mathrm{closed}}\rvert$	0.000274	Average numerical gap between the optimiser variant and the exact SEM2Linear covariance solution.
Maximum $\lvert\hat{\beta}_{\mathrm{MLE}} - \hat{\beta}_{\mathrm{closed}}\rvert$	0.001171	Largest numerical optimisation gap over the 1,000 repeated fits.
Mean $\lvert\hat{\beta}_{\mathrm{MLE}} - \hat{\beta}_{\mathrm{OLS}}\rvert$	0.000274	Average gap between numerical SEM2Linear MLE and the OLS benchmark.
Mean $\lvert\hat{\beta}_{\mathrm{MLE}} - \beta\rvert$	0.018618	Average finite-sample error relative to the true coefficient $\beta = 2$.
Mean LBFGS closure calls	30.0	Average number of objective and gradient evaluations needed by LBFGS per simulated data set.
Maximum LBFGS closure calls	78	Largest observed optimisation effort across the 1,000 simulated data sets.
Mean MLE fitting time per seed	0.0531 s	Average wall-clock time for one numerical SEM2Linear MLE fit.
Repeated simulation seeds	1,000	Number of independent data sets used in the MLE cross-check.

For that fixed sample size, the empirical standard deviation of the fitted coefficients is

$$s(n) = \sqrt{\frac{1}{R-1}\sum_{r=1}^{R}\left(\hat{\beta}_r(n) - \bar{\hat{\beta}}(n)\right)^2}, \qquad \bar{\hat{\beta}}(n) = \frac{1}{R}\sum_{r=1}^{R}\hat{\beta}_r(n).$$

This quantity $s(n)$ is the vertical value plotted in Figure 2.4 and Figure 2.5. Thus each point in the plots corresponds to one sample size n, and its height shows how much the fitted path coefficient varies across repeated simulated data sets of that size.

The expected rate comes from the usual linear-regression error decomposition. In the one-edge model,

$$Y_i = \beta X_i + c\varepsilon_{Y,i},$$

the OLS slope with an intercept satisfies

$$\hat{\beta}(n) - \beta = \frac{\sum_{i=1}^{n}(X_i - \bar{X})c\varepsilon_{Y,i}}{\sum_{i=1}^{n}(X_i - \bar{X})^2}.$$

The numerator is a sum of mean-zero terms and fluctuates at order $\sqrt{n}$, while the denominator grows at order n. Hence

$$\hat{\beta}(n) - \beta = O_p(n^{-1/2}),$$

so the standard deviation of the fitted coefficient should scale approximately as

$$\mathrm{sd}\{\hat{\beta}(n)\} \approx Cn^{-1/2},$$

for some constant $C > 0$. This $n^{-1/2}$ curve is shown as the blue dashed reference line in the two scaling plots.

The closed-form SEM2 estimate is identical to the OLS slope in this calibration study, because both are given by the same covariance ratio,

$$\hat{\beta} = \frac{\widehat{\mathrm{Cov}}(X,Y)}{\widehat{\mathrm{Var}}(X)}.$$

For this reason, the sample-size diagnostic plots OLS and closed-form SEM2 as a single combined series. The numerical SEM2 MLE route is plotted as a separate series. The purpose of this diagnostic is to check the expected $n^{-1/2}$ statistical scaling.

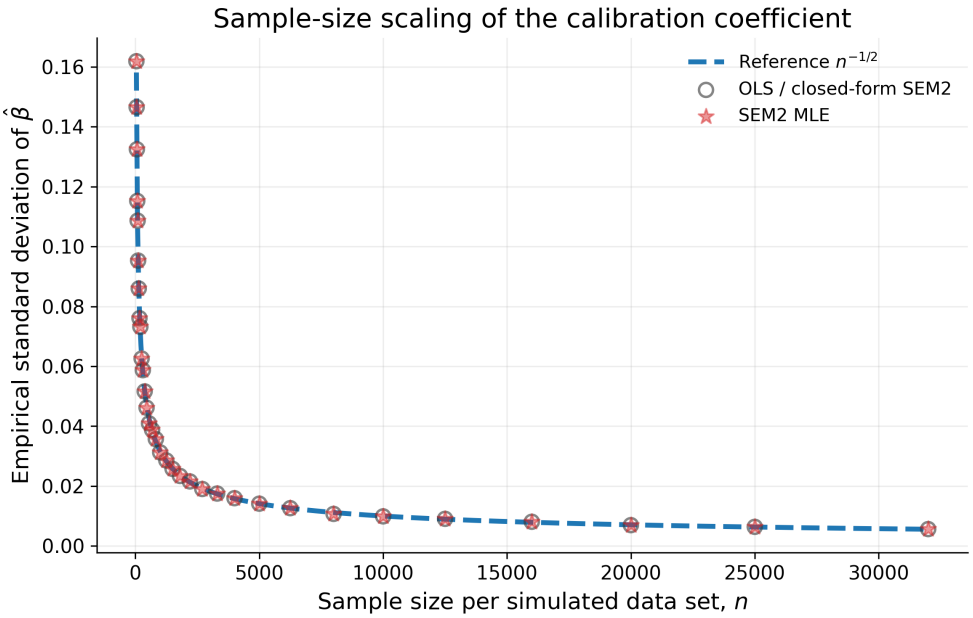


Fig. 2.4 Sample-size scaling diagnostic for the one-edge calibration model on ordinary axes. Each marker corresponds to one sample size n, with the empirical standard deviation of $\hat{\beta}$ computed across repeated simulations. The grey open circles show the combined OLS/closed-form SEM2 estimate, since these two estimates are identical in this model. The red stars show the SEM2 MLE route. The blue dashed curve is the reference $n^{-1/2}$ rate. The fitted methods are plotted as discrete points rather than joined curves because each marker corresponds to a separate simulation setting.

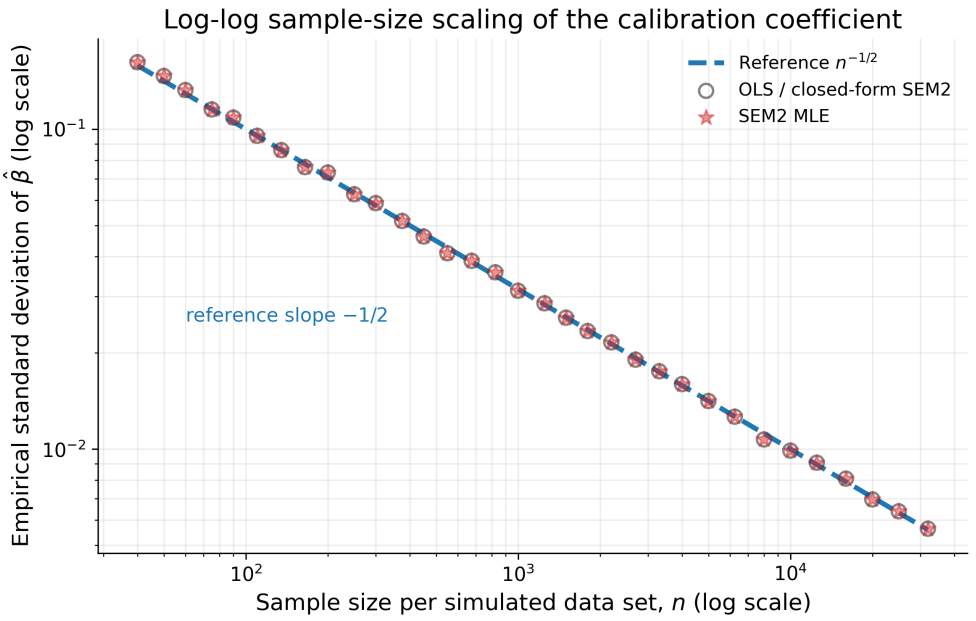


Fig. 2.5 Log-log version of the sample-size scaling diagnostic. On log-log axes, the reference $n^{-1/2}$ rate appears as a straight blue dashed line with slope $-1/2$. The combined OLS/closed-form SEM2 points and the SEM2 MLE points lie close to this reference, which is the expected parametric scaling in the one-edge linear-gaussian model.

Figure 2.4 and Figure 2.5 show the decrease in empirical variability and the expected $n^{-1/2}$ scaling. This basic consistency diagnostic provides an additional check that the simulation behaves as expected.

2.8 Interpretation

The results in Figure 2.2, Figure 2.3, Table 2.2, and Table 2.3 give a consistent conclusion. In the one-edge linear-gaussian calibration model, SEM2 recovers the same path coefficient as ordinary least squares. This is expected theoretically, since both procedures are fitting the covariance relationship

$$\mathrm{Cov}(X,Y) = \beta \mathrm{Var}(X).$$

This chapter should therefore be interpreted as a calibration study rather than a substantive demonstration that SEM2 improves on OLS. The result verifies the coefficient extraction, observed-variable handling, closed-form covariance calculation, and numerical MLE fitting pipeline in the simplest controlled input–response model. The next chapter moves to a classic applied SEM benchmark from social science, where the model contains latent variables, multiple indicators per latent construct, residual covariance terms, and structural paths among latent variables.

Chapter 3

Application Study II: Social-Science Measurement Models

3.1 Applied problem and purpose

Many social-science variables of interest are not directly observed [4]. Concepts such as industrialisation and democracy are instead measured through several imperfect indicators [2, 29]. Structural equation models are useful in this setting because they combine two tasks in one model: a measurement model, which links observed indicators to latent constructs, and a structural model, which relates the latent constructs to one another.

This chapter evaluates SEM2 in this applied social-science setting using the Political Democracy benchmark [1]. The data set contains indicators of industrialisation in 1960 and political democracy in 1960 and 1965. The applied question is not simply whether one observed variable predicts another. Instead, the model asks whether latent industrialisation in 1960 is associated with latent democracy in 1960 and 1965, while accounting for the fact that each latent construct is measured through several observed indicators.

The Political Democracy model is a classical linear-gaussian SEM, so lavaan is used as the external reference implementation. A successful SEM2 result should reproduce a specialised classical SEM software in this regime; the lavaan structural coefficients, measurement loadings, and implied observed covariance structure [33] after converting SEM2 estimates to the same latent-variable scaling convention.

This application is more demanding than the calibration study in Chapter 2. The calibration study contained one observed input, one observed output, and a single structural edge. The Political Democracy benchmark contains latent variables, multiple observed indicators per latent variable, structural paths among latent variables, and residual covariance terms between selected indicators.

3.2 Dataset and model

The Political Democracy data set [1] contains $N = 75$ observations of 11 observed indicators. The indicators measure industrialisation in 1960 and political democracy in 1960 and 1965. The variables x_1, x_2, x_3 are industrialisation indicators in 1960:

$$x_1 = \text{GNP per capita}, \qquad x_2 = \text{energy consumption per capita},$$

$$x_3 = \text{percentage of the labour force in industry}.$$

The variables $y_1, \ldots, y_4$ measure democracy in 1960:

$$y_1 = \text{freedom of the press}, \qquad y_2 = \text{freedom of political opposition},$$

$$y_3 = \text{fairness of elections}, \qquad y_4 = \text{effectiveness of the elected legislature}.$$

The variables $y_5, \ldots, y_8$ measure the same four democracy indicators in 1965.

The model represents these 11 observed indicators using three latent variables:

$$\texttt{ind60}, \qquad \texttt{dem60}, \qquad \texttt{dem65},$$

corresponding to industrialisation in 1960, democracy in 1960, and democracy in 1965. Thus, x_1, x_2, x_3 are indicators of `ind60`; $y_1, \ldots, y_4$ are indicators of `dem60`; and $y_5, \ldots, y_8$ are indicators of `dem65`.

The model has two components. The measurement component links each latent variable to its observed indicators. In schematic form,

$$x_j = \lambda_{xj}\texttt{ind60} + \varepsilon_{xj}, \qquad j = 1, 2, 3,$$

$$y_j = \lambda_{yj}\texttt{dem60} + \varepsilon_{yj}, \qquad j = 1, \ldots, 4,$$

and

$$y_j = \lambda_{yj}\texttt{dem65} + \varepsilon_{yj}, \qquad j = 5, \ldots, 8.$$

The structural component then relates the latent variables:

$$\texttt{dem60} = \alpha_1\texttt{ind60} + \varepsilon_{\texttt{dem60}},$$

$$\texttt{dem65} = \alpha_2\texttt{ind60} + \alpha_3\texttt{dem60} + \varepsilon_{\texttt{dem65}}.$$

The quantities of interest in this application study are therefore the three structural path coefficients:

$$\alpha_1, \qquad \alpha_2, \qquad \alpha_3.$$

Figure 3.1 gives a conceptual path diagram for the Political Democracy SEM. The upper purple-outlined nodes are the latent variables, and the lower green-outlined nodes are the observed indicators. The directed arrows show the measurement links from latent variables to their indicators and the structural paths among the latent variables. The dashed red arcs show the residual covariance terms included in the classical Political Democracy specification.

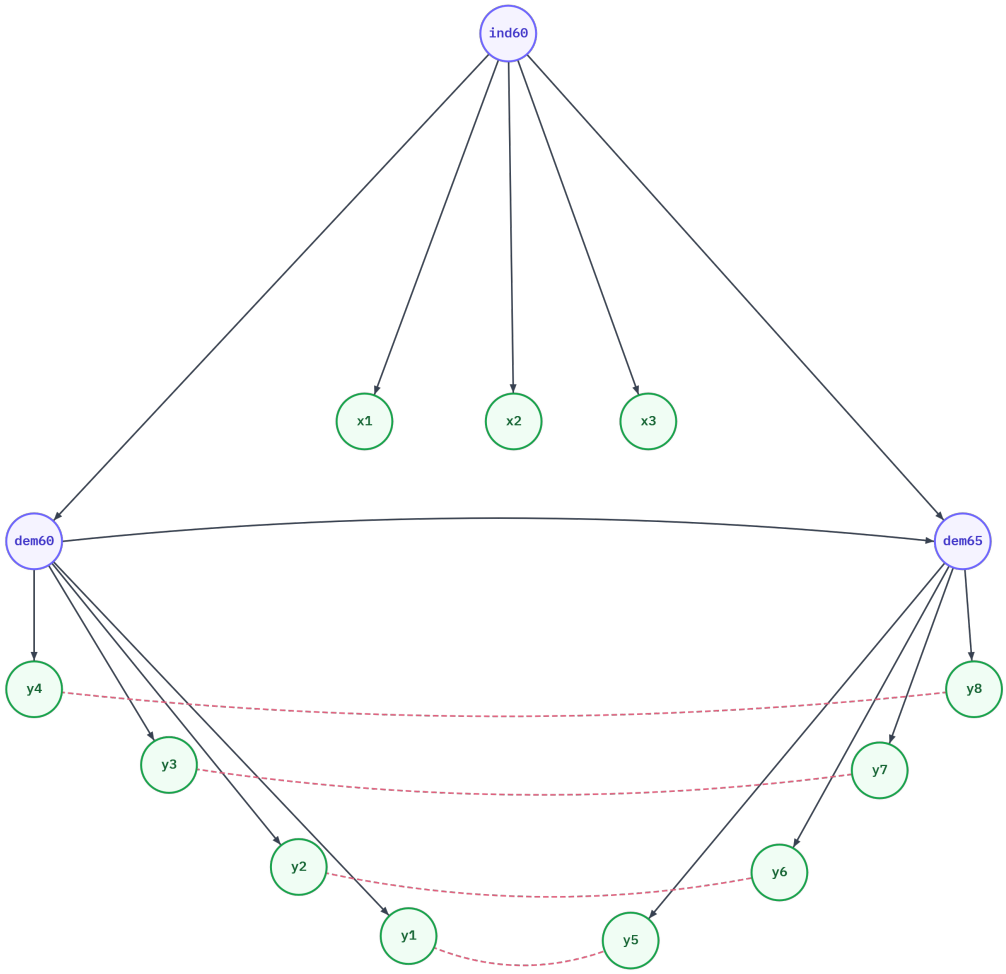


Fig. 3.1 Conceptual path diagram for the Political Democracy SEM. The purple-outlined nodes are the latent variables `ind60`, `dem60`, and `dem65`, while the green-outlined nodes are the observed indicators. The directed dark arrows show both the measurement links from each latent variable to its indicators and the structural paths among the latent variables. The dashed red arcs show the residual covariance terms between selected observed democracy indicators.

Before fitting the SEM, it is useful to inspect the empirical correlation structure of the observed indicators. Figure 3.2 shows the observed correlation matrix for $x_1, x_2, x_3, y_1, \ldots, y_8$. The variables are ordered by their latent-variable groups. Strong within-block correlations support the use of a latent measurement model, while the cross-block correlations motivate the structural paths among `ind60`, `dem60`, and `dem65`.

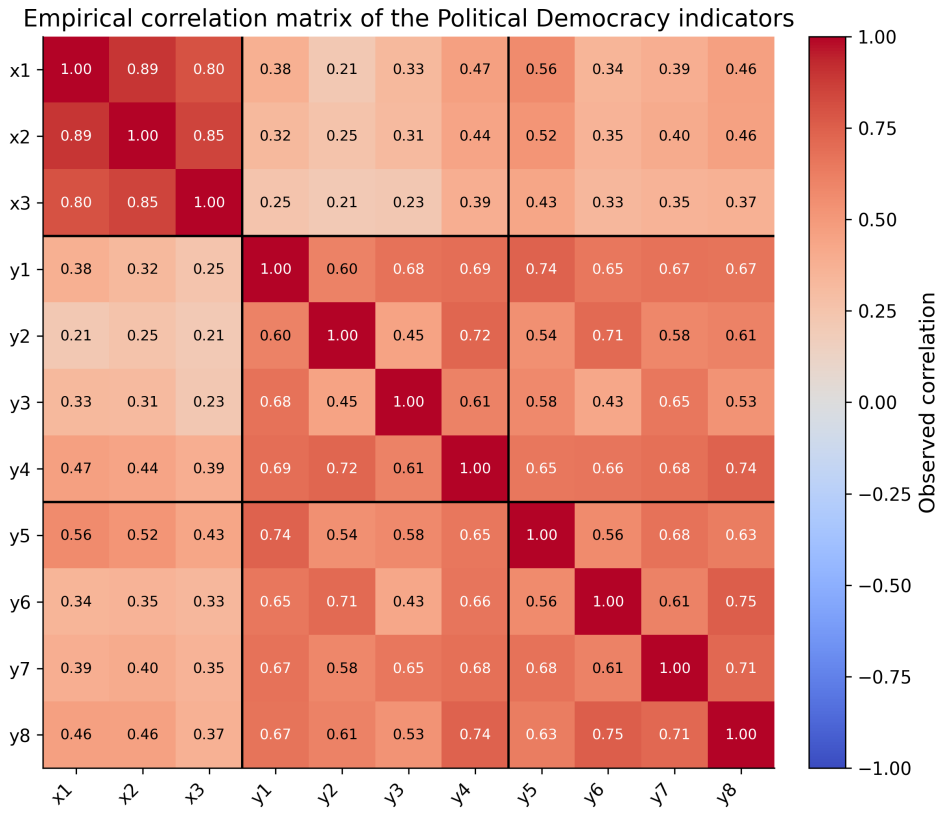


Fig. 3.2 Empirical correlation matrix for the observed Political Democracy indicators. Variables are ordered by their measurement blocks: x_1, x_2, x_3 for `ind60`, $y_1, \ldots, y_4$ for `dem60`, and $y_5, \ldots, y_8$ for `dem65`. This is a data-orientation plot rather than a SEM2 validation result.

This model is a substantially richer test case than the one-edge calibration model. SEM2 must not only estimate a single path coefficient, but also construct latent variables, fit measurement equations, represent the model-implied covariance matrix, and recover structural coefficients under a consistent latent-variable scaling.

3.3 SEM2 specification and lavaan benchmark

Table 3.1 summarises the SEM2 setup and lavaan benchmark. The aim is to compare SEM2 with a trusted classical SEM implementation in a setting where the assumptions are exactly those of a linear-gaussian SEM. The observed variables are $x_1, x_2, x_3, y_1, \ldots, y_8$, while the latent variables are `ind60`, `dem60`, and `dem65`. The structural paths of interest are

$$\texttt{dem60} \sim \texttt{ind60}, \qquad \texttt{dem65} \sim \texttt{ind60}, \qquad \texttt{dem65} \sim \texttt{dem60}.$$

Table 3.1 Experimental setup for the Political Democracy validation test. The experiment checks whether SEM2 reproduces a classical lavaan linear-Gaussian latent-variable SEM fit.

Component	Specification
Dataset	Political Democracy data set with 75 observations and observed indicators $x_1, x_2, x_3, y_1, \ldots, y_8$.
Latent variables	`ind60`, `dem60`, and `dem65`.
Measurement model	x_1, x_2, x_3 measure `ind60`; $y_1, \ldots, y_4$ measure `dem60`; $y_5, \ldots, y_8$ measure `dem65`.
Structural paths of interest	`dem60` $\sim$ `ind60`, `dem65` $\sim$ `ind60`, and `dem65` $\sim$ `dem60`.
Residual covariance terms	y_1 y_5, y_2 y_4, y_2 y_6, y_3 y_7, y_4 y_8, and y_6 y_8.
SEM2 fitting method	SEM2Linear with a centred Gaussian log-likelihood, Adam optimiser and the closed-form model-implied covariance matrix.
Benchmark	lavaan maximum-likelihood fit of the same SEM.
Evaluation	Structural coefficient error, measurement-loading error, and observed-correlation fit diagnostics.

The external benchmark is a lavaan maximum-likelihood fit of the same linear-gaussian SEM. Table 3.2 reports the main lavaan fit measures. This table also provides a basic check that the classical SEM specification is reasonable for this data set. The model chi-square statistic compares the covariance matrix implied by the fitted SEM with the empirical covariance matrix. The test gives $\chi^2 = 38.125$ on 35 degrees of freedom, with $p = 0.329$. Thus, this diagnostic does not provide strong evidence that the classical Political Democracy SEM is badly misspecified. The lavaan fit is therefore a suitable external reference for checking whether SEM2 reproduces the standard linear-gaussian SEM solution.

Table 3.2 lavaan fit summary for the Political Democracy benchmark. Here N is the number of observations; χ^2 is the model chi-square statistic comparing the model-implied and empirical covariance matrices; df denotes the corresponding degrees of freedom; and the p-value is for the chi-square test. CFI is the Comparative Fit Index and TLI is the Tucker-Lewis Index; both are incremental fit indices, with values closer to one indicating better fit. RMSEA is the Root Mean Square Error of Approximation and SRMR is the Standardized Root Mean Square Residual; both are residual-misfit indices, with values closer to zero indicating better fit.

N	χ^2	df	p-value	CFI	TLI	RMSEA	SRMR
75	38.125	35	0.329	0.995	0.993	0.035	0.044

The SEM2 formula follows the same structure as the lavaan model. The variables `ind60`, `dem60`, and `dem65` are latent variables. The observed indicators are generated by linear measurement equations. The residual covariance terms in the lavaan specification are

represented in SEM2 by shared exogenous noise variables such as `e15`, `e24`, and `e68`. This allows the SEM2 formula to represent the same covariance structure while using the same structural-equation syntax as the simpler calibration example. The SEM2 formula used for the experiment is shown in Listing 3.1.

```
ex1,ex2,ex3,ey1,ey2,ey3,ey4,ey5,ey6,ey7,ey8,edem60,edem65 ~
   Normal(0,1);
e15,e24,e26,e37,e48,e68 ~ Normal(0,1);

ind60 ~ Normal(0,1);
dem60 <- ind60 + edem60;
dem65 <- ind60 + dem60 + edem65;

x1 <- ind60 + ex1;
x2 <- ind60 + ex2;
x3 <- ind60 + ex3;

y1 <- dem60 + ey1 + e15;
y2 <- dem60 + ey2 + e24 + e26;
y3 <- dem60 + ey3 + e37;
y4 <- dem60 + ey4 + e24 + e48;

y5 <- dem65 + ey5 + e15;
y6 <- dem65 + ey6 + e26 + e68;
y7 <- dem65 + ey7 + e37;
y8 <- dem65 + ey8 + e48 + e68;

unobs(ex1,ex2,ex3,ey1,ey2,ey3,ey4,ey5,ey6,ey7,ey8,
      edem60,edem65,e15,e24,e26,e37,e48,e68,
      ind60,dem60,dem65)
```

Listing 3.1 SEM2 specification for the Political Democracy model. The observed variables are `x1`, `x2`, `x3`, and `y1`–`y8`. The latent variables `ind60`, `dem60`, and `dem65`, together with the exogenous noise terms, are declared unobserved. Shared exogenous noise variables encode the residual covariance terms used in the lavaan specification.

The exogenous terms are split into two groups for readability. The variables `ex1`, ..., `ey8`, `edem60`, and `edem65` are ordinary residual noise terms for individual measurement and structural equations. The variables `e15`, `e24`, `e26`, `e37`, `e48`, and `e68` are shared residual noise

terms. They are included in pairs of observed indicators to reproduce the residual covariance terms in the lavaan model, such as `y1 ~~ y5` and `y6 ~~ y8`. This is a representational device: it lets SEM2 express the same covariance structure using structural equations and shared exogenous noise variables.

Because the scales of latent variables are arbitrary, SEM2 coefficients must be converted to the same scaling convention as lavaan before numerical comparison. lavaan uses first-indicator scaling: the loading of x_1 on `ind60`, the loading of y_1 on `dem60`, and the loading of y_5 on `dem65` are fixed to one. SEM2 learns these marker loadings directly. Therefore, for a structural path from latent variable A to latent variable B, the SEM2 coefficient is converted by

$$\hat{\alpha}_{\text{lavaan scale}} = \hat{\alpha}_{\text{SEM2 raw}} \frac{\hat{\lambda}_B}{\hat{\lambda}_A},$$

where $\hat{\lambda}_A$ and $\hat{\lambda}_B$ are the SEM2 marker loadings for the parent and child latent variables. All SEM2 coefficients reported below use this lavaan-compatible scale.

Listing 3.1 is longer than the one-edge calibration formula because this benchmark tests more of the package where SEM2 must parse the latent measurement model, the structural regressions among latent variables, and the selected residual covariance terms. A successful fit should therefore agree with lavaan not only on the three structural paths, but also on the measurement loadings and the fitted observed covariance structure.

Figure 3.3 shows the corresponding SEM2 computational graph. Compared with the conceptual SEM diagram in Figure 3.1, this graph exposes the exogenous noise variables used by the implementation. The ordinary noise variables provide residual variation for individual equations, while the shared noise variables encode the residual covariances from the lavaan specification.

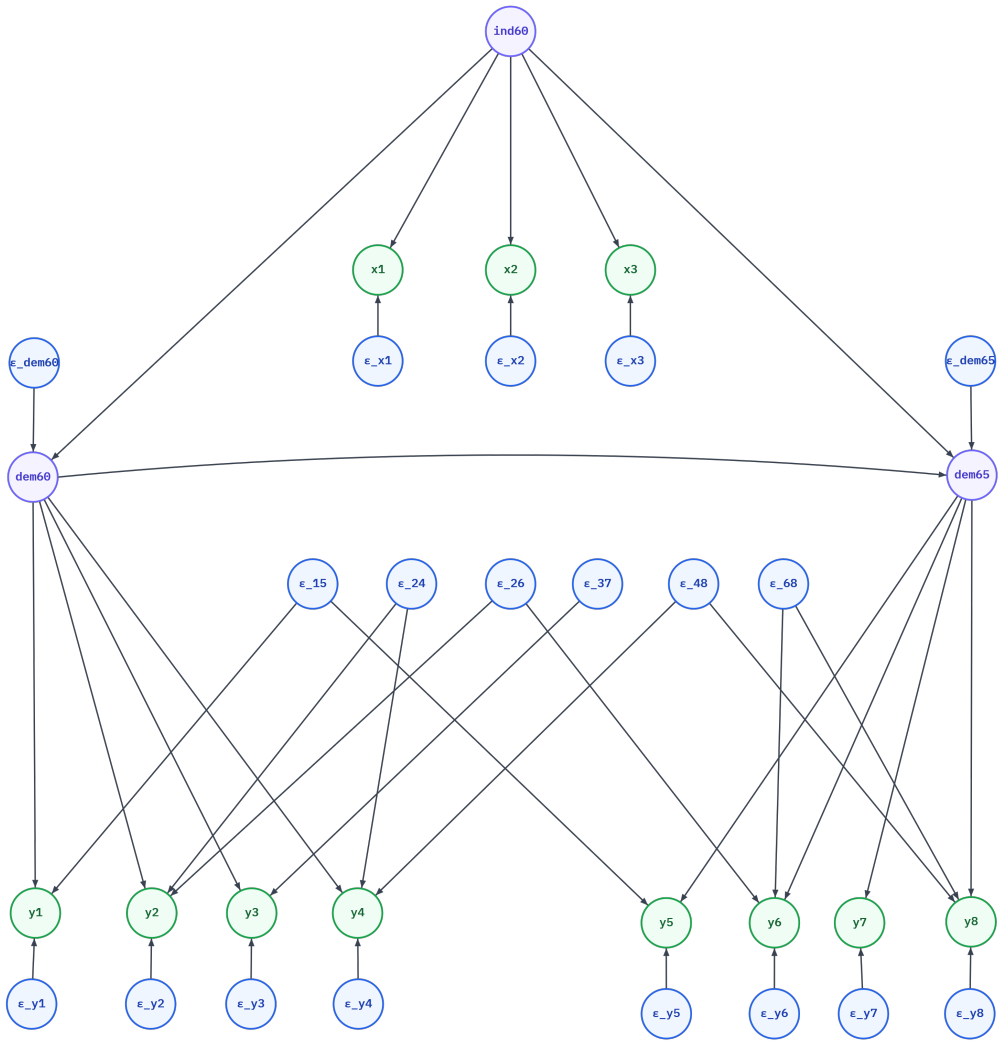


Fig. 3.3 Detailed SEM2 computational graph for the Political Democracy benchmark. Green nodes are observed indicators, blue nodes are latent or exogenous variables, and the directed edges encode the SEM2 structural equations. The shared exogenous variables $e_{15}, e_{24}, e_{26}, e_{37}, e_{48}, e_{68}$ represent the residual covariance terms in the lavaan model. This figure is more detailed than the conceptual SEM diagram because it shows how the residual covariance structure is implemented using shared noise variables.

3.4 Structural coefficient recovery

The first result is the structural coefficient comparison. Table 3.3 compares the SEM2 estimates with the lavaan reference estimates on the three latent-variable regressions. Figure 3.4 shows the same comparison visually, with lavaan 95% confidence intervals and SEM2 point estimates.

The lavaan estimates are shown with 95% confidence intervals, while SEM2 is shown as point estimates. This reflects the role of the experiment: lavaan provides the mature classical SEM reference, including standard errors, while SEM2 is being checked for agreement in point estimation. Extending SEM2 with Hessian-based, bootstrap, or sandwich standard errors is a natural future software development, but is not required for the point-estimation validation in this chapter.

Table 3.3 Political Democracy structural coefficient comparison. SEM2 is compared with the lavaan maximum-likelihood reference fit on the three main structural paths among the latent variables.

Structural path	lavaan	SEM2Linear	Absolute difference	lavaan 95% CI
`dem60 ~ ind60`	1.4830	1.4825	0.000467	[0.7007, 2.2653]
`dem65 ~ ind60`	0.5723	0.5726	0.000283	[0.1386, 1.0061]
`dem65 ~ dem60`	0.8373	0.8363	0.000995	[0.6446, 1.0301]

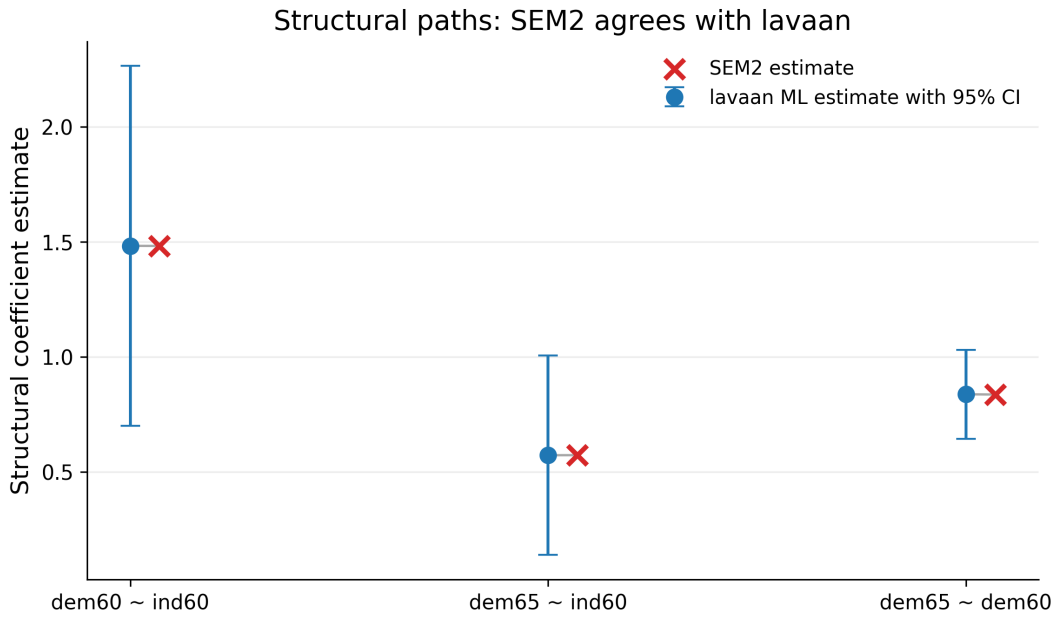


Fig. 3.4 Structural coefficient comparison for the Political Democracy model. Blue points and intervals show the lavaan maximum-likelihood estimates and 95% confidence intervals. Red crosses show the SEM2 estimates after converting to the same first-indicator latent-variable scaling. Agreement on these three paths is the main validation target for this application study.

The agreement in Table 3.3 is very close. The SEM2 estimate differs from the lavaan reference by less than 0.001 on all three structural paths. This indicates that SEM2 recovers the same latent structural regressions as lavaan once the same scaling convention is used.

3.5 Measurement-model recovery

The second result checks the measurement part of the model. Table 3.4 compares the lavaan and SEM2 measurement loadings, while Figure 3.5 visualises these loadings with lavaan confidence intervals. This matters because matching only the three structural regressions would not verify that SEM2 has recovered the latent-variable measurement model.

The first loading for each latent variable has no lavaan confidence interval because it is fixed to one for identification. These fixed marker loadings set the scale of `ind60`, `dem60`, and `dem65`; they are not estimated from the data. The remaining loadings are estimated and therefore have lavaan confidence intervals.

Table 3.4 Measurement-loading comparison for the Political Democracy model. The first indicator for each latent variable is fixed to one under the lavaan scaling convention.

Loading	lavaan estimate	SEM2Linear	Absolute difference
`x1` ← `ind60`	1.0000	1.0000	0.0000
`x2` ← `ind60`	2.1804	2.1805	0.0001
`x3` ← `ind60`	1.8185	1.8185	0.0000
`y1` ← `dem60`	1.0000	1.0000	0.0000
`y2` ← `dem60`	1.2567	1.2565	0.0003
`y3` ← `dem60`	1.0577	1.0573	0.0004
`y4` ← `dem60`	1.2648	1.2630	0.0018
`y5` ← `dem65`	1.0000	1.0000	0.0000
`y6` ← `dem65`	1.1857	1.1863	0.0006
`y7` ← `dem65`	1.2795	1.2796	0.0001
`y8` ← `dem65`	1.2659	1.2647	0.0013

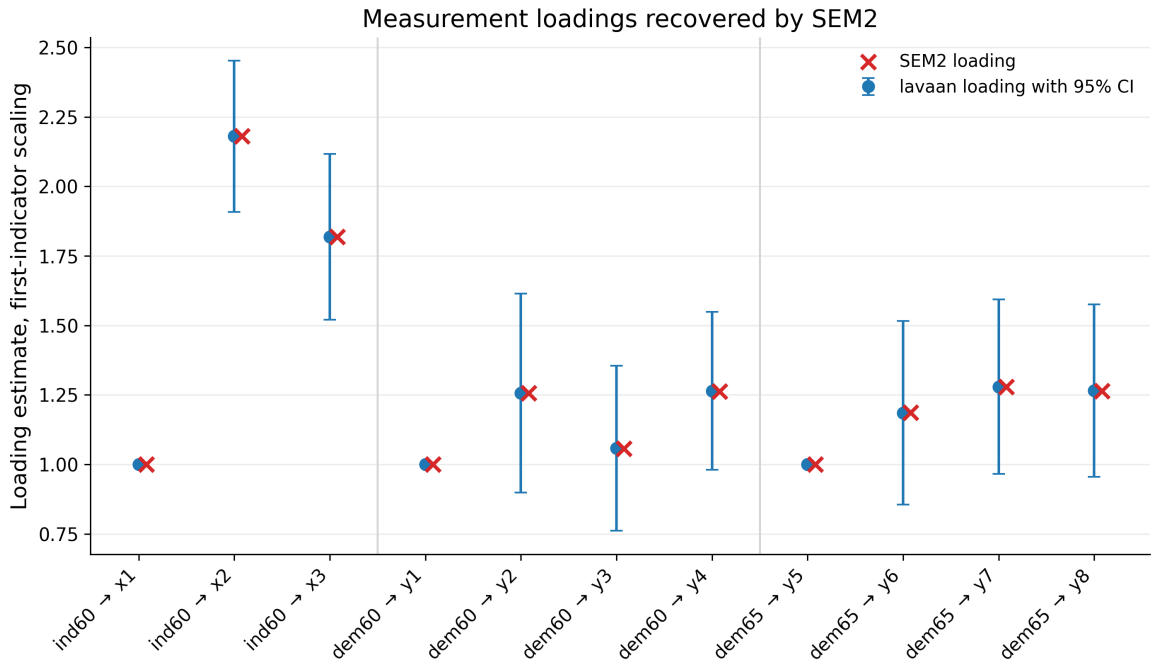


Fig. 3.5 Measurement-loading comparison between lavaan and SEM2. The lavaan first-indicator scaling fixes the first loading for each latent variable to one. SEM2 loadings are converted to the same scaling before comparison. Agreement here shows that SEM2 recovers the measurement component of the latent-variable SEM, not only the structural regressions among latent variables.

The loading comparison shows that SEM2 reproduces the lavaan measurement model to high numerical accuracy. The first loading for each latent variable is fixed to one under the lavaan scaling convention, and the remaining loadings are recovered by SEM2 after converting to that same convention.

3.6 Observed covariance and optimisation diagnostics

The third result checks the fitted covariance structure at the observed-variable level. There are 11 observed variables in the model, so there are

$$\binom{11}{2} = 55$$

unique off-diagonal observed correlations. The diagonal correlations are all equal to one and are therefore omitted from the diagnostic plot. Figure 3.6 compares these 55 empirical observed correlations with the lavaan- and SEM2-implied correlations. Each point corresponds to one pair of observed variables. Points close to the diagonal indicate that the fitted model reproduces the observed correlation structure.

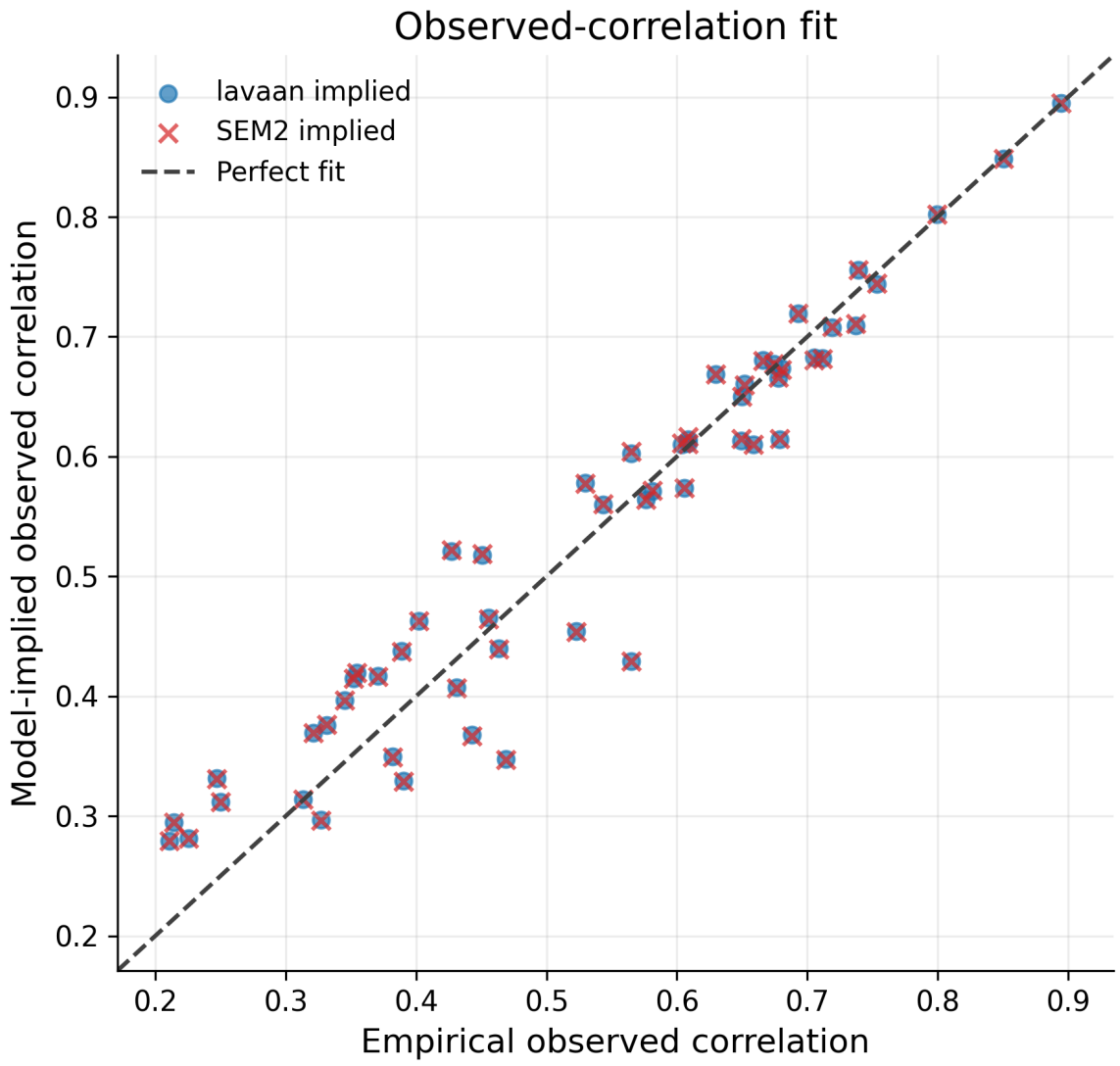


Fig. 3.6 Observed-correlation fit for the Political Democracy model. Each point corresponds to one of the 55 unique off-diagonal observed correlations. The x-axis is the empirical observed correlation, while the y-axis is the model-implied observed correlation from lavaan or SEM2. Points close to the diagonal indicate that the fitted model reproduces the observed covariance structure.

Table 3.5 summarises the correlation-fit diagnostic numerically. The lavaan and SEM2 diagnostics are essentially identical, and the SEM2 implied correlations match the lavaan implied correlations up to numerical precision.

Table 3.5 Observed-correlation fit diagnostics for the Political Democracy model. Diagnostics are computed over the off-diagonal entries of the observed correlation matrix. There are 11 observed variables in the model, so there are $\binom{11}{2} = 55$ unique off-diagonal observed correlations.

Diagnostic	lavaan	SEM2Linear
RMSE versus empirical correlations	0.0488	0.0488
Maximum absolute error versus empirical correlations	0.1355	0.1355
RMSE versus lavaan implied correlations	–	0.0000
Maximum absolute difference versus lavaan implied correlations	–	0.0000
Number of off-diagonal correlations	55	

Figure 3.7 provides a more direct view of the same point by plotting the SEM2–lavaan difference in implied observed correlations. The differences are very small across the observed-variable correlation matrix, which confirms that the two fitted models imply the same observed covariance structure.

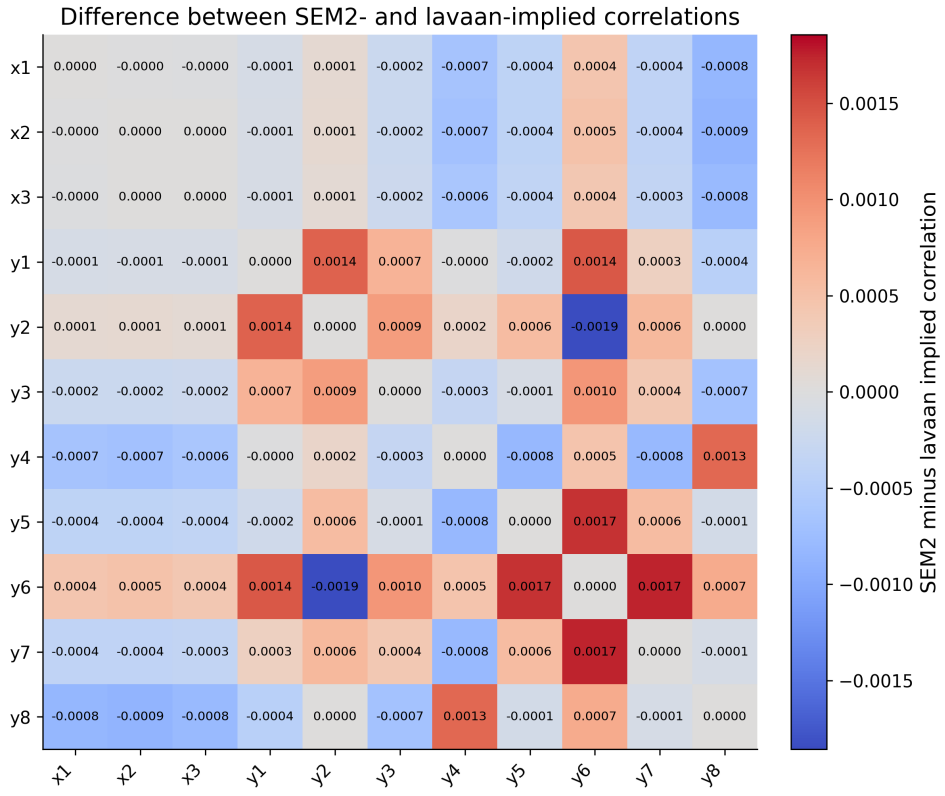


Fig. 3.7 Difference between SEM2- and lavaan-implied observed correlations. Each entry shows the SEM2 implied correlation minus the lavaan implied correlation for a pair of observed variables. Values close to zero indicate that the two implementations recover the same fitted observed correlation structure.

As a final implementation diagnostic, Figure 3.8 shows the SEM2 coefficient trajectories during optimisation. The coloured curves are the three structural coefficients during training, converted to the lavaan scaling convention, and the dotted horizontal lines are the corresponding lavaan reference estimates from Table 3.3. The early iterations show movement from the random initialisation. The important feature is not the early overshoot, but the later stabilisation near the lavaan reference values. This plot is therefore used only to check that the numerical optimisation path reaches the classical SEM solution; the main statistical evidence remains the coefficient, loading, and covariance comparisons above.

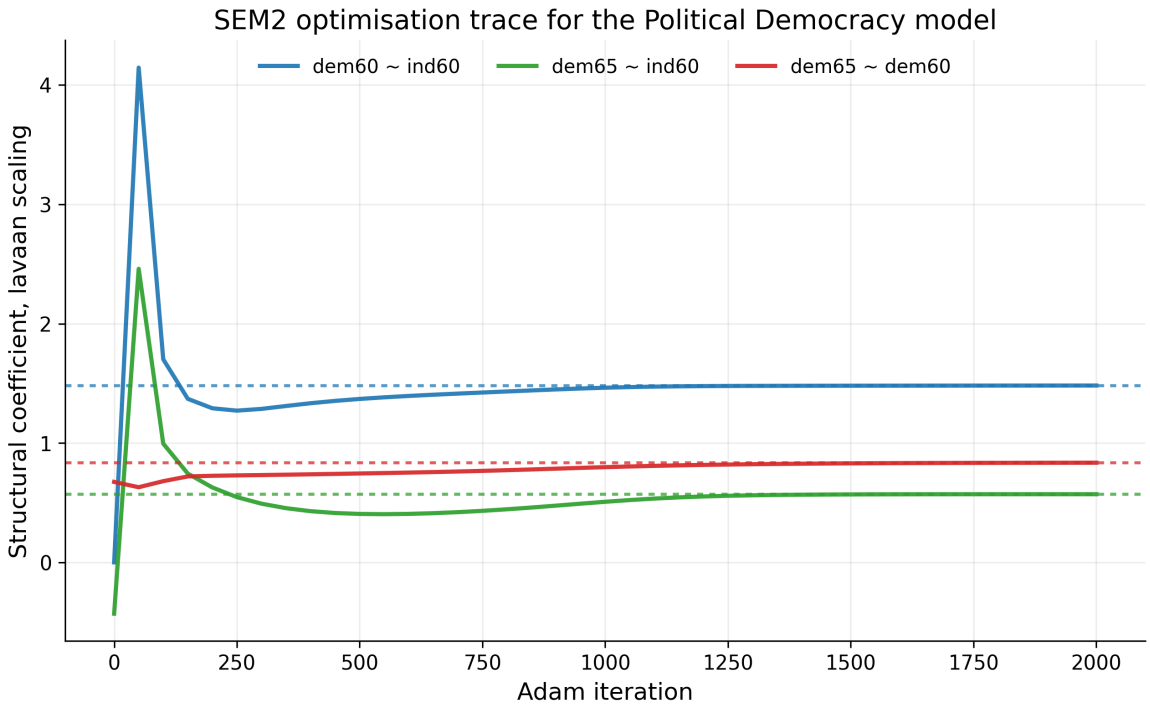


Fig. 3.8 SEM2 optimisation trace for the Political Democracy benchmark. The plot shows deterministic full-batch gaussian log-likelihood training behaviour. The coloured curves show the three structural coefficients during optimisation after conversion to the lavaan scaling convention. The dotted horizontal lines are the lavaan reference estimates from Table 3.3. The early iterations reflect movement from the random initialisation, while the later iterations stabilise near the lavaan values. This figure is an optimisation diagnostic rather than the main statistical validation result.

3.7 Interpretation

The Political Democracy application validates SEM2 in a substantially richer setting than the one-edge calibration study. The model contains latent variables, measurement equations, structural regressions among latent variables, and selected residual covariance terms. Agreement with lavaan therefore tests more than a single regression coefficient: it checks the

formula parser, latent-variable construction, model-implied covariance calculation, gaussian likelihood, coefficient extraction, and latent-variable scaling conversion.

The evidence is layered : Table 3.3 and Figure 3.4 show that SEM2 recovers the three main structural path coefficients from the lavaan fit. Table 3.4 and Figure 3.5 show that the measurement loadings are also recovered. Figure 3.6, Table 3.5, and Figure 3.7 then show that the fitted observed correlation structure is effectively the same as the lavaan-implied structure.

This result should be interpreted as a calibration test in a real social-science SEM benchmark rather than as a claim that SEM2 improves on specialised SEM software. lavaan is a mature implementation for classical linear-gaussian SEMs and provides standard errors and confidence intervals as part of its inference framework. SEM2 is used here for point-estimation validation. The lavaan confidence intervals in the coefficient and loading plots therefore quantify uncertainty for the classical reference fit, whereas the SEM2 estimates are shown as point estimates. Extending SEM2 with standard errors is a natural future software development.

The calibration chapter verified the simplest observed-variable path coefficient and this Political Democracy chapter shows that the same SEM2 machinery extends to a realistic social-science latent-variable SEM with measurement equations and residual covariances. The next application moves from linear-gaussian social-science SEM to independent component analysis, where the structural equations remain linear but the latent variables are non-gaussian.

Chapter 4

Application Study III: Engineering Signal Separation

4.1 Applied problem and purpose

In many engineering systems, sensors tend to record a mixture of several latent sources, which mask the quantities of interest [14, 18, 38]. For example, two microphones placed in the same room may each record different linear combinations of the same speakers [19, 30]. The statistical task is then to recover the original sources from the observed mixtures without observing the sources directly.

Independent component analysis (ICA) is the classical linear formulation of this source-separation problem [10, 21]. The observed vector is modelled as a linear transformation of statistically independent, non-gaussian latent variables. This chapter evaluates whether SEM2 can represent and fit such a model using the same structural-equation language used in the preceding applications.

The ICA application differs from the Political Democracy model in two important ways. First, the latent variables are non-gaussian, so their information is not contained in the covariance matrix alone. Second, the target is not one structural path coefficient. The aim is to recover the complete mixing structure and, through its inverse, the latent source values. FastICA [20] is used as the specialised benchmark because it is designed specifically for linear ICA. The SEM2 fit is evaluated against the same simulated data and the known latent sources.

The experiment uses two latent sources, two observed mixtures, no additive measurement noise, and correctly specified source families. This produces a transparent test of whether the SEM2 graph and likelihood machinery can recover a non-gaussian latent-variable model.

4.2 Data-generating process

The simulator uses the square ICA model

$$X = A_{\text{true}} S,$$

where

$$S = \begin{pmatrix} S_1 \\ S_2 \end{pmatrix}, \qquad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}.$$

The latent sources are mutually independent and follow

$$S_1 \sim \text{Laplace}\left(0, \frac{1}{\sqrt{2}}\right), \qquad S_2 \sim \sqrt{\frac{3}{5}}\, t_5.$$

The scale factors are chosen so that both sources have mean zero and variance one. The first source is sharply peaked with exponential tails, while the second has heavier polynomial tails. Thus, the two sources are both non-gaussian but have visibly different distributional shapes.

The true mixing matrix is

$$A_{\text{true}} = \begin{pmatrix} 1.00 & 0.65 \\ -0.45 & 1.15 \end{pmatrix}.$$

Consequently, the observed variables satisfy

$$X_1 = 1.00 S_1 + 0.65 S_2, \qquad X_2 = -0.45 S_1 + 1.15 S_2.$$

Neither observed variable is itself one of the original sources. Each is a different linear mixture of both sources. The fitting methods observe only the paired measurements (X_1, X_2); the simulated values of (S_1, S_2) are retained only to evaluate recovery.

Figure 4.1 shows the structural interpretation. Every source points to every observed mixture, and the four arrow coefficients are the entries of the mixing matrix.

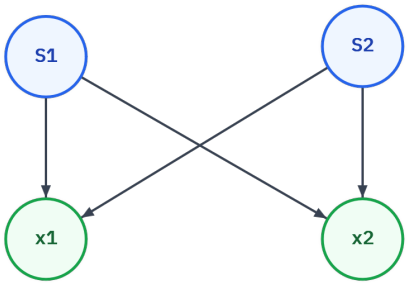


Fig. 4.1 Two-source ICA represented as a SEM2 graph. The upper nodes S_1 and S_2 are independent non-gaussian latent sources, while the lower nodes X_1 and X_2 are observed mixtures. The four directed-edge coefficients form the mixing matrix A.

Figure 4.2 shows a representative sample from the two latent distributions. Figure 4.3 then shows the data after applying A_{true}. The elongated mixture cloud and the similar-looking observed marginals illustrate why inspecting X_1 and X_2 directly does not reveal the original source variables.

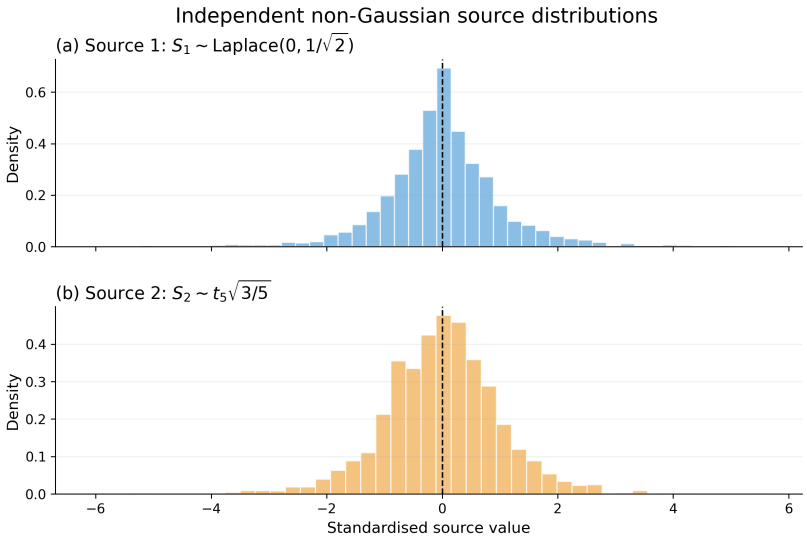


Fig. 4.2 True independent source distributions in the representative ICA data set. Panel (a) shows the unit-variance Laplace source and panel (b) shows the unit-variance Student-t_5 source. The sources have different non-gaussian shapes, but neither is observed by the fitting methods.

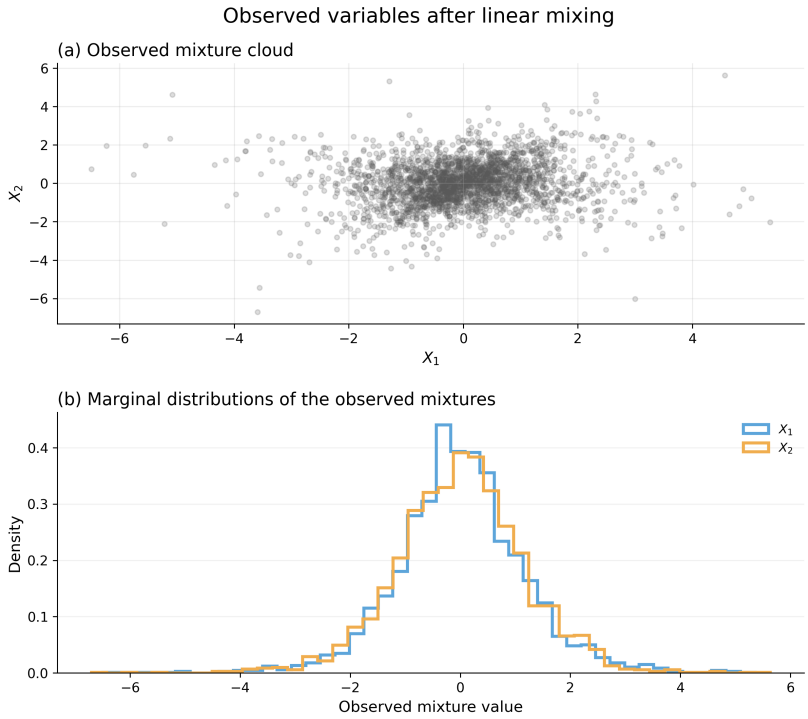


Fig. 4.3 Observed variables after linear mixing. Panel (a) shows the joint mixture cloud (X_1, X_2) and panel (b) shows the two observed marginal distributions. The source-separation task is to recover the independent variables in Figure 4.2 using only these observed mixtures.

4.3 SEM2 specification and likelihood

The fitted model uses the SEM2 formula in Listing 4.1. The source distributions are declared explicitly, while the coefficients connecting the sources to the mixtures are left free. The declaration `unobs(s1,s2)` marks the source variables as latent, so the observed data supplied to the model consist only of `x1` and `x2`.

```
s1 ~ Laplace(0.0,0.7071);
s2 ~ StudentT(5.0,0.0,0.7746);
x1 <- s1 + s2;
x2 <- s1 + s2;
unobs(s1,s2)
```

Listing 4.1 SEM2 specification for the two-source ICA fit. The source variables `s1` and `s2` are latent. The unspecified coefficients in the equations for `x1` and `x2` are learned and together form the estimated mixing matrix.

Let A_θ denote the current 2×2 mixing matrix extracted from the SEM2 coefficient matrix. Because the model is square and A_θ is invertible, define

$$W_\theta = A_\theta^{-1}.$$

For one observed mixture x_i, the source values implied by the current graph are

$$\widehat{s_i} = W_\theta x_i.$$

Independence of the declared sources gives

$$p_S(s) = p_1(s_1)p_2(s_2).$$

Applying the change-of-variables formula therefore gives the observed-data density

$$p_\theta(x_i) = p_1((W_\theta x_i)_1)\, p_2((W_\theta x_i)_2)\, |\det W_\theta|\,,$$

and hence

$$\log p_\theta(x_i) = \log p_1((W_\theta x_i)_1) + \log p_2((W_\theta x_i)_2) + \log |\det W_\theta|\,.$$

The first two terms reward transformations that produce source values consistent with the Laplace and Student-t distributions declared in the SEM2 formula. The determinant term corrects for the change in volume induced by the inverse mixing transformation. Maximising this likelihood therefore learns the coefficients in the SEM2 graph.

FastICA provides the specialised reference fit. It first whitens the observed data and then searches for directions with strong non-gaussianity. The experiment uses unit-variance whitening and the standard `logcosh` contrast. SEM2 is optimised with LBFGS from six random initialisations, and the fit with the largest likelihood is retained.

Table 4.1 Experimental setup for the two-source ICA application. The same 50 simulated data sets are fitted by FastICA and SEM2.

Component	Specification
Applied problem	Recovery of independent latent signals from observed linear mixtures.
Sample size	$n = 2{,}500$ paired observations per simulated data set.
Repeated simulations	50 independent simulation seeds.
Latent sources	Unit-variance Laplace source and unit-variance Student-t_5 source.
True mixing matrix	$A_{\text{true}} = \left(\begin{smallmatrix} 1.00 & 0.65 \\ -0.45 & 1.15 \end{smallmatrix}\right)$.
FastICA benchmark	Unit-variance whitening, `logcosh` contrast, tolerance 10^{-6}, and at most 3,000 iterations.
SEM2 objective	Change-of-variables likelihood induced by the parsed ICA graph and declared source densities.
SEM2 optimiser	LBFGS with strong-Wolfe line search; six random restarts and at most 140 iterations per restart.
Evaluation	Recovered-source distributions, pointwise source correlation, Wasserstein distance, Amari error, mixing-matrix diagnostic, and wall-clock time.

4.4 How source recovery is evaluated

ICA has unavoidable scale, sign, and permutation ambiguities. For example, exchanging the labels of S_1 and S_2 produces an observationally equivalent representation. Multiplying one recovered source by -1 and multiplying its mixing column by -1 also leaves the observed mixtures unchanged. A direct element-by-element comparison of unaligned source estimates would therefore be misleading.

For evaluation, the recovered components are matched to the true sources by maximising the absolute source correlations. Their signs are then adjusted and each component is standardised. This alignment is used only because the true sources are known in the synthetic experiment; it is not information supplied during fitting.

The chapter reports three complementary recovery measures. First, the absolute correlation between each true source and its matched recovered source measures pointwise recovery across observations. A value near one means that individual source values are reconstructed accurately. Second, the one-dimensional Wasserstein distance compares the

recovered and true source distributions. A small value means that the recovered marginal has the correct overall shape, spread, and tails. Third, Amari error evaluates the mixing matrix while respecting ICA's scale and permutation ambiguities. Let

$$G = A_{\text{est}}^{+} A_{\text{true}}, \qquad P = |G|,$$

where A_{est}^{+} is the Moore–Penrose pseudoinverse. For d sources, the Amari error is

$$\text{AE}(A_{\text{est}}, A_{\text{true}}) = \frac{1}{2d}\left[\sum_{i=1}^{d}\left(\frac{\sum_{j=1}^{d} P_{ij}}{\max_j P_{ij}} - 1\right) + \sum_{j=1}^{d}\left(\frac{\sum_{i=1}^{d} P_{ij}}{\max_i P_{ij}} - 1\right)\right].$$

The value is zero when G is exactly a scaled permutation matrix. These metrics are used together: the distribution plots make the result visually interpretable, the source correlations check observation-level recovery, and Amari error checks the learned mixing structure.

4.5 Results

4.5.1 Recovered source distributions

Figure 4.4 provides the main visual result. The grey histograms are the true simulated sources, while the blue and red outlines are the FastICA and SEM2 reconstructions. For both the sharply peaked Laplace source and the heavier-tailed Student-t source, the recovered curves almost completely overlap the truth. The agreement is not restricted to the centre of the distributions: both methods also reproduce the asymmetric finite-sample fluctuations and the observations in the tails of the representative data set.

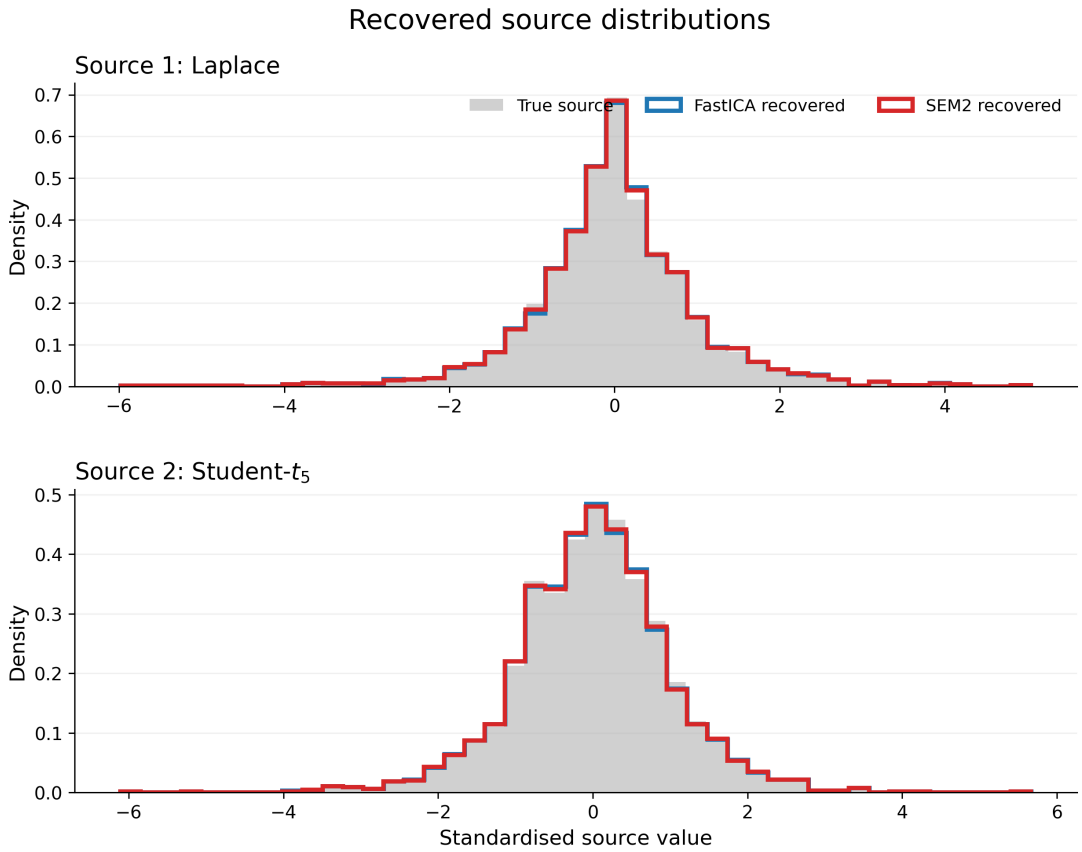


Fig. 4.4 Recovered source distributions in the representative ICA data set. Grey histograms show the true latent sources, blue outlines show FastICA recovery, and red outlines show SEM2 recovery after resolving the source-order and sign ambiguities for evaluation. Close overlap with the grey distribution indicates successful separation of the observed mixtures.

The distributional comparison establishes that the recovered components have the correct marginal laws, but it does not by itself guarantee that the source value for each observation is correct. Figure 4.5 therefore compares the true and recovered values observation by observation. All four point clouds lie tightly along the diagonal. In the representative data set, the source correlations range from 0.9997 to 1.0000, showing that both methods recover the individual latent signals rather than merely producing samples with similar histograms.

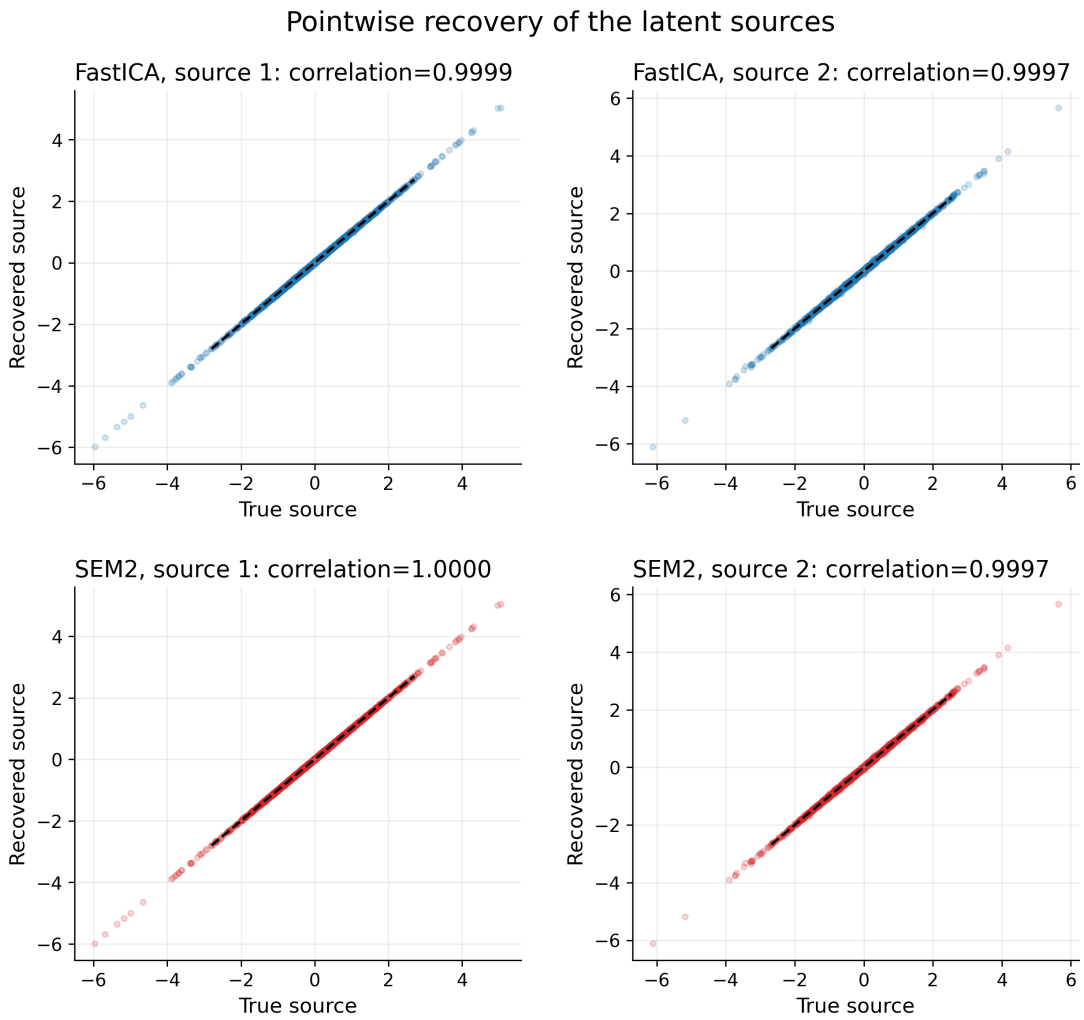


Fig. 4.5 Pointwise recovery of the two latent sources in the representative data set. Each point compares a true source value with the corresponding recovered value. The dashed diagonal represents exact recovery; the tight concentration around this line shows that FastICA and SEM2 reconstruct the source values for individual observations.

4.5.2 Repeated-simulation recovery

A single favourable data set is not sufficient evidence of reliable separation. The experiment is therefore repeated over 50 independently simulated data sets. Figure 4.6 shows the full seed-level results, and Table 4.2 reports the median values.

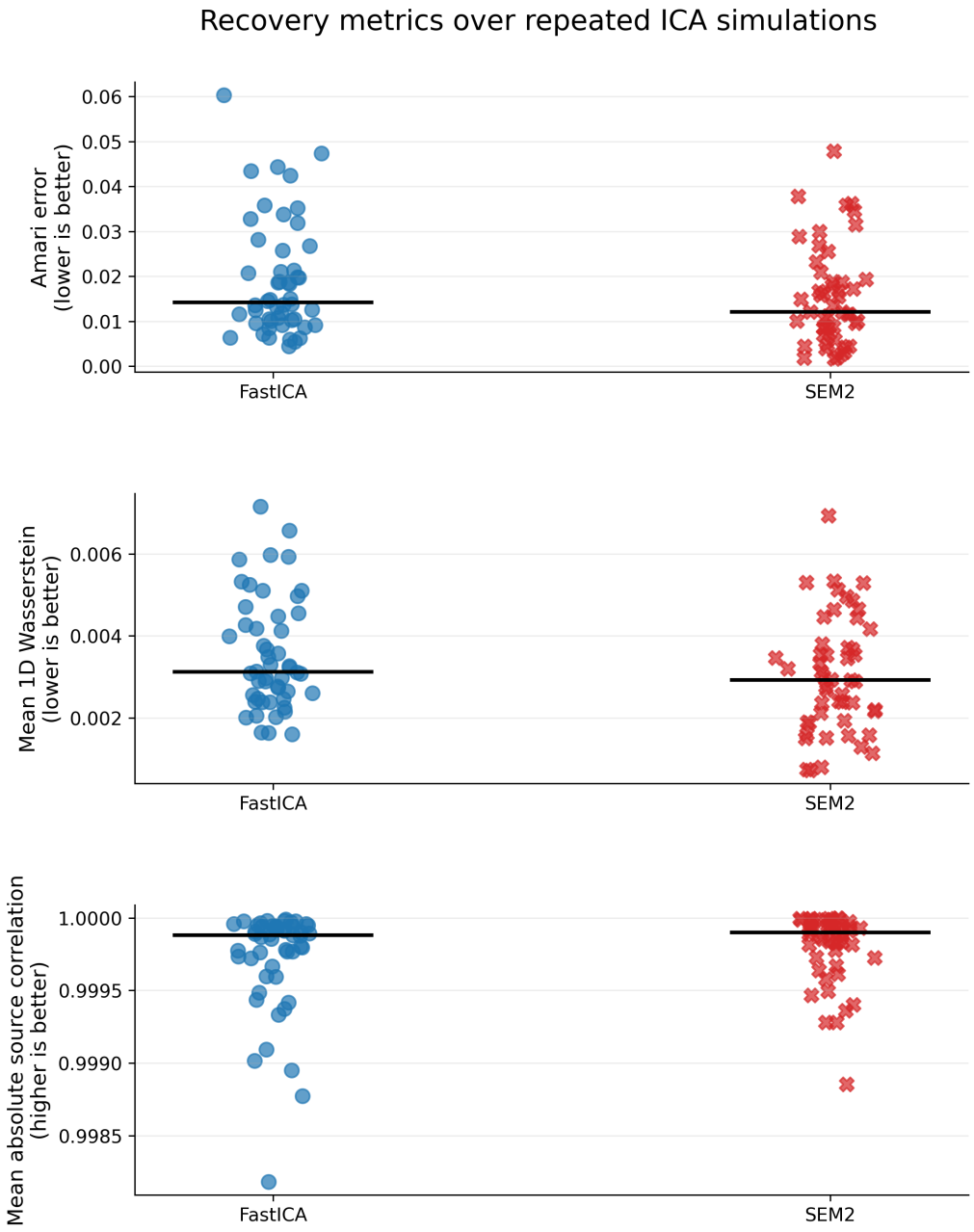


Fig. 4.6 Seed-level recovery metrics over 50 independent ICA simulations. Each marker represents one simulated data set, and the horizontal black segment is the method median. Both methods achieve small matrix and distribution errors and source correlations very close to one.

Table 4.2 Two-source ICA recovery over 50 repeated simulations. Lower values are better for Amari error and Wasserstein distance; higher values are better for absolute source correlation. Runtime is the median wall-clock time for one fitted data set.

Method	Median Amari error	Median Wasserstein	Median absolute source correlation	Median runtime per data set
FastICA	0.0142	0.0031	0.9999	0.0015 s
SEM2	0.0121	0.0029	0.9999	0.2793 s

Both methods recover the model accurately across the repeated simulations. SEM2 has a slightly lower median Amari error and Wasserstein distance (0.0121 versus 0.0142 and 0.0029 versus 0.0031) while both median source correlations round to 0.9999. The differences are small relative to the already low errors, so the appropriate conclusion is that SEM2 matches the specialised FastICA benchmark in this controlled two-source problem.

The clearest disadvantage of SEM2 is computational execution time. Its median fitting time is approximately 0.279 seconds per data set, compared with approximately 0.0015 seconds for FastICA. FastICA is a highly specialised fixed-point algorithm with built-in whitening, whereas the SEM2 result is obtained by repeatedly optimising the graph likelihood from several initialisations. The additional generality of the structural representation therefore carries a substantial runtime cost in this small problem.

4.5.3 Recovery of the mixing structure

Figure 4.7 provides a matrix-level check for the representative data set. After resolving the source-order and sign ambiguities, the row-normalised magnitude of

$$A_{\text{est}}^{+} A_{\text{true}}$$

should be close to the identity matrix. Both heatmaps have diagonal entries equal to one after row normalisation and off-diagonal entries no larger than approximately 0.02. Thus, each recovered component corresponds almost entirely to one true source, with very little cross-contamination from the other source.

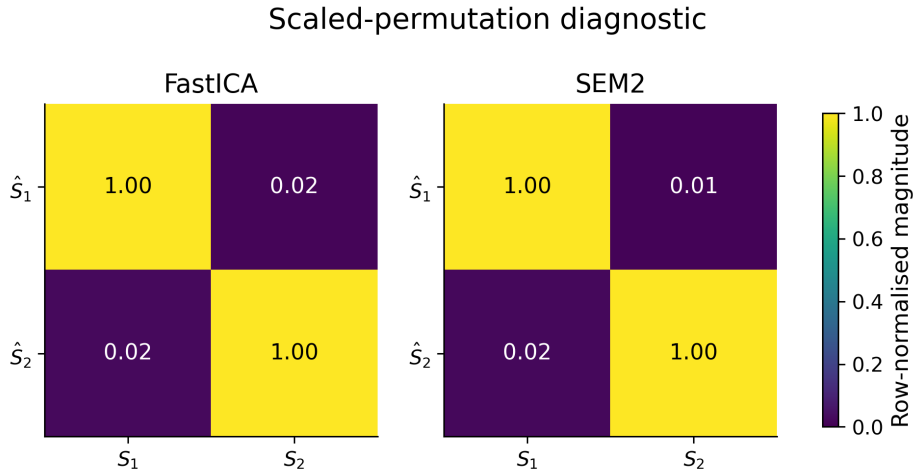


Fig. 4.7 Scaled-permutation diagnostic for the representative ICA data set. Each heatmap shows the row-normalised magnitude of $A_{\text{est}}^{+}A_{\text{true}}$ after resolving the source-order and sign ambiguities. A diagonal pattern indicates that each recovered component is associated with one true latent source.

4.6 Interpretation

This application demonstrates that SEM2 can fit a non-gaussian latent-variable model rather than being restricted to the gaussian covariance setting used in the Political Democracy chapter. The source distributions, mixing equations, and latent-variable declarations are all specified within the SEM2 graph. The fitted likelihood then uses that graph to recover both the mixing matrix and the latent source values.

The evidence is layered: Figures 4.4 and 4.5 show directly that SEM2 recovers both the source marginals and the observation-level latent signals. Table 4.2 and Figure 4.6 show that this recovery is stable over 50 simulated data sets. Figure 4.7 then verifies the corresponding mixing structure. Taken together, these checks give an informative view of how well SEM2 does to achieve similar results as FastICA.

The scope of the claim is however, limited when the sole task is efficient ICA, as FastICA remains the natural specialised method because its runtime advantage is considerable. The contribution of SEM2 is different: the same package-level graph representation used for regression and measurement models can also express and fit this non-gaussian signal-separation problem. The result therefore establishes ICA as a successful third application of the framework while making the computational trade-off explicit.

The next chapter moves from latent-signal recovery to causal policy evaluation. There the structural model is nonlinear, treatment changes at a known threshold, and SEM2 is compared with a specialised regression-discontinuity estimator.

Chapter 5

Application Study IV: Policy Evaluation at a Treatment Threshold

5.1 Applied problem and purpose

Many public policies allocate treatment using a numerical eligibility rule [8, 24, 36]. A scholarship may be awarded above an examination-score threshold [32], a programme may become available below an income threshold [16], or an intervention may begin once a clinical score crosses a specified value [15]. Regression discontinuity designs (RDDs) use this assignment rule to estimate a local treatment effect at the threshold [22, 26] .

Let R denote the running variable and suppose treatment is assigned by the sharp rule

$$D = \mathbb{1}\{R \geq 0\}.$$

The causal quantity of interest is the discontinuity in the conditional outcome mean at the cutoff:

$$\tau_{\mathrm{RD}} = \lim_{r \downarrow 0} \mathbb{E}[Y \mid R = r] - \lim_{r \uparrow 0} \mathbb{E}[Y \mid R = r].$$

The identification argument is local. If the untreated response mechanism varies smoothly through the threshold, any discontinuous change in the conditional mean at $R = 0$ can be attributed to treatment assignment.

This chapter evaluates SEM2 in a policy-threshold problem. The experiment is controlled using synthetic data so that the true cutoff effect and the complete cutoff outcome distributions are known. However, the design is chosen to mimic an applied RDD task: estimating a local policy effect when the conditional mean is nonlinear and when the outcome distribution may become irregular near the treatment threshold.

The classical reference is `rdrobust`, fitted using a local-quadratic point estimator with cubic bias correction [6]. A fixed local-quadratic regression is included as a transparent polynomial comparison. SEM2 is fitted as two side-specific nonlinear structural models, one below and one above the cutoff, using Sinkhorn divergence [12, 17, 31] to match the

observed and generated local distributions. The main question is whether SEM2 can estimate the cutoff contrast competitively while additionally recovering the outcome distributions on both sides of the threshold.

5.2 RDD designs

Both experiments use the same running variable and the same nonlinear conditional mean. For observation i,

$$R_i \sim \text{Unif}(-1,1), \qquad D_i = \mathbb{1}\{R_i \geq 0\},$$

and

$$Y_i = \mu(R_i) + \tau_{\text{RD}} D_i + \varepsilon_i.$$

The smooth baseline response is

$$\mu(r) = 0.50r + 0.80r^2 - 0.40r^3 + \sin(2r),$$

and the treatment discontinuity is fixed at

$$\tau_{\text{RD}} = 1.$$

Since $\mu(r)$ is continuous at zero, the true conditional-mean jump is exactly one in both experiments. The two designs differ only in the distribution of the disturbance ε_i.

5.2.1 Simple Non-Linear RDD problem

The first design is a controlled nonlinear benchmark with homoskedastic gaussian noise:

$$\varepsilon_i = 0.25 Z_i, \qquad Z_i \sim N(0,1).$$

Consequently, the cutoff distributions are

$$Y \mid R = 0^- \sim N(0, 0.25^2),$$

and

$$Y \mid R = 0^+ \sim N(1, 0.25^2).$$

This problem tests whether the three methods recover a known discontinuity when the conditional mean is nonlinear but the local noise distribution is well behaved.

5.2.2 Heavy Tail RDD problem

The second design retains the same conditional mean and the same treatment effect but deliberately makes the local outcome distribution difficult near the cutoff. Define

$$\sigma(r) = 0.16 + 0.90 \exp\left\{ -\left(\frac{r}{0.075}\right)^2 \right\}.$$

The scale therefore rises sharply in a narrow neighbourhood around $R = 0$ and returns toward 0.16 away from the cutoff. The error is

$$\varepsilon_i = \sigma(R_i) Z_i,$$

where

$$Z_i \sim N(0,1) \quad \text{for } R_i < 0,$$

and

$$Z_i \sim \frac{t_{2.5}}{\sqrt{2.5/(2.5-2)}} \quad \text{for } R_i \geq 0.$$

The right-side Student-t disturbance is scaled to unit variance. Thus, both sides have mean-zero noise and the true conditional-mean discontinuity remains one, but the treated-side outcome distribution is much heavier-tailed. Importantly, the stress is concentrated near the threshold, where RDD estimators obtain most of their identifying information.

5.3 SEM2 specification and classical reference

The classical reference is `rdrobust` with polynomial order $p = 2$, bias-correction order $q = 3$, a triangular kernel, and MSE-optimal data-driven bandwidth selection. The fixed local-quadratic comparison uses the same polynomial order with a fixed bandwidth of $h = 0.20$ on both sides.

SEM2 is fitted separately below and above the threshold. Within each side-specific local window, the running variable and outcome are standardised. The same nonlinear SEM2 formula is then fitted on each side:

```
eps_r, eps_y ~ Normal(0,1);
Z <- n(eps_r);
Y <- n(Z, eps_y);
unobs(eps_r,eps_y)
```

Listing 5.1 Side-specific SEM2 specification for the RDD experiment. The latent variables `eps_r` and `eps_y` generate the observed standardised running variable `Z` and observed standardised outcome `Y`. The symbol `n(...)` denotes a learned nonlinear structural mechanism. The same formula is fitted independently below and above the cutoff.

The declaration `unobs(eps_r,eps_y)` makes the exogenous noises latent. Since `Z` and `Y` are not included in `unobs(...)`, they are treated as observed variables. The first nonlinear equation learns the marginal distribution of the standardised running variable within the local window. The second learns the conditional outcome mechanism from the running variable and an independent outcome-noise source.

For each side, the model generates samples

$$(\widetilde{Z},\widetilde{Y}) \sim P_{\theta,\pm},$$

and the parameters are learned by minimising the Sinkhorn divergence between the observed and generated pairs in that local window. The two fitted models are then evaluated at the standardised value corresponding to the original cutoff $R = 0$. This produces generated samples from

$$\widehat{P}_{-}(Y \mid R = 0^{-}) \quad \text{and} \quad \widehat{P}_{+}(Y \mid R = 0^{+}).$$

The SEM2 cutoff-effect estimate is the difference between their generated means:

$$\widehat{\tau}_{\text{SEM2}} = \widehat{\mathbb{E}}[Y \mid R = 0^{+}] - \widehat{\mathbb{E}}[Y \mid R = 0^{-}].$$

Figure 5.1 shows the structural graph. The left and right regions at the discontinuity have the same graph but are fitted to different local samples and therefore learn different structural mechanisms.

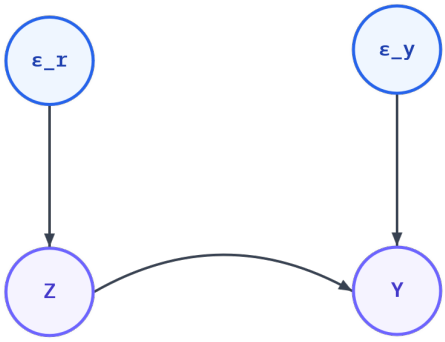


Fig. 5.1 Side-specific SEM2 graphs for the RDD experiment. Blue nodes are latent exogenous noise variables and purple nodes are observed standardised variables. Separate models are fitted below and above the cutoff. At $R = 0$, the generated left and right outcome distributions are used to compute the SEM2 estimate of the discontinuity.

Table 5.1 summarises the final experiment settings. These fitting configurations selected for the comparison benchmark methods were based on the best performing settings after extensive ablation studies. For SEM2, however, the substantially higher computational cost limited the scope of the evaluation. The fitting objective was therefore chosen based on a quick preliminary comparison between Sinkhorn distance and Energy Distance, in which Sinkhorn performed slightly better and was consequently selected for the results presented here. Although a more comprehensive ablation study and hyperparameter sweep would have been desirable, the available time and computational budget did not permit a thorough optimisation of SEM2. As a result, its reported performance likely understates its full potential; nevertheless, it achieved competitive results despite the relatively naive configuration, as seen later.

Table 5.1 Experimental setup for the policy-threshold study.

Component	Specification
Running variable	$R \sim \text{Unif}(-1,1)$.
Treatment rule	$D = \mathbb{1}\{R \geq 0\}$.
True cutoff effect	$\tau_{\text{RD}} = 1$.
Sample size	$n = 2{,}000$ observations per simulated data set.
Repeated simulations	50 independent seeds per design.
Representative data set	Seed 2.
Classical reference	`rdrobust`, $p = 2$, $q = 3$, triangular kernel, MSE-optimal bandwidth.
Polynomial comparison	Fixed local quadratic, $h = 0.20$.
SEM2 architecture	Two hidden layers of width 64 with `tanh` activations for each nonlinear node.
SEM2 objective	Sinkhorn divergence on the local observed/generated (Z,Y) pairs.
SEM2 optimiser	Adam, learning rate 7.5×10^{-4}, cosine learning-rate schedule.
SEM2 fitting budget	Up to 3,500 steps for the representative fits and 1,700 steps for each repeated-simulation fit.
Sinkhorn regularisation	Blur parameter 0.20.
Cutoff evaluation	10,000 generated outcomes per side.

5.4 Conditional-mean recovery

Figure 5.2 compares the fitted conditional means over the full running-variable range. The true simulator curve is drawn in yellow. The red curve is the `rdrobust` local-quadratic fit, the green curve is the fixed local-quadratic fit, and the purple curve is the SEM2 generated conditional mean. Each legend reports the estimated discontinuity from the representative data set.

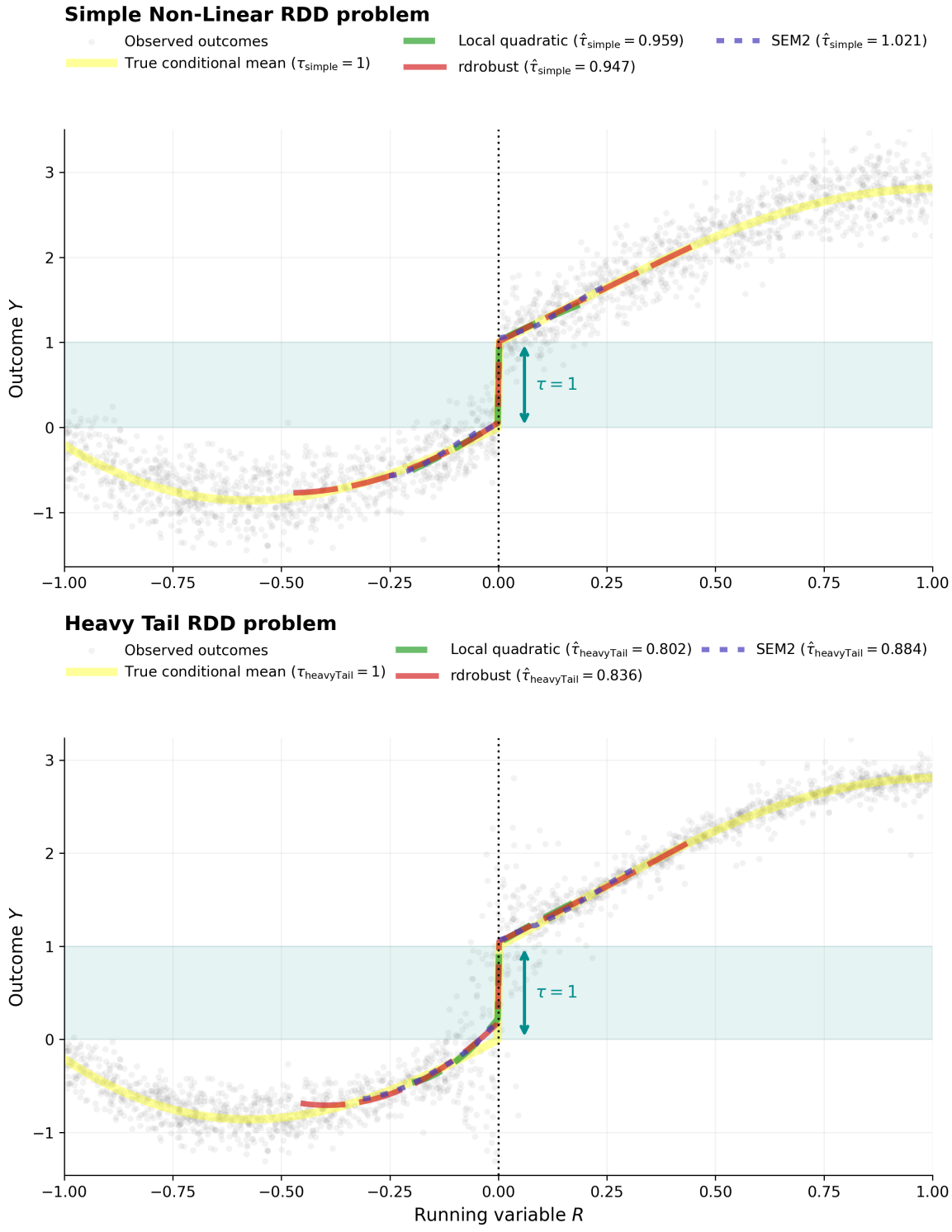


Fig. 5.2 Fitted conditional means for the simple and heavy tail non-linear RDD problems. The solid yellow curve is the true conditional mean, the red curve is `rdrobust`, the green curve is the fixed local-quadratic fit, and the purple curve is SEM2. The vertical black dotted line marks the treatment cutoff and the blue interval $\tau = 1$ is the target of interest.

In the Simple Non-Linear problem, the representative estimates are

$$\widehat{\tau}_{\text{rdrobust}} = 0.947, \qquad \widehat{\tau}_{\text{quadratic}} = 0.959, \qquad \widehat{\tau}_{\text{SEM2}} = 1.021.$$

All three methods recover the discontinuity closely. The classical fits follow the local mean well inside their estimation windows, while the SEM2 generated mean nearly coincides with the true curve around the threshold.

In the Heavy Tail problem, the representative estimates are

$$\widehat{\tau}_{\text{rdrobust}} = 0.836, \qquad \widehat{\tau}_{\text{quadratic}} = 0.802, \qquad \widehat{\tau}_{\text{SEM2}} = 0.884.$$

The near-cutoff heavy-tailed observations make all three estimates less accurate on this particular data set. SEM2 is closest to the true value among the three, although the figure should be interpreted together with the repeated-seed comparison rather than as evidence from a single seed.

5.5 Repeated-simulation cutoff-effect accuracy

Figure 5.3 shows the empirical distributions of

$$\widehat{\tau} - \tau_{\text{RD}}$$

over 50 independently simulated data sets for each design. The horizontal zero line represents exact recovery of the true discontinuity.

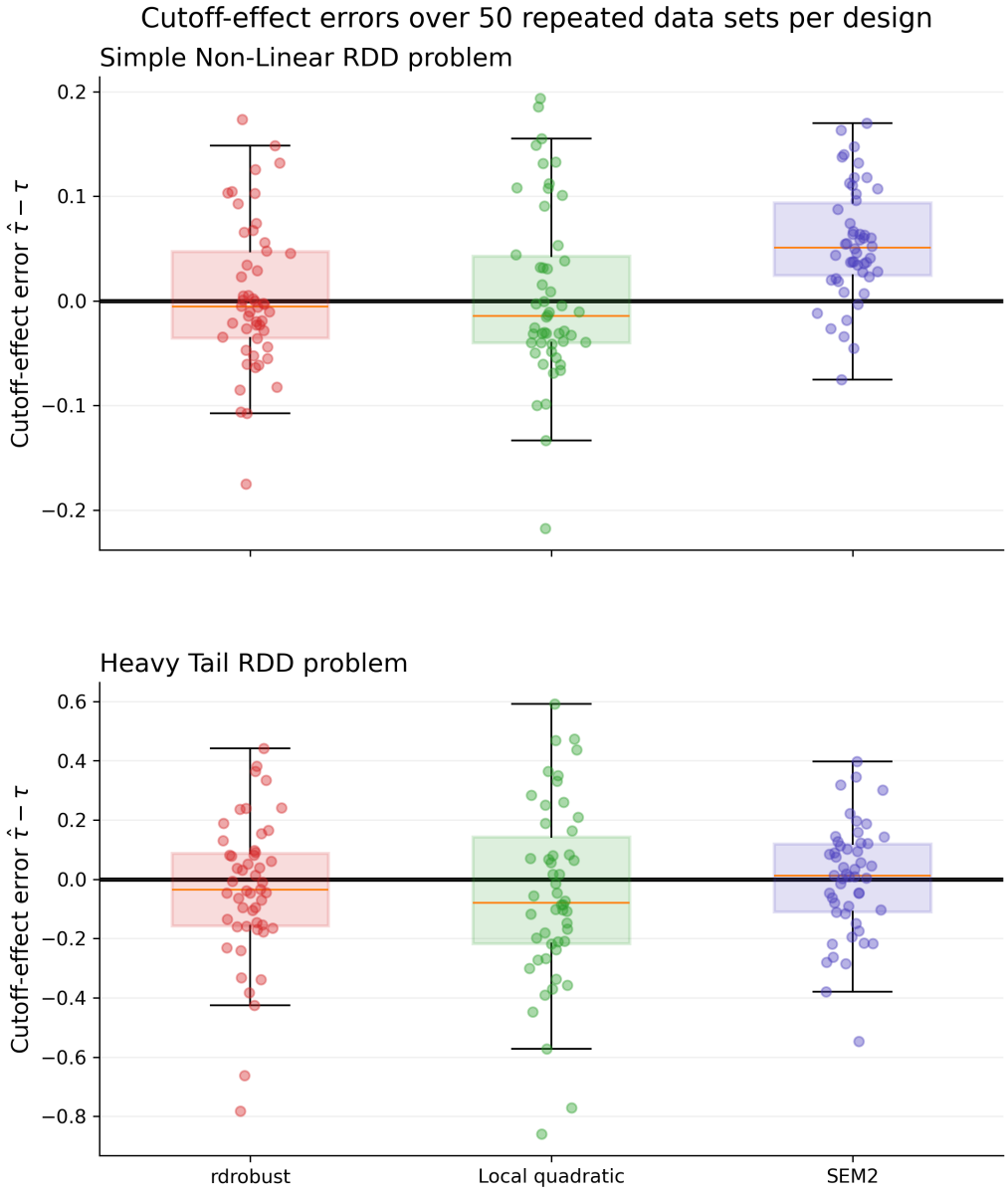


Fig. 5.3 Cutoff-effect errors over 50 repeated simulations for each RDD problem. Each point represents one simulated data set, and the boxplots summarise the empirical error distribution. The three methods are `rdrobust`, fixed local quadratic, and SEM2.

Table 5.2 gives the numerical summaries. In the Simple Non-Linear problem, `rdrobust` has the smallest RMSE, followed closely by SEM2 and the fixed local-quadratic fit. SEM2 has a small positive bias in this regular design. In the Heavy Tail problem, SEM2 has the lowest RMSE and is essentially unbiased across the 50 repetitions, whereas both local-polynomial methods are more variable and negatively biased.

Table 5.2 Cutoff-effect accuracy and computational cost over 50 repeated simulations. Bias is the mean of $\widehat{\tau} - \tau_{\mathrm{RD}}$, MAE is mean absolute error, RMSE is root mean squared error, and runtime is the median wall-clock time per simulated data set.

Design	Method	Bias	MAE	RMSE	Median runtime
Simple Non-Linear	`rdrobust`	+0.004	0.053	0.070	0.048 s
	Fixed local quadratic	+0.005	0.064	0.083	0.0002 s
	SEM2	+0.055	0.064	0.077	109.17 s
Heavy Tail	`rdrobust`	−0.035	0.177	0.240	0.048 s
	Fixed local quadratic	−0.050	0.243	0.307	0.0002 s
	SEM2	−0.001	0.144	0.186	116.66 s

The clean-design result is consistent with the role of specialised RDD estimators: when the conditional mean is smooth and the errors are regular, a local-polynomial method is statistically efficient and extremely fast. The Heavy Tail design gives a different result. The large, highly variable observations are concentrated near the threshold, so they receive substantial local weight. SEM2's distribution-matching fit is more stable for the mean contrast in this problem setting, producing a lower RMSE than either local-polynomial comparison.

5.6 SEM2's generated outcome distributions at the cutoff

A local-polynomial RDD estimator targets the conditional-mean discontinuity. SEM2 instead returns two generative structural models and therefore produces complete fitted outcome distributions at the threshold. Figures 5.4 and 5.5 compare the true simulator distributions with the SEM2-generated distributions.

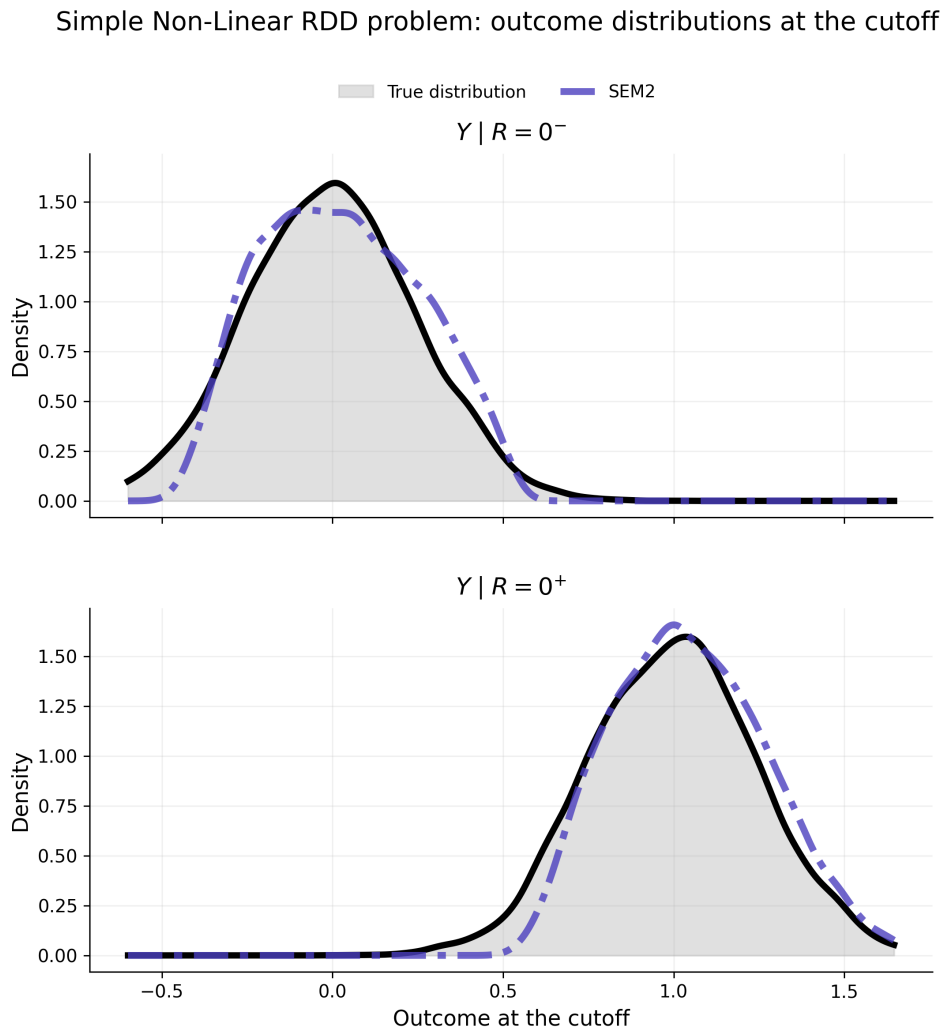


Fig. 5.4 True and SEM2-generated outcome distributions at the cutoff for the Simple Non-Linear RDD problem. The upper panel shows $Y \mid R = 0^-$ and the lower panel shows $Y \mid R = 0^+$.

In the Simple Non-Linear problem, the fitted distributions are close to the simulator on both sides. The Wasserstein distances are 0.032 on the top and 0.048 on the bottom. The SEM2 standard deviations, 0.229 and 0.226, are also close to the true values, 0.249 and 0.250.

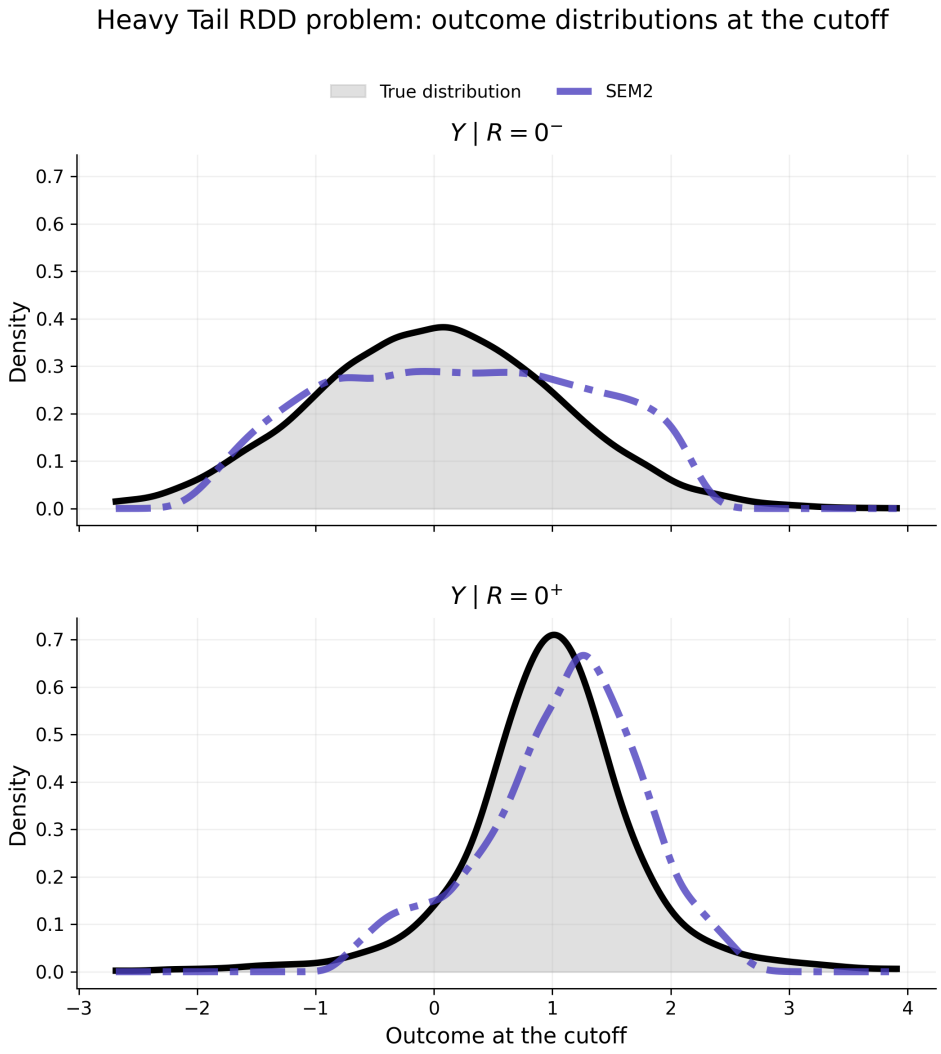


Fig. 5.5 True and SEM2-generated outcome distributions at the cutoff for the Heavy Tail RDD problem. The upper panel shows $Y \mid R = 0^-$ and the lower panel shows $Y \mid R = 0^+$. The right-side SEM2 fit captures part of the increased dispersion but remains less heavy-tailed than the simulator.

The Heavy Tail problem is more difficult. For one side, SEM2 reproduces the scale of the outcome distribution reasonably well but shifts its mean upward. For the other, SEM2 captures a wider distribution than in the simple experiment but remains under-dispersed:

$$\mathrm{sd}_{\mathrm{SEM2}}(Y \mid R = 0^+) = 0.678, \qquad \mathrm{sd}_{\mathrm{true}}(Y \mid R = 0^+) = 1.064.$$

The Wasserstein distance is 0.186 and it looks like the model captures the main location shift and some of the local distributional change, but it does not fully reproduce the most extreme Student-t tail.

The distributional output is nevertheless useful because it supports quantities that are unavailable from the mean contrast alone, including conditional variances, quantiles, and tail probabilities. These generated outcome distributions describe outcome variability under the fitted structural model but they are not for standard errors for $\widehat{\tau}_{\text{SEM2}}$ as formal uncertainty quantification for the causal contrast would require an additional inferential procedure, such as repeated refitting under a bootstrap.

5.7 Interpretation

The RDD application demonstrates two distinct properties of SEM2. First, the side-specific nonlinear models can estimate a local conditional-mean discontinuity. In the Simple Non-Linear design, the specialised `rdrobust` estimator remains the strongest method, as expected under regular local-polynomial conditions. In the Heavy Tail design experiment, SEM2 gives the smallest repeated-simulation RMSE among the three fitted methods and is approximately unbiased.

Second, SEM2 estimates a richer object than the local mean alone. The fitted structural models generate outcome distributions on both sides of the threshold, making it possible to examine distributional changes that accompany the mean discontinuity. This feature is particularly relevant in applications where a policy may affect not only the average outcome but also its spread or tail risk.

The additional flexibility comes with clear limitations. SEM2 requires approximately two minutes per simulated data set in this experiment, compared with a fraction of a second for the local-polynomial methods. The Heavy Tail distribution is only partially recovered, and SEM2 currently provides no formal standard error for the estimated cutoff contrast. The conclusion is therefore this application shows that SEM2 can provide a competitive local mean estimate under a deliberately difficult distributional design while also supplying an interpretable generative description of the local outcome mechanisms.

Chapter 6

Discussion, Conclusion and Future Directions

6.1 Discussion

This report evaluated SEM2 against external statistical targets and across the application studies, the same formula-driven graph representation was used for a controlled calibration model, a latent-variable social-science problem, non-gaussian source separation, and a nonlinear policy-threshold problem. These models would ordinarily be analysed using separate specialised tools, but SEM2 represents them through a single unified probabilistic-programming interface and exposes their coefficients, moments, latent structures, generated outcomes, and intervention contrasts through a common API.

The main strength of this approach is its combination of generality and model awareness. When exact structure is available, SEM2 exploits it: linear models use differentiable matrix algebra and linear-gaussian models use an exact observed-data likelihood. When a closed-form density is unavailable, the same graph can instead be sampled and fitted through a distributional discrepancy. The application chapters show why this distinction matters: classical linear calculations are deterministic and stable, whereas nonlinear generative fitting provides richer distributional output but requires substantially more computation and more careful objective design.

The external comparisons also clarify the role of a general-purpose solver. SEM2 closely reproduces the statistical objects targeted by OLS, `lavaan`, and FastICA in the various scenarios considered here. In the RDD study, SEM2 additionally produces complete fitted outcome distributions on both sides of the cutoff, rather than only a local mean contrast. This makes quantities such as conditional variances, quantiles, and tail probabilities accessible from the same fitted structural model.

However, SEM2's generality does not remove the advantages of specialised methods. `lavaan` already provides mature standard errors and confidence intervals. FastICA is substantially faster for the source-separation problem for which it was designed. `rdrobust` provides

specialised bandwidth selection and bias-corrected inference at negligible computational cost relative to SEM2. Although, the results demonstrate that SEM2 can recover the intended targets in representative regimes; they do not establish uniform superiority across all SEMs or causal designs. SEM2 therefore currently offers its greatest advantage when a common structural representation or a full generative distribution is scientifically useful, rather than when only a classical estimate is required.

6.2 Future directions

The first priority is uncertainty quantification. For linear-gaussian models, hessian-based standard errors and delta-method transformations could be added to the likelihood framework. Bootstrap or sandwich procedures could provide more robust alternatives. For nonlinear causal contrasts, a full refitting bootstrap is the most immediate route, although analytic theory is needed to account jointly for data sampling, optimisation, model selection, and intervention functionals.

The second priority is computational efficiency. non-linear distributional fitting would benefit from faster GPU batching, improved early stopping, model-specific initialisation, and more efficient loss function approximations. More automatic operator selection could also use the declared graph structure to choose exact likelihood, minimum-distance, or simulation-based fitting without requiring the user to manage these distinctions manually.

Finally, the application range should be widened. Important extensions include mixed discrete-continuous graphs, noisy and higher-dimensional source-separation models, more real world policy dataset problems, and proxy-variable models for hidden confounding. Proximal causal learning [35] is a particularly interesting next application because it would test whether the same graph language can represent epidemiological problems in which relevant confounders are latent but imperfect proxies are observed.

6.3 Conclusion

Report 1 showed that a difficult partial-identification problem could be recast as computation over structural models. Report 2 developed that proof of concept into a model-aware and externally validated software framework. Its central contribution is not a replacement for every specialised estimator, but a common structural language in which apparently different applied problems from various domains can be fitted, sampled, and interrogated.

The results show that SEM2 can reproduce classical solutions and additionally can retain a generative interpretation when moving beyond those settings. This establishes

SEM2 as a flexible foundation for further work on semiparametric structural models, while also identifying the principal remaining challenges: computational cost, formal uncertainty quantification, and robust learning in more complex latent and nonlinear systems.

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